
Redox-decoupled electrolysis for direct air capture of CO₂

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Supplementary Discussions

Discussion 1: Current efficiency (CE) limitation via H⁺ or OH⁻ crossover through cation exchange membrane (CEM)

During electrochemical operation, the total ion flux through CEM was calculated by (E1):

$$J_{\text{total}} = \frac{I}{AnF} \quad (\text{E1})$$

Where I is the electrolyzer operating current density (A), A is the geometric area (cm²), n is the number of electrons transferred per mole of product formation, and F is the faraday's constant (96485 C mol⁻¹). The calculated total ion flux induced by electromigration was calculated to range from 518 to 2073 nmol cm⁻² s⁻¹ at current densities between 50 and 200 mA cm⁻².

The decrease in CE is attributed to two possible factors: undesired H⁺ crossover and OH⁻ crossover through CEM.

We first identified a threshold H⁺ concentration of 0.009M (~pH 2) in the CO₂ liberation solution at which a decline in CE begins to occur (Supplementary Fig. 4). This effect, however, is mitigated by the buffering capacity from CO₃²⁻ present in CO₂ liberation solution. According to inorganic carbon equilibrium analysis, complete CO₂ release is achieved below pH 4¹. Therefore, we selected a cutoff pH of 3 for the redox-decoupled configuration, which ensures full CO₂ gas release while maintaining a high CE (Supplementary Fig. 5).

We then quantified the OH⁻ crossover flux through CEM under simulated extreme conditions, using a pH 14 cathode compartment. The measured OH⁻ crossover flux ranged from 82 to 391 nmol cm⁻² s⁻¹ under current densities from 50 to 200 mA cm⁻², identifying it as a primary contributor to the observed decline in CE (Supplementary Fig. 6).

The normalized OH⁻ molarity is calculated by (E2, E3) and the OH⁻ crossover flux is estimated by (E4):

$$c\text{OH}_{\text{crossover}}^- = 10^{(\text{pH}_{\text{anolyte}} - 14)} \times \frac{V_{\text{anolyte}}}{1000} \quad (\text{E2})$$

$$\text{normalized OH}_{\text{crossover}}^- = \frac{c\text{OH}_{\text{crossover}}^-}{A_{\text{anode}}} \quad (\text{E3})$$

$$J_{\text{OH}_{\text{crossover}}^-} = \frac{\text{normalized OH}_{\text{crossover}}^-}{t_{\text{operation}}} \quad (\text{E4})$$

where $c\text{OH}_{\text{crossover}}^-$ is the total amount of OH⁻ crossover from cathode compartment through CEM (mole), the $\text{pH}_{\text{anolyte}}$ is the pH measurement after calibration at the anolyte, V_{anolyte} is the total

anolyte volume, A_{anode} is the geometric area of the anode catalyst (cm^2), and $t_{\text{operation}}$ is the total electrolysis operation time (s).

In conclusion, K^+ serves as primary charge carriers by maintaining a weakly alkaline environment at the P1 anode during the electrochemical operation. However, as the concentration of OH^- increases over time, inevitable OH^- crossover through the CEM, leading to a gradual decline in CE.

The local pH near the anode catalyst, induced by OH^- crossover, was estimated to range from 9.7 to 10.3 under current densities of 50 to 200 mA cm^{-2} , respectively, analyzed by in-situ Raman analysis (Supplementary Figs. 7–9). These values reflect conditions without any buffering contribution from CO_3^{2-} . In the practical redox-decoupled configuration, the local pH near the anode is expected to be slightly higher due to the presence CO_3^{2-} . Therefore, the pH near the anode is estimated to remain above 10 during the entire electrochemical operation at current densities above 50 mA cm^{-2} .

Discussion 2: Theoretical and experimental analysis of oxygen evolution reaction (OER)

The theoretical full-cell voltage for the hydrogen evolution reaction (HER) and OER in the electrolyzer were analyzed under standard conditions. The operating temperature was at 298 K, with both product and reactant gas pressures maintained at 1 atm, and the electrolyte containing 0.25M K_2SO_4 . The steady-state pH values of catholyte and anolyte were assumed to be 14 and 7, respectively.

HER at the electrolyzer cathode (E5):



OER at the electrolyzer anode (E6):



The theoretical half-cell potential was calculated from the Gibbs free energy difference between reactants and products at standard conditions (E7):

$$E^0 = -\frac{\Delta G^0}{nF} \quad (\text{E7})$$

where E^0 is the standard half-cell reduction potential, ΔG^0 is the change in Gibbs free energy, F is the Faraday's constant (96485 C mol^{-1}), and n is the number of electrons transferred per mole product generated (2 for HER and 4 for OER).

The Nernst equation was applied to calculate the voltage induced by non-standard concentrations of reactants and products (E8):

$$E = E^0 - \frac{RT}{nF} \ln \frac{\prod \alpha_p^\nu}{\prod \alpha_r^\nu} \quad (\text{E8})$$

where R is the universal gas constant ($8.314 \text{ J mol}^{-1} \text{ K}^{-1}$), $\prod \alpha_p^\nu$ is the product of activity of product with respective stoichiometric coefficients, $\prod \alpha_r^\nu$ is the product of activity of reactant with respective stoichiometric coefficients.

The theoretical full-cell voltage was calculated based on cathode and anode potential (E9):

$$E_{\text{full-cell}} = E_{\text{cath}} - E_{\text{an}} \quad (\text{E9})$$

The theoretical full-cell voltage for HER and OER was calculated to be -1.64 V , excluding contributions from activation overpotentials, concentration, and ohmic losses.

During P1 operation, OER experiences a sharp increase when a full-cell voltage exceeds -2 V ([Supplementary Fig. 10](#)). In the galvanostatic charge-discharge experiments of the TEMPO-S flow battery, the coulombic efficiency exceeded 99% at current densities ranging from 50 to 200 mA cm^{-2} , indicating minimal contribution from the OER under the tested conditions ([Supplementary Fig. 72](#)).

Discussion 3: In-situ Raman spectroscopy configuration

In-situ Raman spectra were performed using a customized flow cell configuration to investigate the local pH near the anode catalyst ([Supplementary Fig. 7](#)). The electrocatalytic active area was $1 \text{ cm} \times 1 \text{ cm}$. An 80 nm sputtered gold film on hydrophilic carbon electrode was placed in the working compartment, facing the objective lens. A PtC on hydrophilic carbon electrode was installed in the counter compartment. A Nafion 115 cation exchange membrane separated the working and counter compartment. Various rubber gaskets were used to prevent liquid leakage. Both working compartment and counter compartment were operated at a flow rate of 10 sccm. Liquid circulation in the working compartment was driven by hydrostatic pressure resulting from a height difference between the inlet and outlet reservoirs. A constant current was applied to initiate OH^- electromigration. The working electrolyte consisted of 20 mL 0.1 M TEMPO-S in 1 M KHCO_3 , and the counter electrolyte was 50 mL 1 M KOH. Raman spectra were collected using a $\times 63$ water-immersion objective lens with an exposure time of 1 seconds. A Renishaw Raman spectrometer equipped with a 532 nm excitation laser with a 1200 l mm^{-1} diffraction grating. The power of laser was kept at 50 mW with 1 s exposure time for spectral acquisition. Each spectrum was recorded by the accumulation of 60 scans.

The local pH at the P1 anode electrode was analyzed (Supplementary Fig. 9). Specifically, local pH was estimated based on CO_3^{2-} and HCO_3^- concentrations. The initial working electrolyte was 0.1 M TEMPO-S in 1 M KHCO_3 . During the electrochemical operation, OH^- crossover through HER counter compartment to TEMPO oxidation working compartment via diffusion and electromigration. This OH^- chemically reacts with HCO_3^- to form CO_3^{2-} (E10–E13). The Raman spectra obtained the HCO_3^- and CO_3^{2-} peaks at 1014 cm^{-1} and 1067 cm^{-1} , respectively. The HCO_3^- and CO_3^{2-} concentrations were calibrated based on calibration curves (Supplementary Fig. 8).



$$K = \frac{[\text{CO}_3^{2-}][\text{H}^+]}{[\text{HCO}_3^-]} \quad (\text{E11})$$

$$\text{DIC} = [\text{CO}_3^{2-}] + [\text{HCO}_3^-] \quad (\text{E12})$$

$$[\text{H}^+] = \frac{K[\text{HCO}_3^-]}{\text{DIC} - [\text{HCO}_3^-]} = \frac{K[\text{DIC} - [\text{CO}_3^{2-}]]}{[\text{CO}_3^{2-}]} \quad (\text{E13})$$

Where the K is the stoichiometric equilibrium constant (1.437×10^{-9} M), DIC represents the dissolved inorganic carbon concentration (1 M), $[\text{CO}_3^{2-}]$ and $[\text{HCO}_3^-]$ were obtained from the Raman spectra.

Discussion 3: DFT Simulation

DFT was applied to explore the interactions between the molecules and the carbon (graphene) electrode. All DFT calculations were carried out in the Vienna ab initio simulation package with a plane wave pseudo-potential implementation^{2,3}. The exchange-correlation functional was Perdew-Burke-Ernzerhof (PBE) functional and the electron-ion interactions were described by projector augmented wave potentials^{4,5}. The kinetic cut-off energy of 450 eV was used for the plane-wave expansion. The $1 \times 1 \times 1$ Monkhorst-Pack k-point mesh was used for geometry optimization, with the convergence criteria of electronic and ionic iterations being 10^{-5} eV and $0.02\text{ eV}\text{\AA}^{-1}$, respectively. The long-range Van de Waals interactions were described by the zero-damping DFT-D3 method⁶. A $15\text{ \AA}$ vacuum space was set among the periodic surfaces in the z-direction to decouple the interaction between them. Note that the side chain of the viologen molecule carries charges of -2 . To avoid the intricacies arising from net charges during DFT calculation, two protons were applied to neutralize the side chain.

The binding energy is calculated by (E14):

$$E_b = E_a - (E_s + E_m) \quad (\text{E14})$$

where E_a , E_s , and E_m are the energies of an adsorbed system, bare carbon surface, and isolated viologen molecules, respectively.

In this study, The DFT-optimized geometries of carbon electrode with adsorbed molecules were examined ([Supplementary Figs. 16–20](#)).

All the calculations of TEMPO derivatives were performed using the Gaussian 16 programs ([Supplementary Fig. 22](#)). All the structures were fully optimized with the B3LYP method and Ahlrichs' split-valence def2-SVP basis set in 1,4 dioxane solvent employing the Polarizable Continuum Model (PCM)^{7–10}. Grimmes's DFT-D3 dispersion correction was used to describe the van der waals interaction¹¹. Vibrational frequency calculations were performed to ensure that a local minimum has no imaginary frequency.

Discussion 4: Influence of TEMPO substituents on charge distribution, ion pairing, and net charge effects

The charge distribution effect:

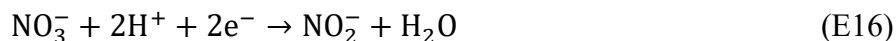
The hydroxyl (-OH) group acts as an electron-donating group (EDG), whereas the sulfate (-OSO₃) and phosphate (-OPO₃) groups function as electron-withdrawing group (EWG). EWGs pull electron density away from the TEMPO skeleton, increasing its electrophilicity and shifting the redox potential toward more positive values ([Supplementary Fig. 49](#))¹². The incorporation of EWGs reduces the electron density on the alicyclic ring of nitroxyl radicals, which not only enhances the electron transfer rate but also improves the stability of the oxidized species (TEMPO⁺) by suppressing side reactions^{13–15}. DFT results indicated that TEMPO-S exhibited the highest Gibbs free energy change (ΔG) for side reactions compared to TEMPO-OH and TEMPO-P, indicating superior stability of TEMPO-S ([Supplementary Fig. 50](#)).

The ion pairing effect:

To investigate this, we conducted CV experiments comparing chloride (Cl⁻) and sulfate (SO₄²⁻) as counterions for TEMPO derivatives ([Supplementary Fig. 51](#)). The results showed that for the TEMPO-OH case, Cl⁻ facilitates slightly faster diffusivity than SO₄²⁻, indicated by higher current density. Specifically, TEMPO-OH⁺ has a +1 net charge and will form a large Van der Waals radius in the presence of the SO₄²⁻ counterion, increasing its mass transfer resistance. In comparison, TEMPO-S reaction kinetics are similar under both Cl⁻ and SO₄²⁻.

In this work, we sought to utilize additional potassium chemicals to enhance CO₂ liberation solution ionic conductivity to reduce system overpotential. We screened common economically feasible potassium chemicals, including potassium chloride (KCl), potassium nitrate (KNO₃), and potassium sulfate (K₂SO₄). In the redox-decoupled system, hydrophilic carbon electrodes were employed for TEMPO redox reactions. We sought to avoid possible chloride oxidation reactions

with a redox potential of 1.36 V vs. SHE and avoid possible nitrate reduction reactions with a redox potential of 0.43 V vs. SHE (E15, E16)^{16,17}. As a result, we selected K₂SO₄ chemical for the supporting electrolyte.



The net charge effect:

The net charge of TEMPO / TEMPO⁺ plays a critical role in maintaining sufficient local reactant concentration near the electrode surface, thereby influencing reaction kinetics (Supplementary Fig. 52)¹⁸. Experiments were conducted to elucidate the effects of electrostatic interactions and diffusion-controlled cases. For electrostatic attraction case, the full-cell voltage remains independent of the liquid flow rate (Supplementary Fig. 53). In contrast, in diffusion-controlled cases, increasing the liquid flow rate enhances mass transport, thereby improving the full-cell voltage (Supplementary Fig. 53). For electrostatic repulsion case—TEMPO-P⁺ carries a net charge of −1—electrostatic forces repel the negatively charged species from the reduction electrode in P2. This repulsion limits the reactant's concentration and consequently limits the reaction rate, similar to the behavior observed with ferrocyanide (Supplementary Fig. 15 and Supplementary Fig. 52).

Discussion 5: Randles-Sevcik equation and Nicholson's method to estimate D_{TEMPO}, D_{TEMPO⁺}, and k⁰

Randles-Sevcik equation was applied to estimate TEMPO and TEMPO⁺ diffusion coefficients as well as Nicholson's method was used to calculate electron transfer rate constants, using following equations (E17–E19):

$$i_p = (2.69 \times 10^5) \cdot n^{3/2} \cdot A \cdot D^{1/2} \cdot C_0 \cdot v^{1/2} \quad (\text{E17})$$

$$\Psi = \frac{-0.6288 + 0.0021\Delta E_p}{(1 - 0.017\Delta E_p)} \quad (\text{E18})$$

$$\Psi = k^0 \cdot \left(\frac{\pi \cdot D_0 \cdot n \cdot F}{R \cdot T} \right)^{-1/2} \cdot v^{1/2} \quad (\text{E19})$$

Where i_p represents the peak redox current, obtained from (Supplementary Tables 4–6) (A cm^{−2}), n is the number of electrons involved in redox reaction (1), A is the glassy carbon electrode active surface area (0.071 cm²), C_0 is the bulk concentration of species (mol cm^{−3}), v is the CV scan rate (V s^{−1}), ΔE_p is the peak potential separation, obtained from (Supplementary Tables 1–3).

Discussion 6: TEMPO permeability test

The permeability of TEMPO-S across the conventional Nafion 115 and the cast SPEEK were measured using an H-cell configuration (**Supplementary Fig. 62**). This setup consisted of a donating compartment, containing 0.1 M TEMPO-S in aqueous solution, which initially oxidized to 50% TEMPO-S: 50% TEMPO-S⁺ ratio. And a receiving compartment initially filled with DI water, with each compartment initially holding 15 mL of solution and sealed with the parafilm. Magnetic bars were placed in both compartments to simulate the fluid convection.

The migration of the TEMPO-S was monitored over a week, with daily measurements taken from the receiving compartment. A 100 μL sample was collected each day, subsequently diluted, and analyzed using UV-visible spectrophotometry to quantify the concentration of TEMPO-S that permeated through the membranes.

The permeability of TEMPO-S was estimated using Fick's law (E20):

$$P_{\text{TEMPO-S}} = \frac{\Delta \ln \left(1 - \frac{2C_i}{C_0} \right) \left(\frac{V_0 L}{2A} \right)}{\Delta t} \quad (\text{E20})$$

Where $P_{\text{TEMPO-S}}$ represents the TEMPO-S permeability ($\text{cm}^2 \text{ s}^{-1}$), C_i is the TEMPO-S concentration detected in the receiving compartment, C_0 is the initial TEMPO-S concentration in the donating compartment (0.1 M), L is the initial solution volume (15 mL), A is the effective membrane surface area (1.13 cm^2), L is the membrane thickness (cm), and Δt is the time interval for sample collection (s).

Discussion 7: Redox-decoupled system minimum and practical energy consumption

The minimum energy consumption was estimated from the cyclic voltammetry analysis (E21, E22):

$$E_{\text{min}} = E_{\text{P1}} + E_{\text{P2}} = \Delta E_{\text{HER/HOR}} + \Delta E_{\text{TRR/TOR}} \quad (\text{E21})$$

$$W_{\text{min}} = \frac{E_{\text{min}} \times 96.49 \text{ kJ/mol} \cdot \text{eV}}{\eta_{\text{CE}}} \quad (\text{E22})$$

where $\Delta E_{\text{HER/HOR}}$ represents the voltage difference between hydrogen reduction and oxidation peak (0.25 V, obtained from **Supplementary Fig. 69**), and $\Delta E_{\text{TRR/TOR}}$ represents the voltage difference between TEMPO-S reduction and oxidation peak (0.04 V, obtained from **Supplementary Fig. 69**) and η_{CE} is the current efficiency of the redox-decoupled system (%).

The practical energy consumption was calculated based on two distinct electrolyzers, P1 and P2:

For carbonate:

In the CO_3^{2-} post-capture solution, two OH^- are required to capture one CO_2 molecule, and two H^+ are needed to release one CO_2 molecule (E23). Consequently, two electrons are involved in the electrochemical process. The energy consumption of the redox-decoupled system for capturing 1 tonne of CO_2 was calculated (E24):



$$w_{\text{CO}_2} = \frac{W_{\text{CO}_2}}{m_{\text{CO}_2}} = \frac{2 \cdot F \cdot (V_{\text{P1}} + V_{\text{P2}})}{M_{\text{CO}_2} \cdot \eta_{\text{CE}}} \quad (\text{E24})$$

where V_{P1} and V_{P2} represent the practical voltages required for the capture solution formation process (P1) and the CO_2 liberation formation process (P2) (obtained from [Supplementary Fig. 70](#)), M_{CO_2} is the molar mass of CO_2 (44 g mol^{-1}).

For bicarbonate:

In the HCO_3^- post-capture solution, one OH^- is needed to capture one CO_2 molecule, and one H^+ is required to release the captured one CO_2 molecule (E25). Therefore, one electron is involved in the electrochemical process. The energy consumption of the redox-decoupled system to capture 1 tonne of CO_2 was calculated (E26):



$$w_{\text{CO}_2} = \frac{W_{\text{CO}_2}}{m_{\text{CO}_2}} = \frac{F \cdot (V_{\text{P1}} + V_{\text{P2}})}{M_{\text{CO}_2} \cdot \eta_{\text{CE}}} \quad (\text{E26})$$

Discussion 8: Customized crystallizing air capture experimental details

A customized crystallizing air capture unit was employed to conduct CO_2 capture experiments under atmospheric conditions ([Supplementary Fig 73](#)). The setup consisted of three polypropylene (PP) lines, each with a diameter of 1.5 mm, woven into a structured array. These PP lines were partially immersed in the regenerated capture solution to enable OH^- wicking via capillary action. This configuration locally concentrates OH^- at the surface of the PP lines, facilitating CO_2 capture and subsequent precipitation of K_2CO_3 . To eliminate the influence of ambient humidity, the entire system was enclosed within a customized container. Compressed air with a controlled relative

humidity of 3% was supplied from the building system. The airflow velocity near the PP line was regulated to 0.4 m s^{-1} to simulate natural convection conditions and was measured using an anemometer (BTMETER BT-856A).

Theoretical CO_2 captured via DAC:

The DAC experiments were conducted using three different regenerated capture solutions after stability experiment, each with an initial pH of 13.97 ± 0.01 . The theoretical amount of CO_2 captured ($c\text{CO}_{2 \text{ cap,theoretical}}$) and K_2CO_3 precipitate ($m_{\text{K}_2\text{CO}_3,\text{theoretical}}$) were estimated based on the quantity of OH^- consumed during the CO_2 absorption process (E27–E29).

$$c\text{OH}^-_{\text{consumed}} = 10^{(\text{pH}_{\text{before}}-14)} \times \frac{V_{\text{before}}}{1000} - 10^{(\text{pH}_{\text{after}}-14)} \times \frac{V_{\text{after}}}{1000} \quad (\text{E27})$$

$$c\text{CO}_{2 \text{ cap,theoretical}} = \frac{c\text{OH}^-_{\text{consumed}}}{2} \quad (\text{E28})$$

$$m_{\text{K}_2\text{CO}_3,\text{theoretical}} = \frac{M_{\text{K}_2\text{CO}_3} \cdot c\text{OH}^-_{\text{consumed}}}{2} \quad (\text{E29})$$

where $c\text{OH}^-_{\text{consumed}}$ is the total quantity of OH^- consumed during the CO_2 absorption process (mol), $\text{pH}_{\text{before}}$ and pH_{after} are the pH of capture solution before and after DAC experiments, V_{before} and V_{after} are the volume of capture solution before and after DAC experiments (ml), and $M_{\text{K}_2\text{CO}_3}$ is the molar mass of K_2CO_3 (138.2 g mol^{-1}) (Supplementary Table 8).

The corresponding mass of K_2CO_3 precipitate was measured by a high-resolution mass balance (E30). The precipitate efficiency was estimated based on experimental mass of precipitate and theoretical mass (E31):

$$m_{\text{precipitate}} = m_{\text{line,after}} - m_{\text{line,before}} \quad (\text{E30})$$

$$\eta_{\text{precipitate}} = \frac{m_{\text{precipitate}}}{m_{\text{K}_2\text{CO}_3,\text{theoretical}}} \quad (\text{E31})$$

where $m_{\text{line,after}}$ is the mass of PP lines after the DAC experiment (mg), $m_{\text{line,before}}$ is the mass of PP lines before the DAC experiment (mg) (Supplementary Table 8). The small difference is attributed to handling/transfer losses (e.g., minor detachment and loss of trace K_2CO_3 crystals during collection), and likely not to process inefficiency.

Discussion 9: Current efficiency calculation

Faraday's law was used to calculate the theoretical hydroxide production rates in the capture solution and proton production rate in the CO_2 liberation solution (E32, E33):

$$r_{\text{OH}^-} = \frac{I_{\text{process 1}}}{n \cdot F} \quad (\text{E32})$$

$$r_{\text{H}^+} = \frac{I_{\text{process 2}}}{n \cdot F} \quad (\text{E33})$$

During the experiment, the pH of the capture solution was monitored. The recorded pH was then converted according to the pH calibration curve, the actual produced hydroxide concentration was calculated, and system current efficiency can be obtained using equation (E34):

$$\eta_{\text{CE}} = \frac{C_{\text{OH}^-, \text{produced}}}{C_{\text{OH}^-, \text{theoretical}}} \quad (\text{E34})$$

Discussion 10: CO₂ release analysis

During the 7-hour redox-decoupled system operation, a 0.05 M K₂CO₃ precipitates (obtained from the customized crystallizer) were transferred to the CO₂ liberation solution prior to each round. Gaseous CO₂ was released over the experiment as H⁺ continuously react with CO₃²⁻. The released CO₂ gas flow rate was monitored by a flowmeter (Omega FMA-LP1615A) with a customized vacuum pump connected to the outlet, ensuring a complete release of CO₂ at the end of the experiment.

The total amount of CO₂ released during the eDAC operation was calculated from the following equations (E35):

$$n_{\text{CO}_2\text{-released}} = \frac{P \cdot Q_{\text{released}} \cdot t}{R \cdot T} \cdot x_{\text{CO}_2} \quad (\text{E35})$$

Where Q_{released} is the volumetric flow rate of total outlet gas (mL min⁻¹), x_{CO_2} is the CO₂ molar concentration (97%), and $n_{\text{CO}_2\text{-released}}$ is the total amount of CO₂ released (mmol).

Discussion 11: H₂ gas circulation system analysis and setup

A detailed analysis of theoretical H₂ production rate at the P1 cathode, experimental H₂ production rate at the P1 cathode, and experimental H₂ consumption rate at the P2 anode were analyzed in this study ([Supplementary Table 9](#)).

The theoretical H₂ production rate (sccm) was calculated from the following equations (E36 – E38)



$$r_{\text{H}_2, \text{theoretical}} = \frac{I}{nF} \quad (\text{E37})$$

$$V_{\text{H}_2, \text{theoretical}} = \frac{nRT}{P} \quad (\text{E38})$$

Where I is the electrolyzer current density (A m^{-2}), n is the number of electrons transferred per mole of H_2 produced (2), T is the electrolyzer operating temperature (298 K), and P is the atmospheric pressure (101325 pa).

The experimental H_2 production rate at the P1 cathode was measured using a bubble meter and was averaged over five independent measurements.

The experimental H_2 consumption rate was determined using a water displacement method. Specifically, the outlet of the H_2 supply line was connected to a sealed water reservoir chamber. The minimum H_2 flow rate required to sustain system operation was identified by gradually reducing the H_2 inflow until a steady water level was observed. This flow rate was measured using a bubble meter and averaged over five independent measurements.

Based on our analysis, the experimental H_2 production rate at the P1 cathode was within 0.5% of the theoretical value, indicating a near 100% faradaic efficiency for HER. An additional 2% of H_2 was required to maintain effective H_2 circulation between P1 and P2.

In the current lab-scale setup, H_2 gas was supplied by an external pressurized cylinder. In future pilot-scale systems, two design strategies can be considered to enable H_2 circulation between P1 and P2. First, H_2 generated from the HER can be collected in a storage cylinder and subsequently redirected to the anode for HOR. Alternatively, in a well-sealed system, introducing a slight H_2 excess (~2%) can increase the internal pressure sufficiently to drive gas flow toward the P2 anode for HOR.

Discussion 12: Redox-decoupled system techno-economic analysis (TEA):

The TEA was conducted based on the redox-decoupled system configuration. The following assumptions were applied:

1. The system is designed for an annual DAC capacity of 100 tCO₂, operating continuously for 365 days per year, with an overall system efficiency of 90%.
2. All capital cost (CAPEX) is assumed to have a 20-year operational lifetime.
3. A project discount rate of 5% was applied.
4. The electrolyte volume needed is estimated based on the geometric volume of fluid electrolyte compartment with a fraction of 0.5¹⁹. The electrolyzer channel thickness is 0.5 cm.

5. The electrolyzer stack replacement cost was estimated to be 30% of the uninstal cost for 7 years^{20,21}.
6. The electricity used is sourced from an onshore wind power, priced at \$35 MWh⁻¹ ²².
7. The reverse osmosis water price is \$0.92 per m³ ²³.

Project scope:

CAPEX cost includes	Redox-decoupled system, CO ₂ liberation tank, and balance of plants (BOP)
OPEX cost includes	Electrolyzer energy, process water, electrolyte pumping/circulation, gas compression/handing, chemical replacement, stack replacement, and others

Summary of redox-decoupled system cost:

CAPEX type:	Cost per tCO ₂
Redox-decoupled system	95.4
CO ₂ liberation tank	5.6
BOP	10.1
Net	111.1

OPEX type:	Cost per tCO ₂
Electrolyzer energy	56.4
Process water	4.4
Electrolyte pumping/circulation	5.6
Gas compression/handing	17.2
Chemical replacement	27.0
Stack replacement	51.1
Others	16.2
Net	177.9
Overall cost	289.0

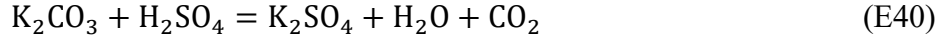
Detailed TEA calculations:

Two moles of OH⁻ are needed to capture 1 mole of CO₂ (E39–E41). The CO₂ capture rate is specified as 12.68 kgCO₂ h⁻¹ to achieve an annual capture of 100 tCO₂ (E42). Converting this rate yields 288.18 molCO₂ h⁻¹ and 576.36 molOH h⁻¹ (E43, E44).

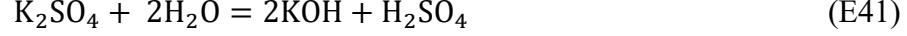
Air contactor:



CO₂ liberation tank:



Redox-decoupled system:



$$\frac{100 \text{ tCO}_2}{1 \text{ yr}} \times \frac{1 \text{ yr}}{8760 \text{ h}} \times \frac{1000 \text{ kg}}{1 \text{ t}} \div 0.9 = 12.68 \left[\frac{\text{kgCO}_2}{\text{h}} \right] \quad (\text{E42})$$

$$\frac{12.68 \text{ kgCO}_2}{1 \text{ h}} \times \frac{1000 \text{ g}}{1 \text{ kg}} \times \frac{1 \text{ mol}}{44 \text{ g}} = 288.18 \left[\frac{\text{molCO}_2}{\text{h}} \right] \quad (\text{E43})$$

$$\frac{288.18 \text{ molCO}_2}{\text{h}} \times 2 = 576.36 \left[\frac{\text{molOH}}{\text{h}} \right] \quad (\text{E44})$$

Faraday's law is applied to calculate the total current required (E45):

$$I = 576.36 \left[\frac{\text{molOH}}{\text{h}} \right] \times \frac{1 [\text{h}]}{3600 [\text{s}]} \times nF = 15447.2 [\text{A}] \quad (\text{E45})$$

The cell voltage is 1.2V for 100 mA cm⁻² operation, the power needed is calculated by using the equation (E46):

$$P = 1.2 [\text{V}] \times 15447.2 [\text{A}] = 18.54 [\text{kW}] \quad (\text{E46})$$

The total electrolyzer area is calculated by using the equation (E47):

$$\text{Area}_{\text{electrolyzer}} = \frac{15447.2 [\text{A}]}{0.1 \left[\frac{\text{A}}{\text{cm}^2} \right]} = 15.45 [\text{m}^2] \quad (\text{E47})$$

Capital recovery factor (CRF) was used to calculate cost per tonne CO₂ for all CAPEX, and it is assumed to have a discount rate of 5% and a material lifetime of 20 years (E48).

$$\text{CRF} = \frac{0.05 \times (1.05)^{20}}{(1.05)^{20} - 1} = 0.08 \quad (\text{E48})$$

CAPEX calculation:

Redox-decoupled system: The system scope includes frame, bipolar plates, assembly and endplates, balance of stack, power supply, and other auxiliary components. According to a

reference study employing a platinum-catalyst and cation exchange membrane electrolyzer, the system cost was reported as \$7,722 m⁻²²¹. Based on this metric, our redox-decoupled configuration, requires a total active area of 15.45 m², corresponding to an estimated total cost of \$119,305 (E49). Electrolyzer cost per tonne CO₂ is calculated to be **\$95.4 tCO₂⁻¹** (E50).

$$\text{Total electrolyzer cost} = 15.45 [\text{m}^2] \times 7722 [\$/\text{m}^2] = 119,305 [\$] \quad (\text{E49})$$

$$\text{Electrolyzer cost}_{\text{per tCO}_2} = 119,305 [\$] \times \frac{0.08}{100 [\text{tCO}_2]} = 95.4 \left[\frac{\$}{\text{tCO}_2} \right] \quad (\text{E50})$$

CO₂ liberation tank: The solution tank consists of a water tank integrated with an electric mixer. The estimated CAPEX for this unit is projected to be \$7,000. Liberation tank cost per tonne CO₂ was calculated to be **\$5.6 tCO₂⁻¹** (E51).

$$\text{Liberation tank cost}_{\text{per tCO}_2} = 7,000 [\$] \times \frac{0.08}{100 [\text{tCO}_2]} = 5.6 \left[\frac{\$}{\text{tCO}_2} \right] \quad (\text{E51})$$

²⁰**BOP:** The BOP is assumed to be 10% of the total CAPEX. BOP cost per tonne CO₂ was calculated to be **\$10.1 tCO₂⁻¹** (E52).

$$\begin{aligned} \text{BOP cost}_{\text{per tCO}_2} &= (119,305 + 7,000) [\$] \times 0.1 \times \frac{0.08}{100 [\text{tCO}_2]} \\ &= 10.1 \left[\frac{\$}{\text{tCO}_2} \right] \end{aligned} \quad (\text{E52})$$

OPEX calculation:

Electrolyzer energy: The redox-decoupled system requires 5.8 GJ of electrical energy per tonne of CO₂ captured. Assuming the electricity price is \$35 per MWh. The electrolyzer energy cost is projected to be **\$56.4 tCO₂⁻¹** (E53).

$$\begin{aligned} \text{Electrolyzer cost}_{\text{per tCO}_2} &= 5.8 \left[\frac{\text{GJ}}{\text{tCO}_2} \right] \times 0.278 \left[\frac{\text{MWh}}{\text{GJ}} \right] \times 35 [\$/\text{MWh}] \\ &= 56.4 \left[\frac{\$}{\text{tCO}_2} \right] \end{aligned} \quad (\text{E53})$$

Process water: Process water is required for air contactor evaporative losses. The water consumption is 4.74 tH₂O tCO₂⁻¹²⁴. Assuming the process water price is \$0.92 per m³, the process water cost is projected to be **\$4.4 tCO₂⁻¹** (E54).

$$\begin{aligned}
\text{Process water cost}_{\text{per tCO}_2} &= 4.74 \left[\frac{\text{tH}_2\text{O}}{\text{tCO}_2} \right] \times 1 \left[\frac{\text{m}^3}{\text{tH}_2\text{O}} \right] \times 0.92 \left[\frac{\$}{\text{m}^3} \right] \\
&= 4.4 \left[\frac{\$}{\text{tCO}_2} \right]
\end{aligned} \tag{E54}$$

Electrolyte pumping and circulation: Electrolyte circulation pumps consume 98 kW of power for a 1MW electrolyzer stack²⁵. Based on this assumption, the pumping energy consumption of 15.9 MWh for an 18.54 kW-scale redox decoupled configuration (E55). The electrolyte pumping and circulation cost is projected to be **\$5.6 tCO₂⁻¹** (E56).

$$\begin{aligned}
\text{Electrolyte pumping} &= 98 \text{ [kW]} \times \frac{18.54 \text{ [kW]}}{1000 \text{ [kW]}} \times 8760 \text{ [h]} \\
&= 15.9 \text{ [MWh]}
\end{aligned} \tag{E55}$$

$$\begin{aligned}
\text{Electrolyte pumping cost}_{\text{per tCO}_2} &= 15.9 \text{ [MWh]} \times \frac{35 \text{ [\$ MWh}^{-1}\text{]}}{100 \text{ [tCO}_2\text{]}} \\
&= 5.6 \left[\frac{\$}{\text{tCO}_2} \right]
\end{aligned} \tag{E56}$$

Gas compression/handing: The release of CO₂ must be compressed to 15 MPa at 45°C for geological sequestration. The gas compression/handing cost is estimated to be **\$17.2 tCO₂⁻¹**²⁴.

Chemical replacement: All chemicals are assumed to be recycled, and the electrolyte volume was estimated based on the geometric volume of fluid electrolyte compartment¹⁹, the total electrolyte volume is estimated to be 154.5 L (E57). The TEMPO prices used in this analysis was \$43.7 mol⁻¹, based on the price from Oakwood chemical. The decay rate in the stability experiment was measured to be 0.046% hr⁻¹, corresponding to 0.4 mol L⁻¹ yr⁻¹ at laboratory scale (E58). When scaled to a 100 tCO₂ scale system, the annual decay is estimated to be 61.8 mol yr⁻¹ (E59). The estimated chemical replacement cost is projected to be **\$27.0 tCO₂⁻¹** (E60).

$$\text{Electrolyte volume} = 15.45 \text{ [m}^2\text{]} \times 0.005 \text{ [m]} \div 0.5 = 154.5 \text{ [L]} \tag{E57}$$

$$\begin{aligned}
\text{Decay rate}_{\text{lab}} &= 0.1 \text{ [mol L}^{-1}\text{]} \times 0.00046 \text{ [hr}^{-1}\text{]} \times 8760 \left[\frac{\text{hr}}{\text{year}} \right] \\
&= 0.4 \text{ [mol L}^{-1}\text{ yr}^{-1}\text{]}
\end{aligned} \tag{E58}$$

$$\text{Decay rate}_{\text{pilot}} = 0.4 \text{ [mol L}^{-1}\text{ yr}^{-1}\text{]} \times 154.5 \text{ [L]} = 61.8 \text{ [mol yr}^{-1}\text{]} \tag{E59}$$

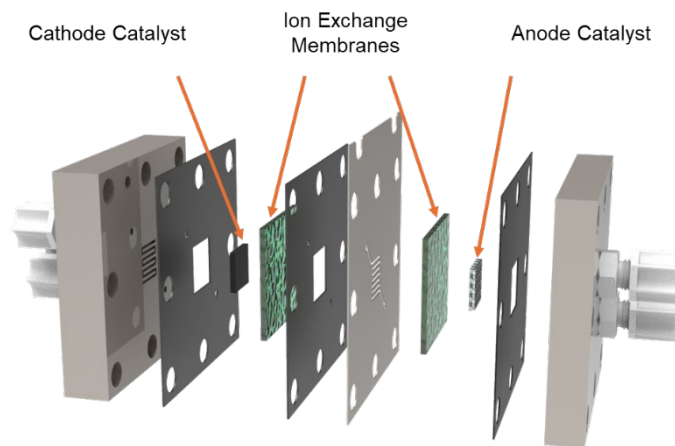
$$\begin{aligned}\text{Chemical replacement cost}_{\text{per tCO}_2} &= 61.8 [\text{mol yr}^{-1}] \times \frac{43.7 [\$ \text{mol}^{-1}]}{100 [\text{tCO}_2]} \\ &= 27.0 \left[\frac{\$}{\text{tCO}_2} \right]\end{aligned}\quad (\text{E60})$$

Stack replacement: The stack replacement cost was assumed to be 30% of the uninstall cost for 7 years²¹. The stack replacement cost is projected to be **\$51.1 tCO₂⁻¹** (E61).

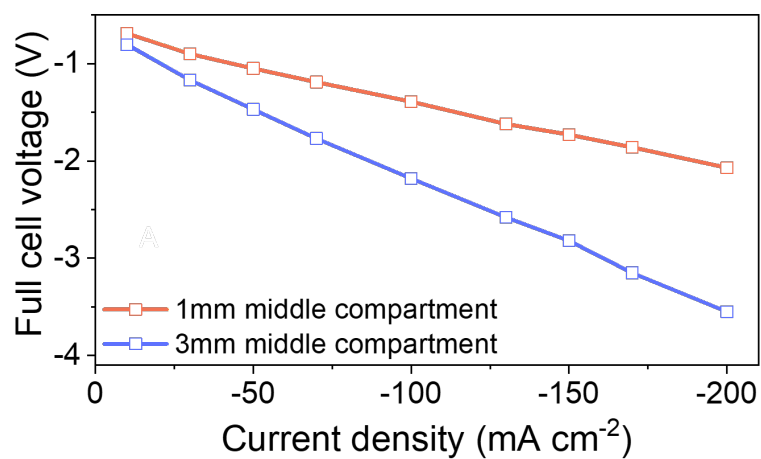
$$\begin{aligned}\text{Stack replacement cost}_{\text{per tCO}_2} &= 119,305 [\$] \times \frac{0.3}{7 \times 100 [\text{tCO}_2]} \\ &= 51.1 \left[\frac{\$}{\text{tCO}_2} \right]\end{aligned}\quad (\text{E61})$$

Others: The other operational costs, including transportation, labor and maintenance, are assumed to be 10% of total OPEX. The other cost is projected to be **\$16.2 tCO₂⁻¹** (E62).

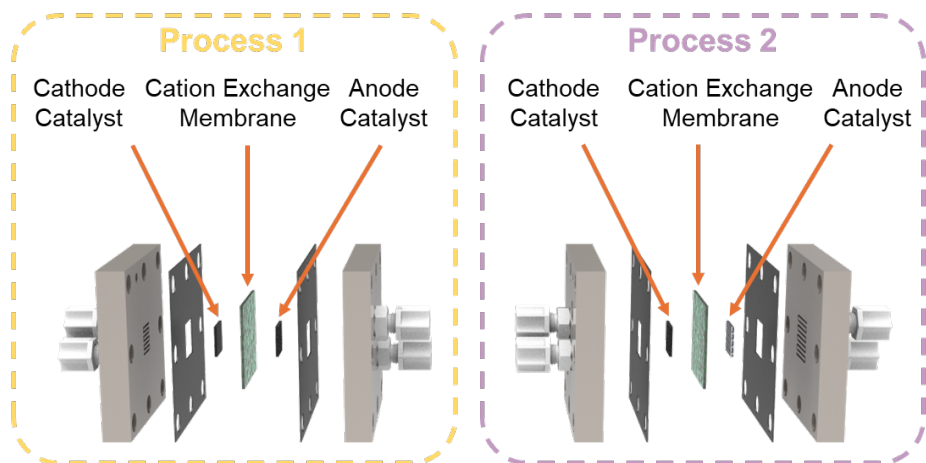
$$\begin{aligned}\text{Other cost}_{\text{per tCO}_2} &= (56.4 + 4.4 + 5.6 + 17.2 + 27 + 51.5) [\$] \times 0.1 \\ &= 16.2 \left[\frac{\$}{\text{tCO}_2} \right]\end{aligned}\quad (\text{E62})$$



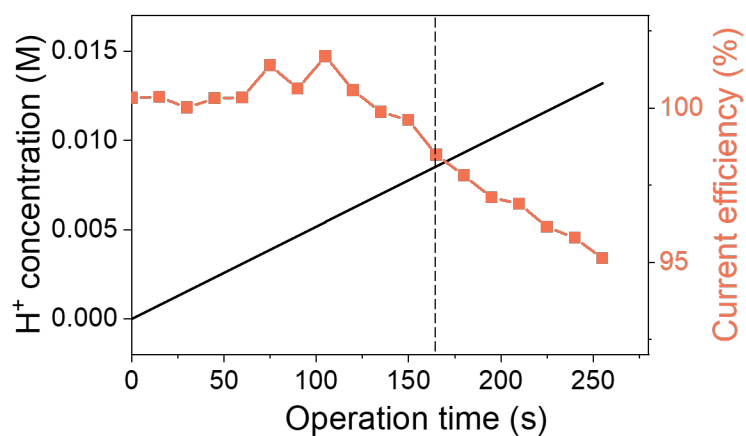
Supplementary Fig. 1: Schematics of the three-compartment pH-swing configuration. Three compartments are separated by two ion exchange membranes. OH^- -based capture solution is generated at the cathode compartment, while CO_2 is liberated at the middle compartment.



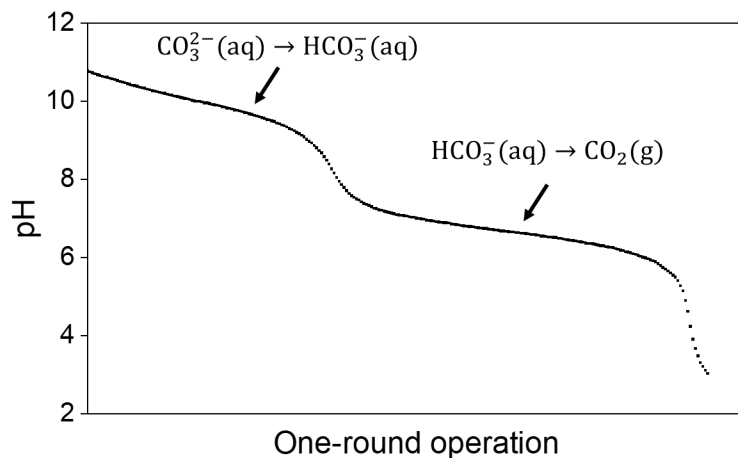
Supplementary Fig. 2: Comparison of the full-cell voltage in three-compartment pH-swing configuration, using 0.5 M K₂CO₃ in 0.25 M K₂SO₄ aqueous electrolyte. The middle compartment thickness of 1 mm and 3 mm.



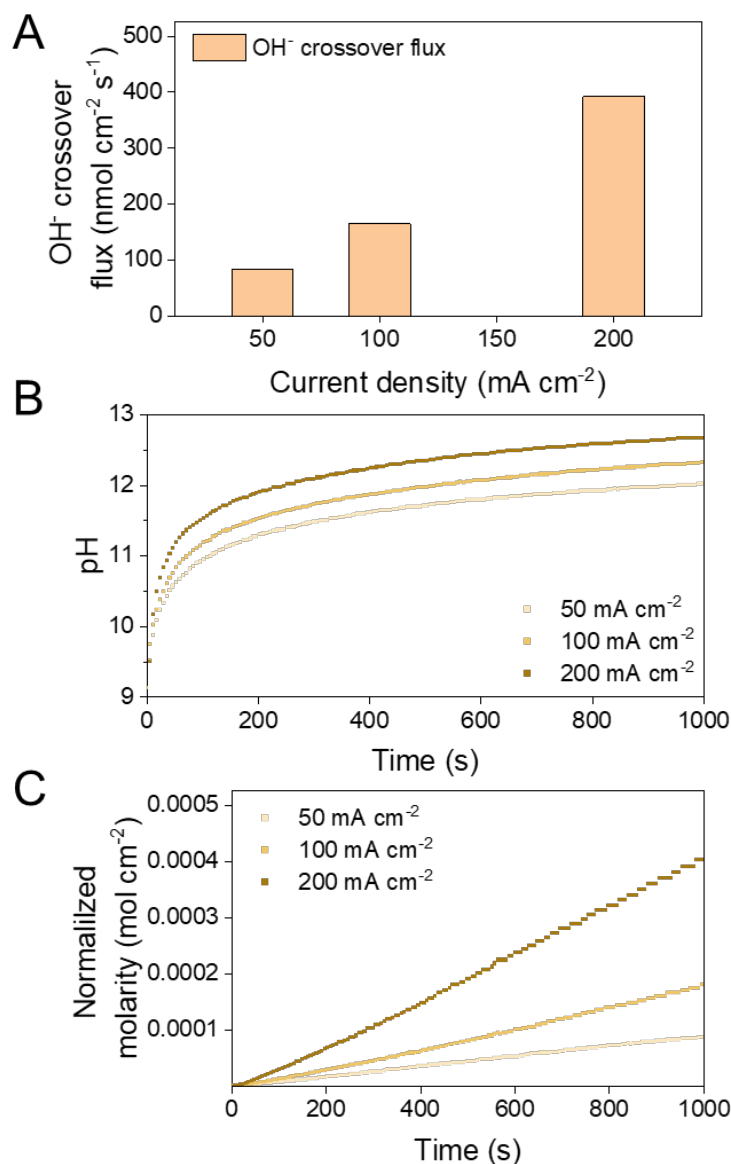
Supplementary Fig. 3: Schematics of the redox-decoupled configuration. Two distinct zero-gap membrane electrode assemblies to generate OH^- and H^+ , respectively. OH^- -based capture solution is generated at the cathode compartment of P1, while CO_2 is liberated at cathode compartment of P2, and collected at the gas-liquid separator.



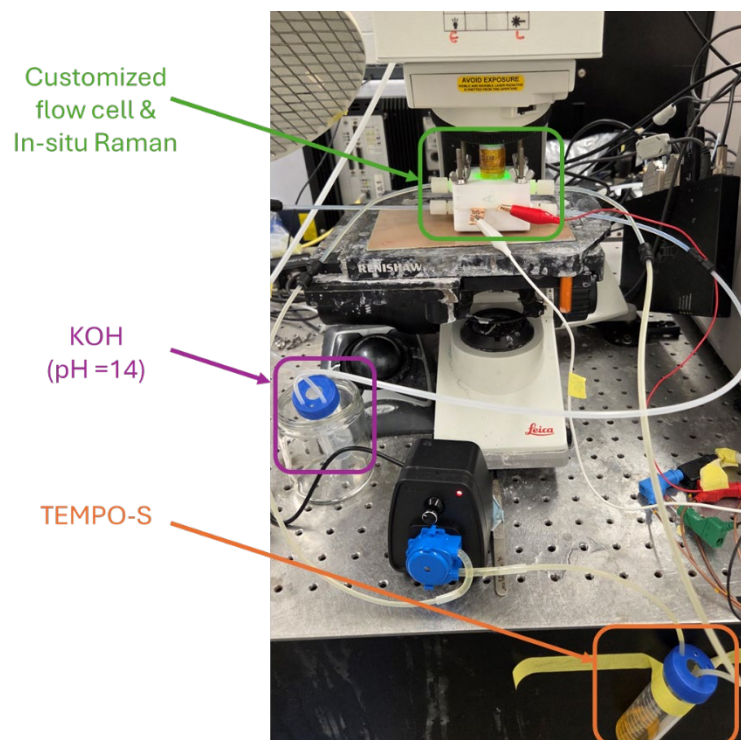
Supplementary Fig. 4: Threshold H⁺ concentration triggering decline in CE. Redox-decoupled configuration at 100 mA cm⁻², using a 50 mM ferricyanide/ 50 mM ferrocyanide in 0.25 M K₂SO₄ aqueous electrolyte. The capture solution volume was 50 mL of Di water, and the CO₂ liberation solution volume was 20 mL. A noticeable decrease in CE was observed when the H⁺ concentration was above 0.009 M.



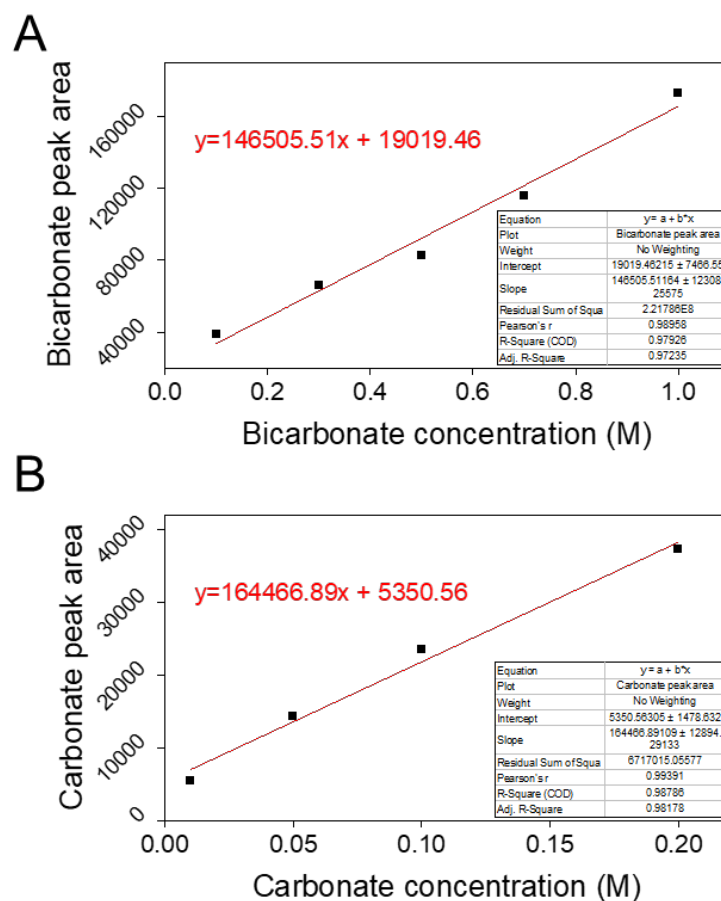
Supplementary Fig. 5: Experimental pH of the CO₂ liberation solution during the redox-decoupled system, using 0.1 M TEMPO-S (50% TEMPO-S: 50% TEMPO-S⁺ ratio) in 0.25 M K₂SO₄ and 0.05 M K₂CO₃ aqueous electrolyte. CO₂ solution pH maintains ~3 at the end of each round, indicating a near-complete conversion of CO₃²⁻ to gaseous CO₂. Two plateau pH regions were observed in the CO₂ liberation solution during one round operation. The first indicates the dissolved CO₃²⁻ converts to HCO₃⁻, and the second indicates the dissolved HCO₃⁻ converts to gaseous CO₂. Negligible released CO₂ was observed during the first plateau. It started to release in the second plateau.



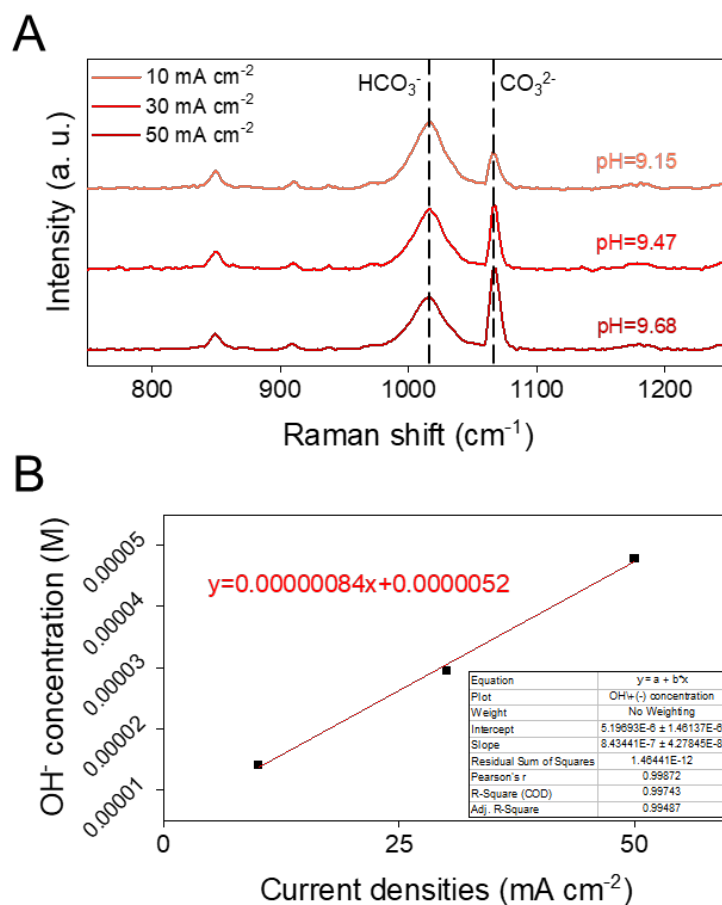
Supplementary Fig. 6: Investigation of OH⁻ crossover through Nafion 115 from a pH 14 cathode compartment under varying current densities (50, 100, and 200 mA cm⁻²). (A) OH⁻ crossover flux profile. (B) Anode compartment pH profile. (C) Anode compartment normalized molarity profile. The estimated OH⁻ crossover flux through Nafion 115 ranged from 82 to 391 nmol cm⁻² s⁻¹ under current densities between 50 and 200 mA cm⁻².



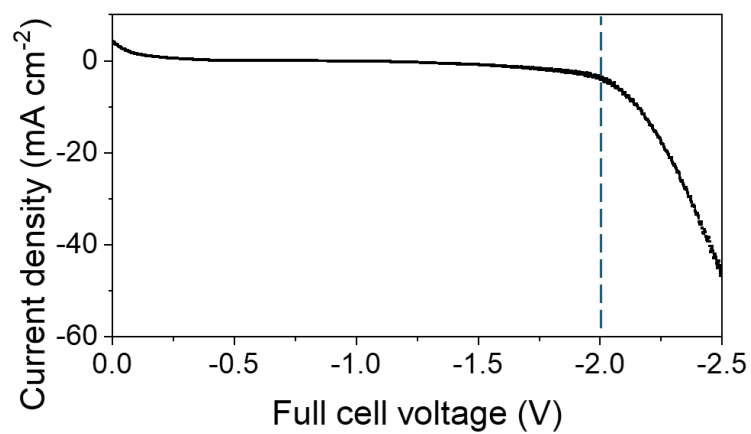
Supplementary Fig. 7: Experimental setup for in-situ Raman configuration. The counter compartment was composed of a 50 mL pH 14 KOH solution and the working compartment was composed of a 20 mL 0.1 M TEMPO-S in 1 M KHCO_3 solution.



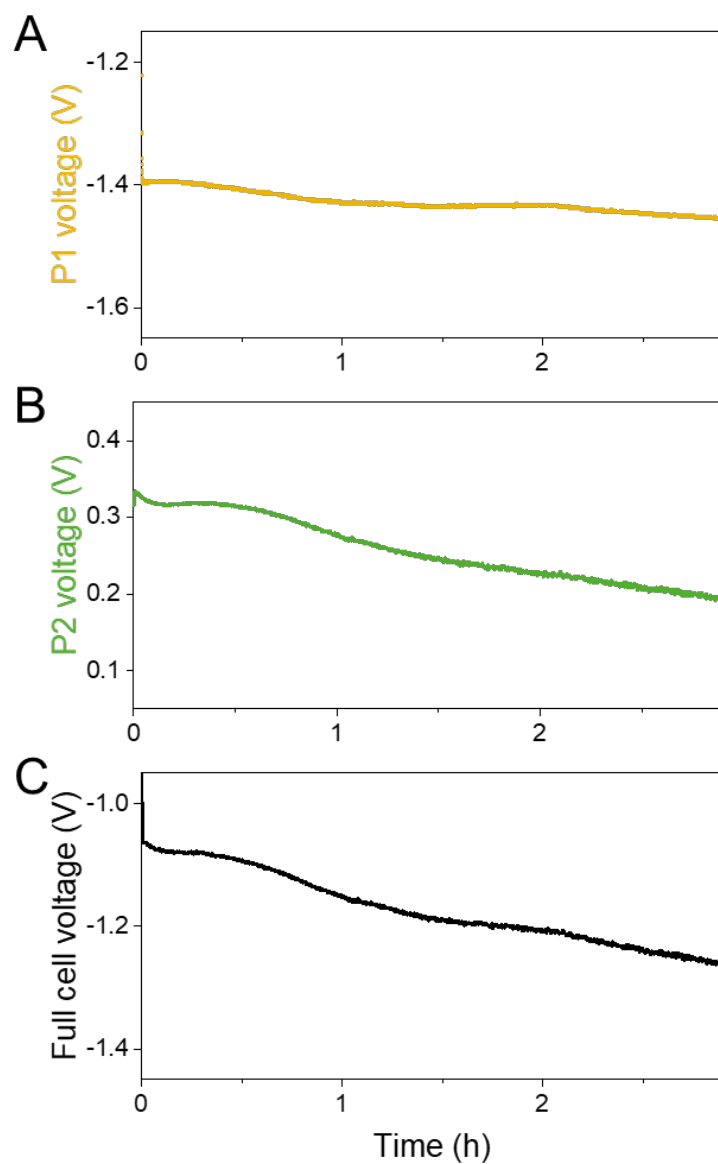
Supplementary Fig. 8: In-situ Raman spectra calibration curves. (A) Correlation between bicarbonate concentration and Raman peak area. **(B)** Correlation between carbonate concentration and Raman peak area.



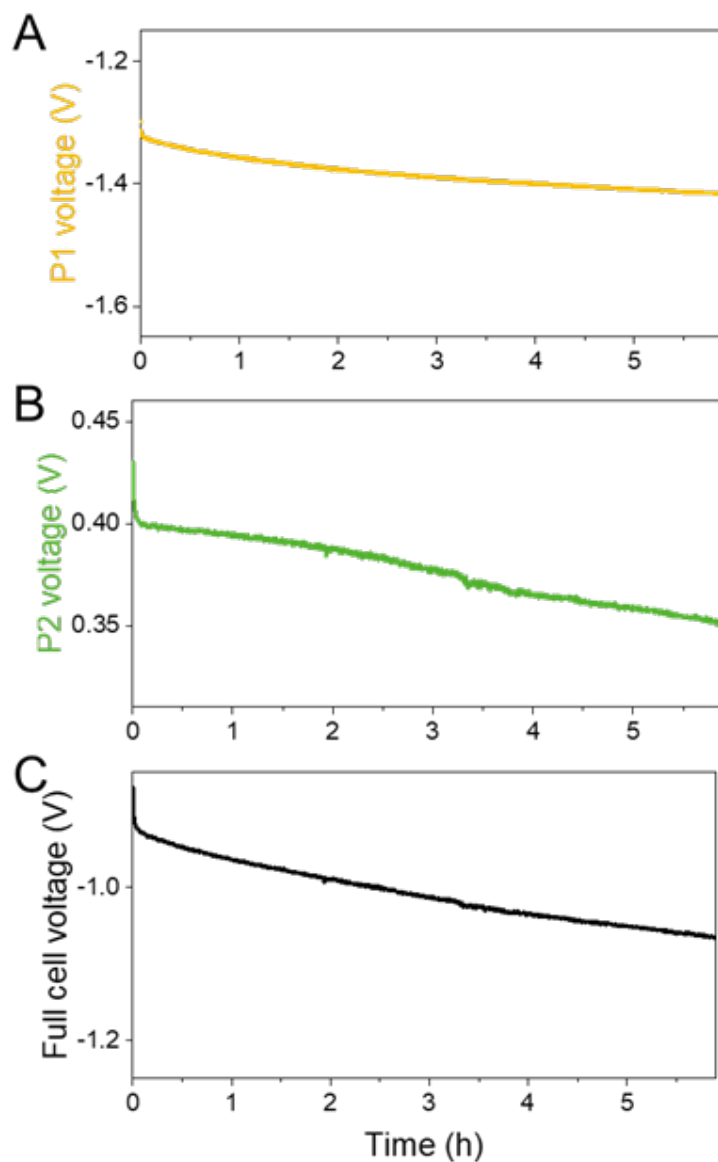
Supplementary Fig. 9: In-situ Raman spectra under different current densities. (A) Local CO₃²⁻ and HCO₃⁻ concentration under current densities of 10, 30, and 50 mA cm⁻². **(B)** Corresponding OH⁻ concentrations, calculated based on CO₃²⁻ and HCO₃⁻ concentrations. Note that the in-situ Raman setup could not be operated at higher currents due to H₂ bubble formation and accumulation in the counter compartment. Local pH values at higher currents were estimated by assuming a linear relationship between OH⁻ crossover flux and current density, yielding local pH values of 9.95, 10.12, 10.26 for current densities of 100, 150, and 200 mA cm⁻², respectively.



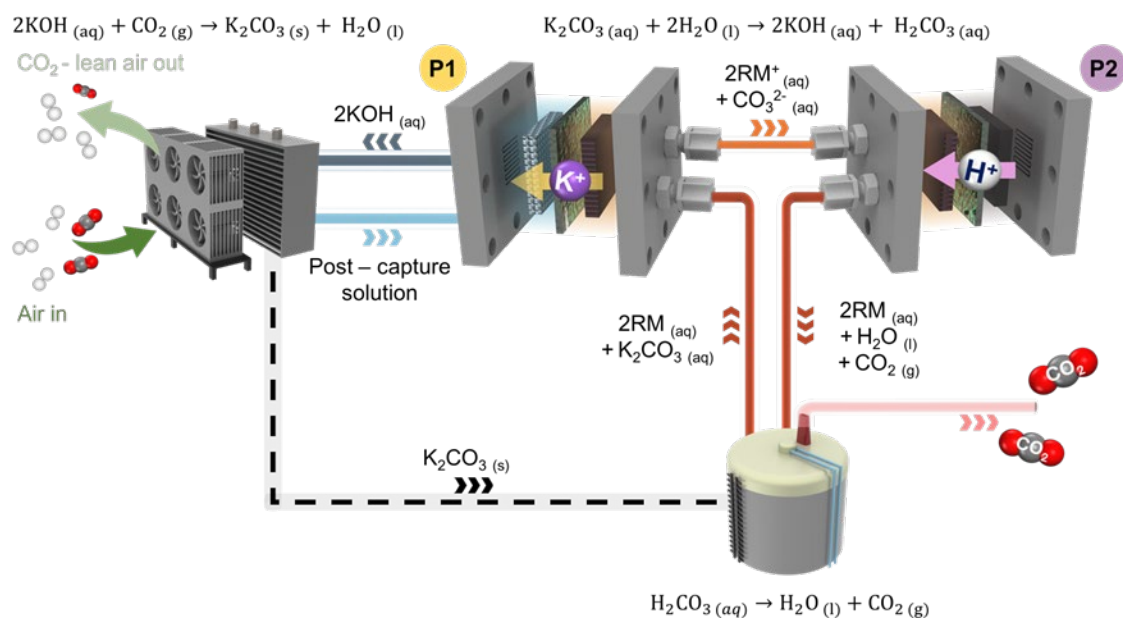
Supplementary Fig. 10: Illustrates the OER during P1 full-cell voltage operation. Linear scan voltammetry analysis reveals a sharp increase in OER at full-cell voltages below -2 V, indicating the cut-off P1 full-cell voltage at -2 V.



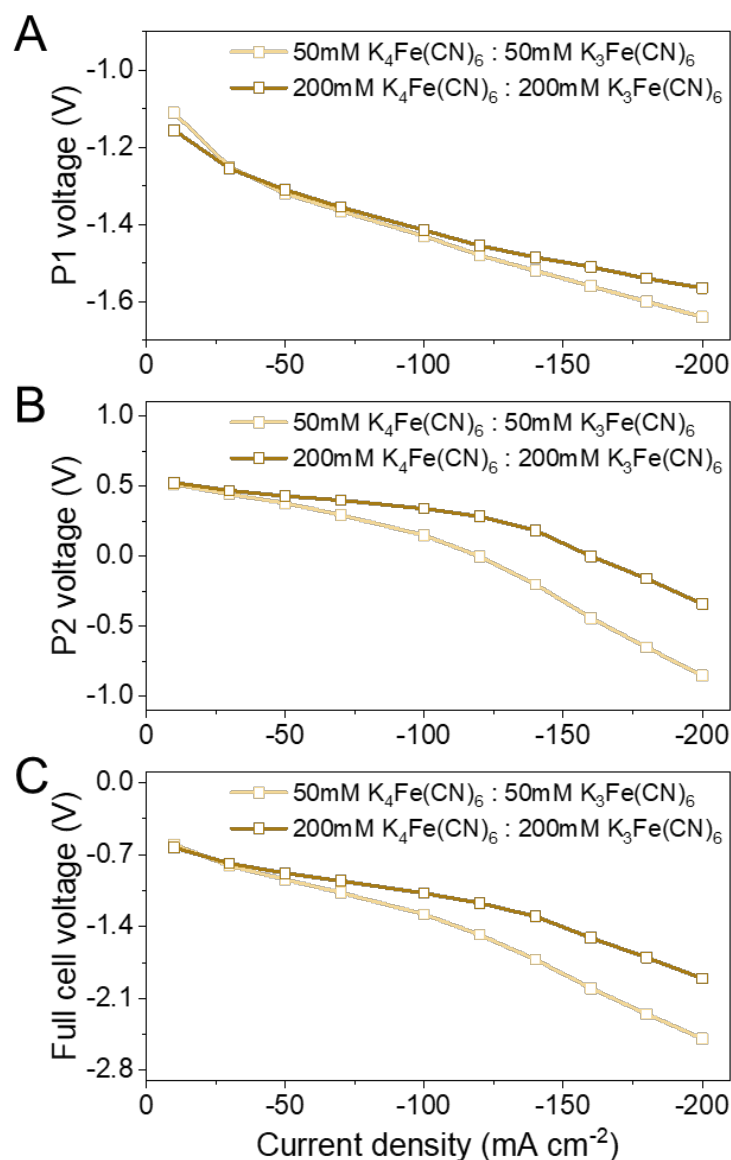
Supplementary Fig. 11: Redox-decoupled configuration at -100 mA cm^{-2} , using 50 mM ferricyanide/ 50 mM ferrocyanide in $0.5 \text{ M K}_2\text{CO}_3$ and $0.25 \text{ M K}_2\text{SO}_4$ aqueous electrolyte. (A) P1 voltage profile. (B) P2 voltage profile. (C) Full-cell voltage.



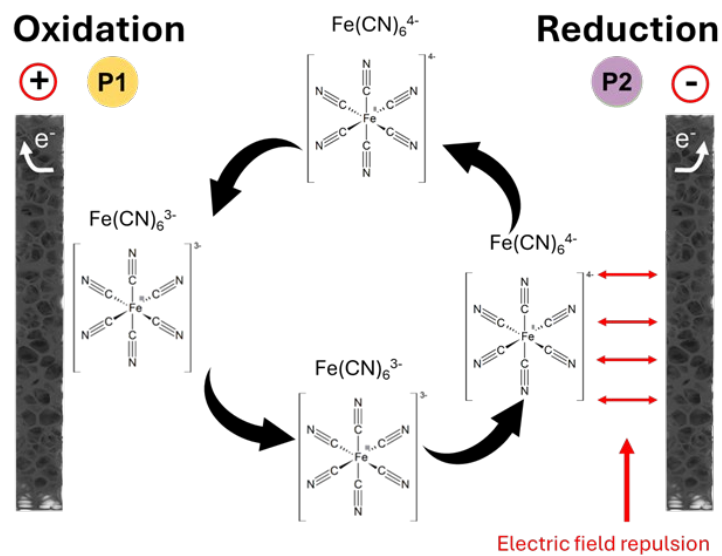
Supplementary Fig. 12: Redox-decoupled configuration at -50 mA cm^{-2} , using 50 mM ferricyanide/ 50 mM ferrocyanide in 0.5 M K_2CO_3 and 0.25 M K_2SO_4 aqueous electrolyte. (A) P1 voltage profile. (B) P2 voltage profile. (C) Full-cell voltage.



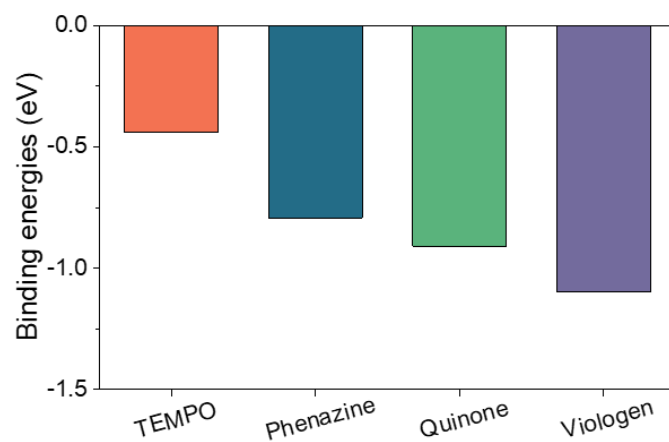
Supplementary Fig. 13: Redox-decoupled configuration for DAC, achieving stable and energy-efficient atmospheric CO₂ capture and release.



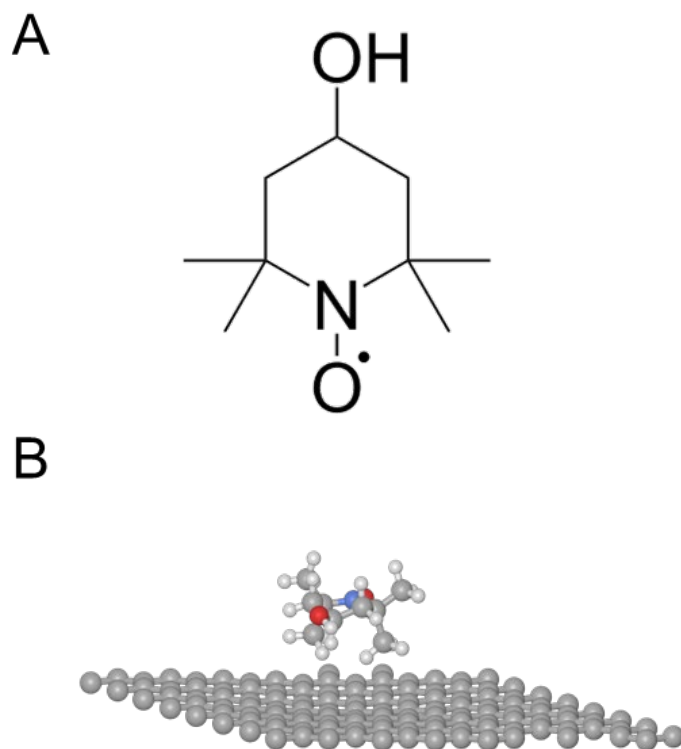
Supplementary Fig. 14: Performance using ferricyanide/ferrocyanide in 0.25 M K_2SO_4 + 0.5 M K_2CO_3 aqueous electrolyte during the redox-decoupled system at varying ferricyanide: ferrocyanide concentration. (A) P1 voltage profile. (B) P2 voltage profile. (C) Full-cell voltage profile.



Supplementary Fig. 15: Schematic of electrochemical behaviour during oxidation and reduction of ferricyanide/ferrocyanide. Illustration of ferricyanide/ferrocyanide with overall charge of -3 and -4 , limiting reduction reaction due to the negatively charged electrostatic repulsion.

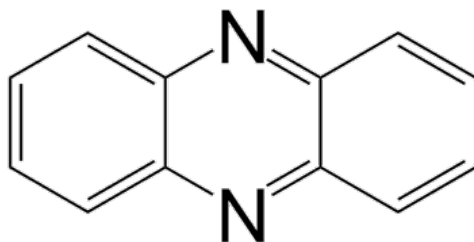


Supplementary Fig. 16: Binding energy of carbon electrode with adsorbed TEMPO, phenazine, quinone, and viologen.

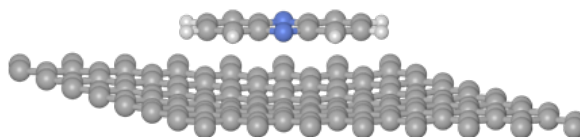


Supplementary Fig. 17: DFT simulation on carbon electrode binding energy with TEMPO molecule. (A) Hydroxyl substituted TEMPO molecular structure that is employed in DFT. **(B)** DFT-optimized geometries of carbon electrode with adsorbed TEMPO molecule.

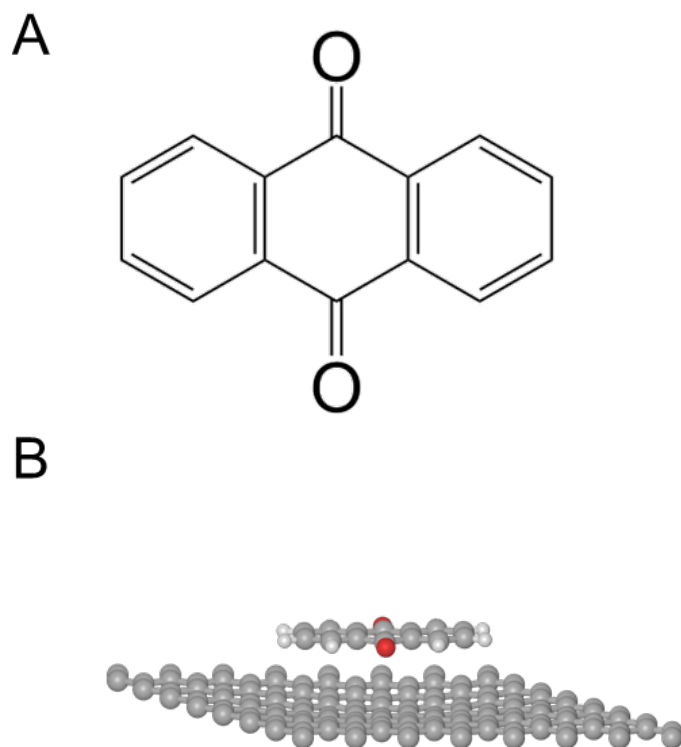
A



B

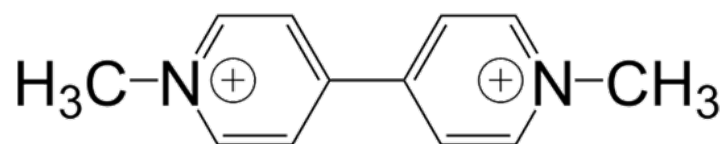


Supplementary Fig. 18: DFT simulation on carbon electrode binding energy with phenazine molecule. (A) Phenazine molecular structure that is employed in DFT. **(B)** DFT-optimized geometries of carbon electrode with adsorbed phenazine molecule.

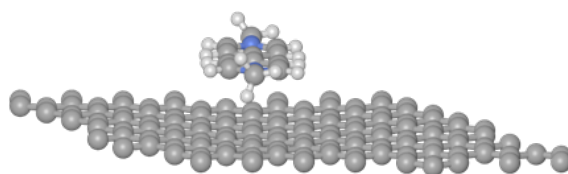


Supplementary Fig. 19: DFT simulation on carbon electrode binding energy with quinone molecule. (A) Anthraquinone molecular structure that is employed in DFT. **(B)** DFT-optimized geometries of carbon electrode with adsorbed quinone molecule.

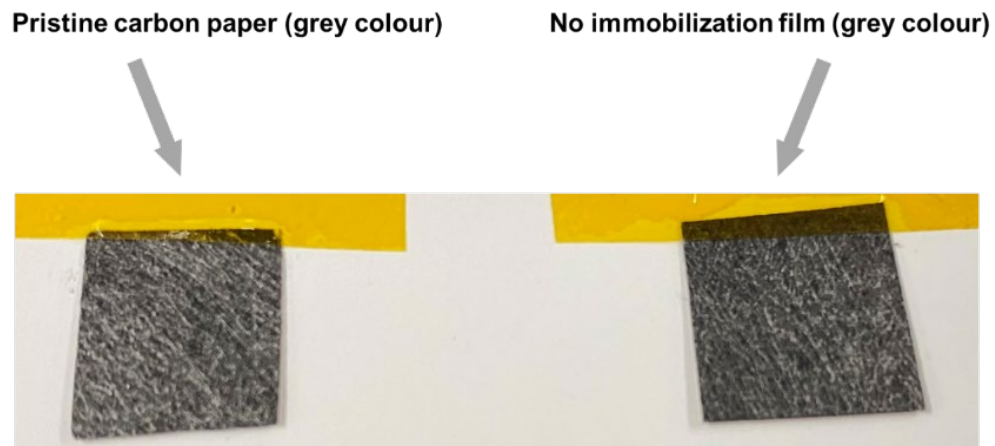
A



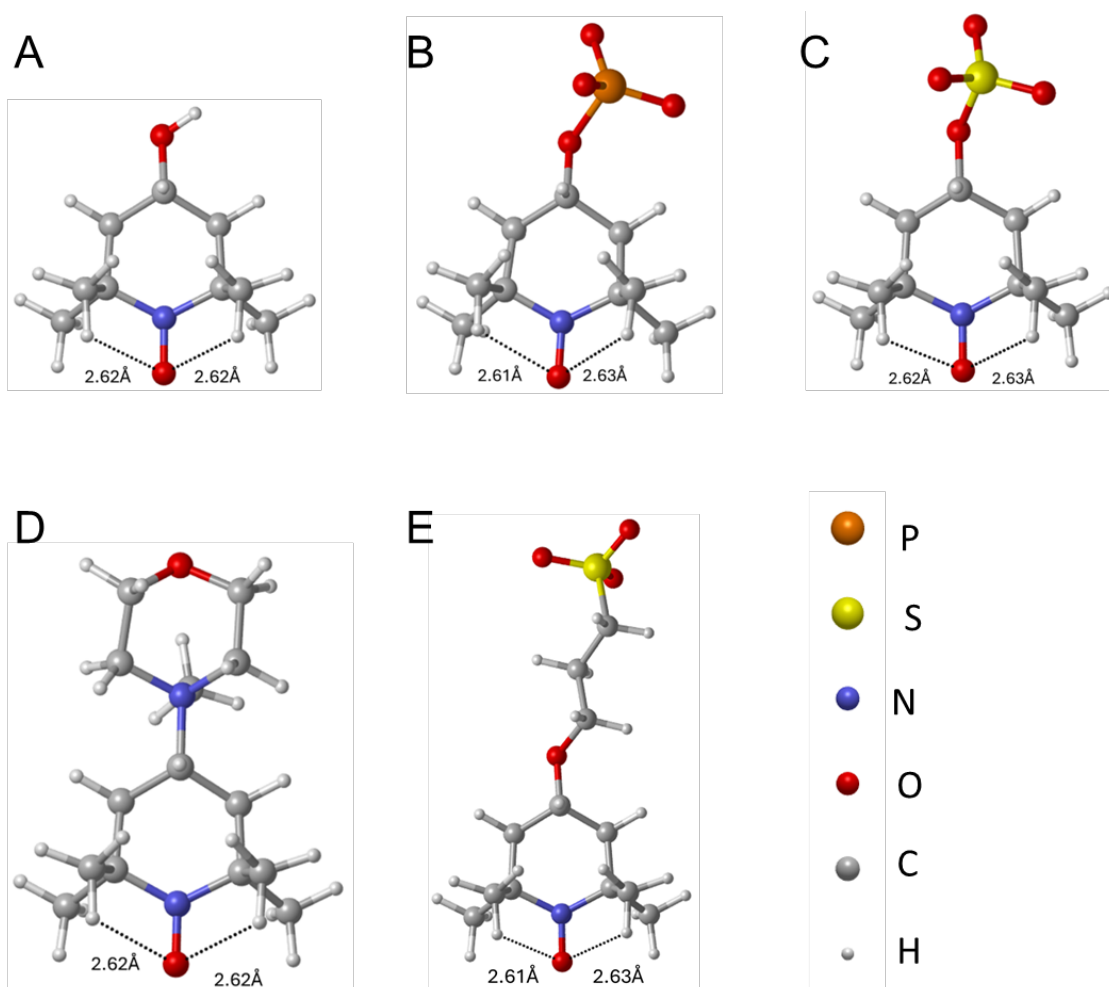
B



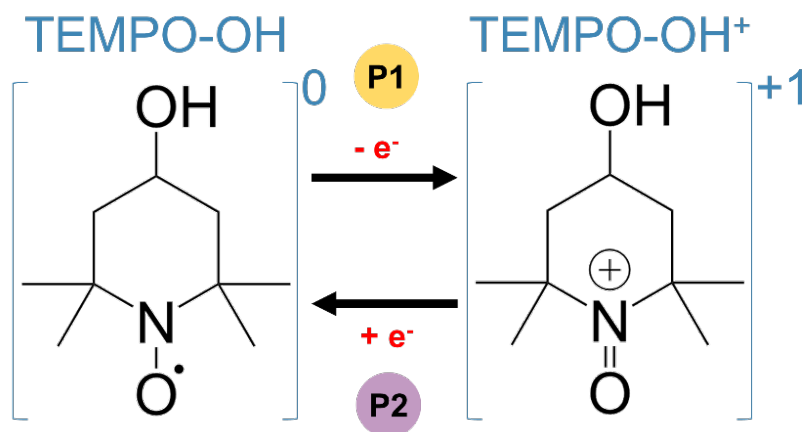
Supplementary Fig. 20: DFT simulation on carbon electrode binding energy with viologen molecule. (A) Methyl viologen molecular structure that is employed in DFT. **(B)** DFT-optimized geometries of carbon electrode with adsorbed viologen molecule.



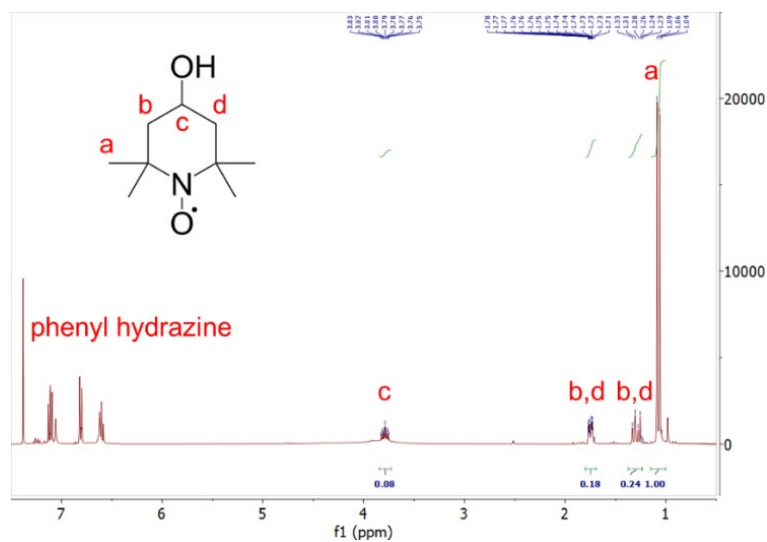
Supplementary Fig. 21: Illustration of the carbon electrode in the H-cell configuration, using a 20 mM TEMPO-S in 0.25 M K₂SO₄ aqueous electrolyte. The pristine carbon electrode (left); no immobilization film formation after electrocatalytic reactions (right).



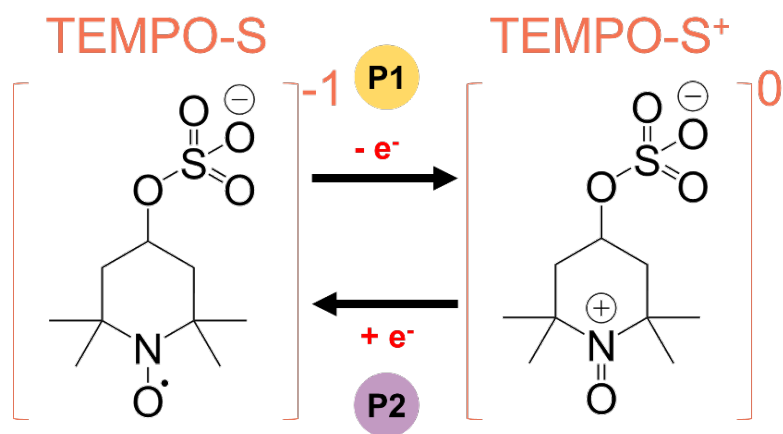
Supplementary Fig. 22: DFT calculations confirmed that the O–H distances of TEMPO derivatives are similar at its equilibrium state due to steric protection arising from the four methyl groups. (A) TEMPO-OH. (B) TEMPO-P. (C) TEMPO-S. (D) TEMPO-MOR. (E) TEMPO-PSS.



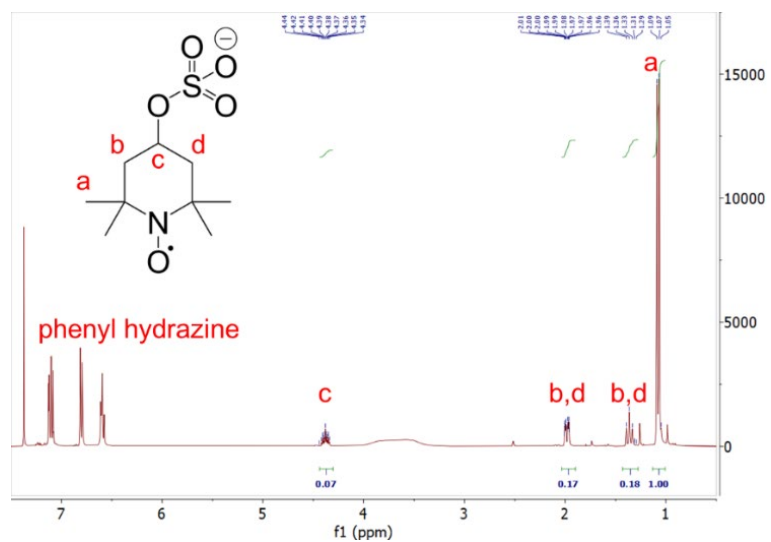
Supplementary Fig. 23: Schematic of the one-electron redox transfer mechanism of TEMPO-OH. This process involves neutral TEMPO-OH (left) and +1 charged TEMPO-OH⁺ (right), undergoing oxidation (P1) and reduction (P2) to store and release electrons, respectively.



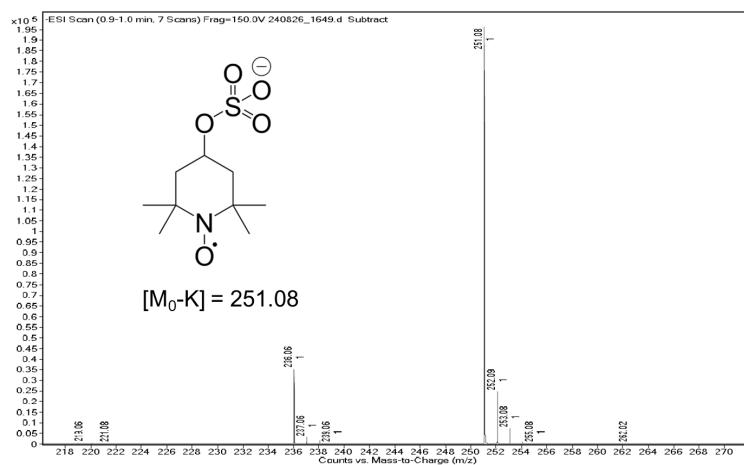
Supplementary Fig. 24: The ^1H NMR spectra of TEMPO-OH in D_2O reveals distinct peaks. The phenyl hydrazine signal around 7 ppm. The labels correlate individual peaks to specific chemical bonds.



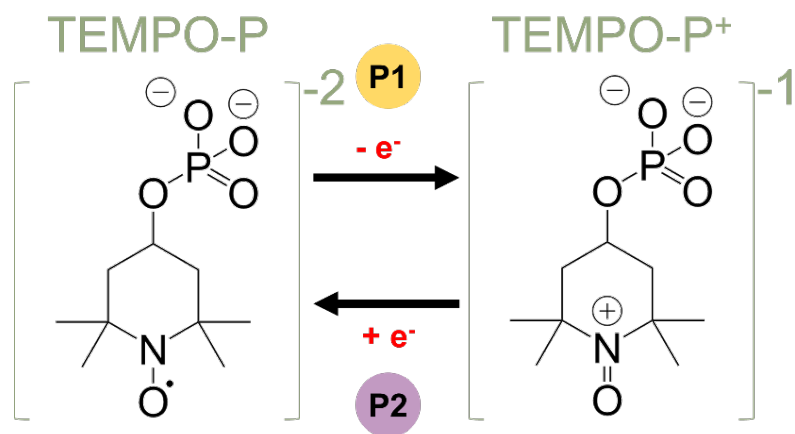
Supplementary Fig. 25: Schematic of the one-electron redox transfer mechanism of TEMPO-S. This process involves -1 charged TEMPO-S (left) and neutral TEMPO-S⁺ (right), undergoing oxidation (P1) and reduction (P2) to store and release electrons, respectively.



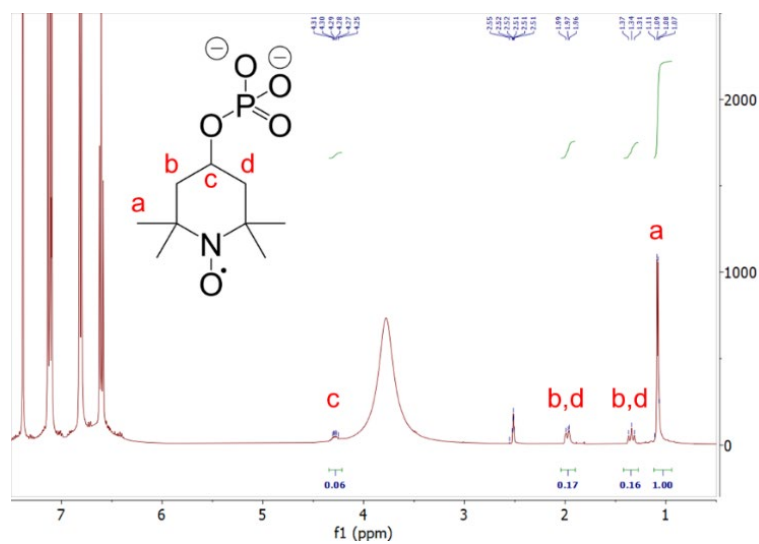
Supplementary Fig. 26: The ^1H NMR spectra of TEMPO-S in D_2O reveals distinct peaks. The phenyl hydrazine signal around 7 ppm. The labels correlate individual peaks to specific chemical bonds.



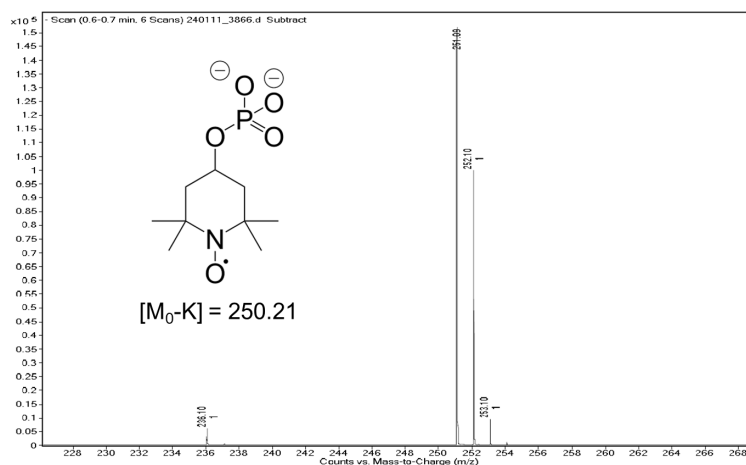
Supplementary Fig. 27: The high-resolution mass spectrometry profile of TEMPO-S. The m/z of calculated $[M_0-K]$: 251.08.



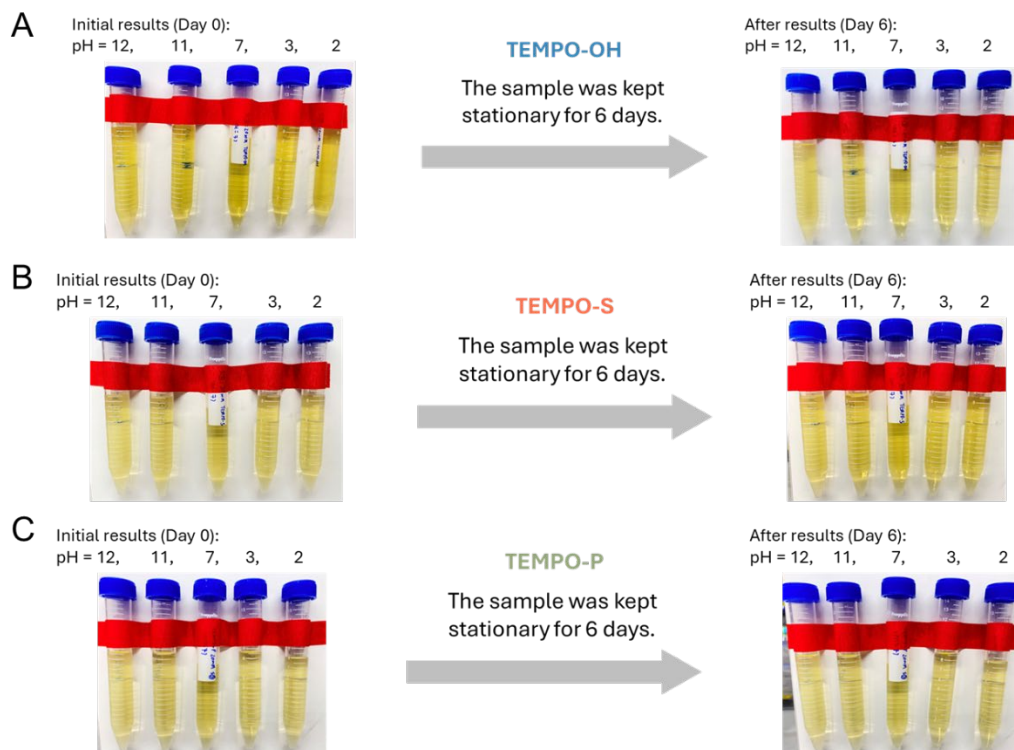
Supplementary Fig. 28: Schematic of the one-electron redox transfer mechanism of TEMPO-P. This process involves -2 charged TEMPO-P (left) and -1 charged TEMPO-P⁺ (right), undergoing oxidation (P1) and reduction (P2) to store and release electrons, respectively.



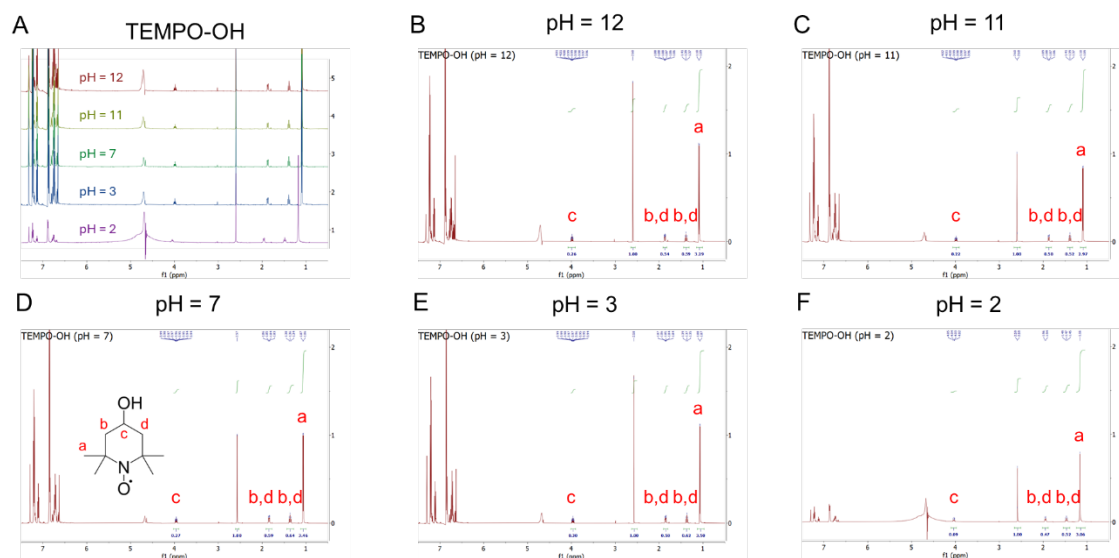
Supplementary Fig. 29: The ^1H NMR spectra of TEMPO-P in D_2O reveals distinct peaks. The phenyl hydrazine signal around 7 ppm. The labels correlate individual peaks to specific chemical bonds.



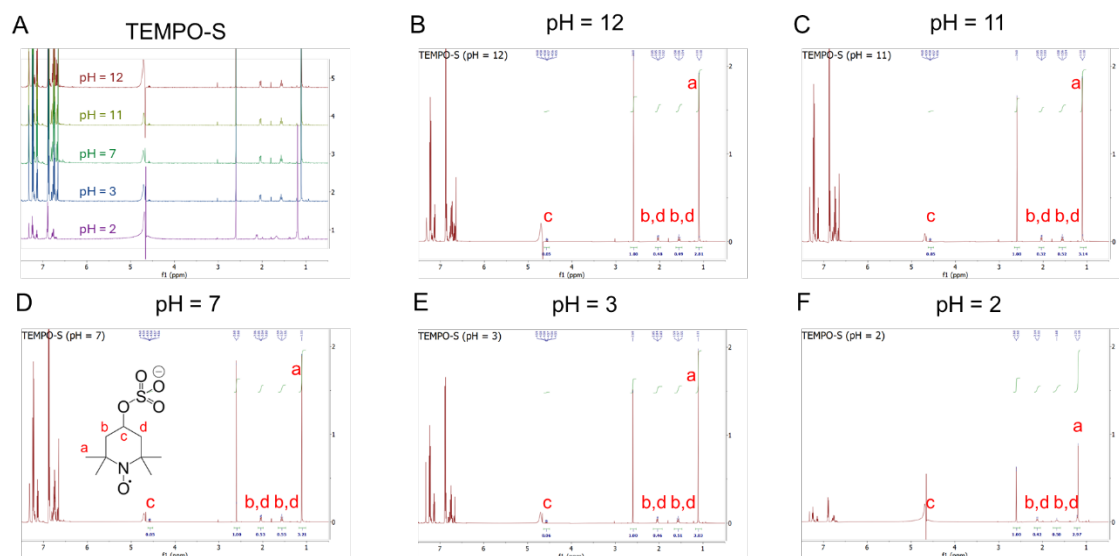
Supplementary Fig. 30: The high-resolution mass spectrometry profile of TEMPO-P. The m/z of calculated $[M_0-K]$: 250.21.



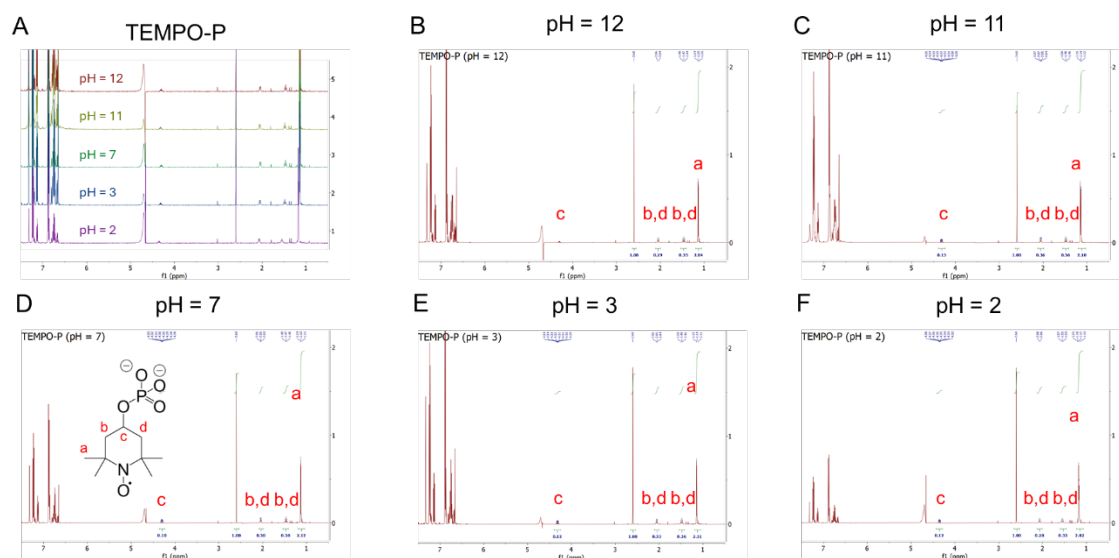
Supplementary Fig. 31: Stability of TEMPO-OH, TEMPO-S, and TEMPO-P under varying pH conditions (pH 12, 11, 7, 3, 2) over 6 days. (A) TEMPO-OH. (B) TEMPO-S. (C) TEMPO-P. All samples are 20 mM TEMPO concentrations. No visible changes were observed for three TEMPO derivatives across the tested pH conditions.



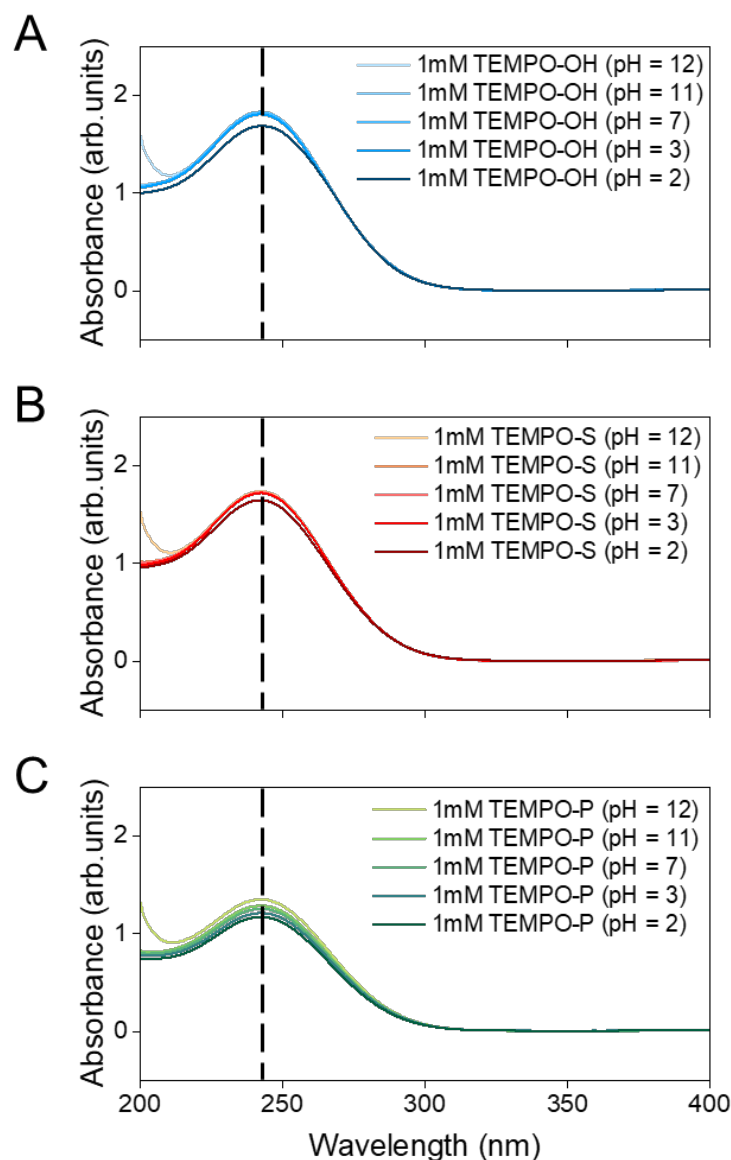
Supplementary Fig. 32: The ^1H NMR spectra of TEMPO-OH under varying pH conditions (pH 12, 11, 7, 3, 2) after 6 days. (A) Summary of spectra for all pH conditions. (B-F) ^1H NMR spectra of TEMPO-OH at pH 12, 11, 7, 3, and 2, respectively. All samples are prepared using 200 μl 20 mM TEMPO-OH in 400 μl D_2O . No additional peaks were detected, indicating the chemical robustness of TEMPO-OH across the tested pH range.



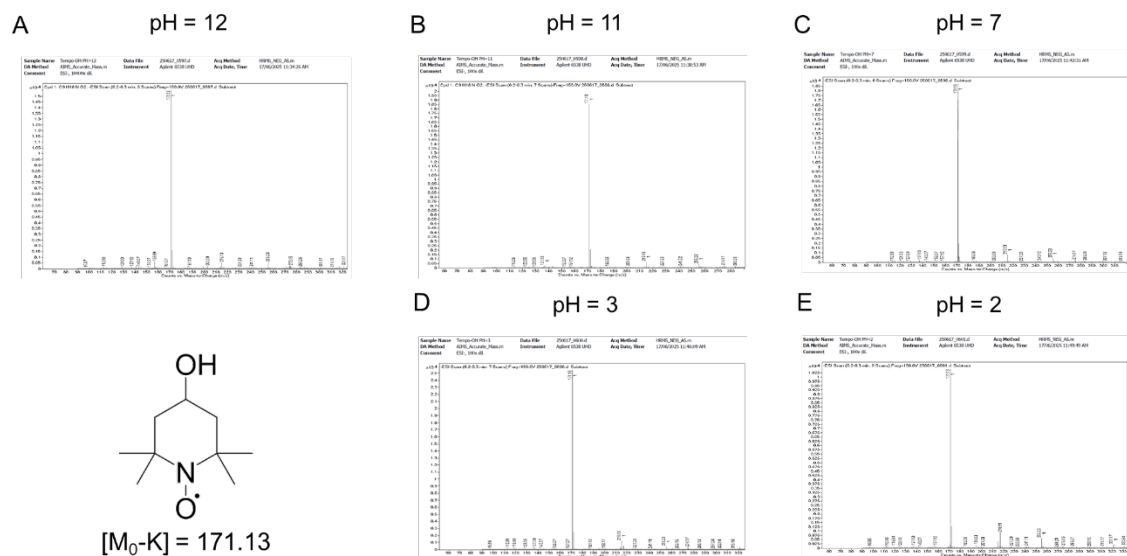
Supplementary Fig. 33: The ^1H NMR spectra of TEMPO-S under varying pH conditions (pH 12, 11, 7, 3, 2) after 6 days. (A) Summary of spectra for all pH conditions. (B-F) ^1H NMR spectra of TEMPO-S at pH 12, 11, 7, 3, and 2, respectively. All samples were prepared using 200 μl 20 mM TEMPO-S in 400 μl D_2O . No additional peaks were detected, indicating the chemical robustness of TEMPO-S across the tested pH range.



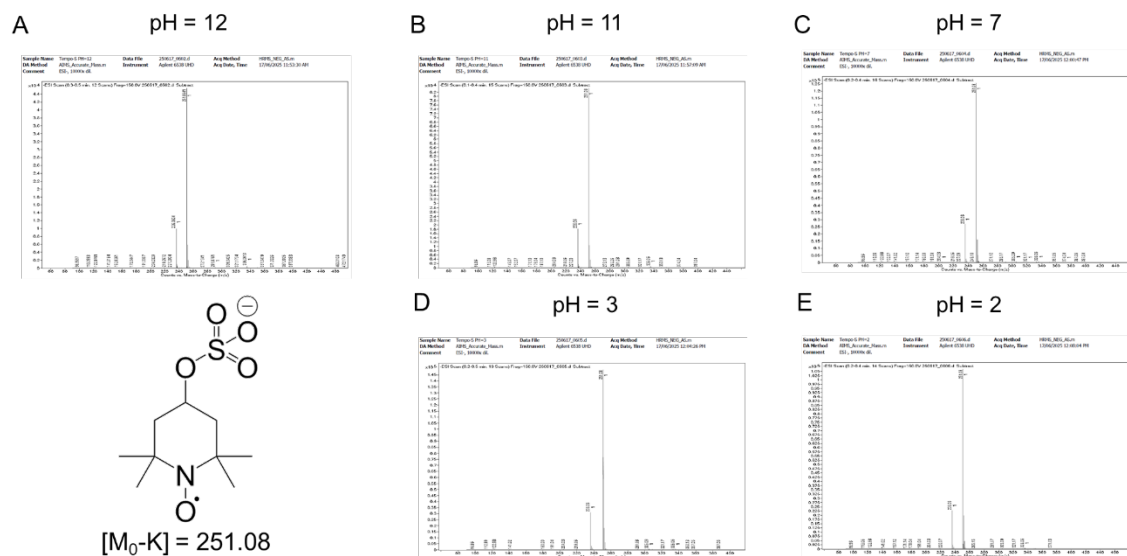
Supplementary Fig. 34: The ^1H NMR spectra of TEMPO-P under varying pH conditions (pH 12, 11, 7, 3, 2) after 6 days. (A) Summary of spectra for all pH conditions. (B-F) ^1H NMR spectra of TEMPO-P at pH 12, 11, 7, 3, and 2, respectively. All samples are prepared using 200 μl 20 mM TEMPO-P in 400 μl D_2O . No additional peaks were detected, indicating the chemical robustness of TEMPO-P across the tested pH range.



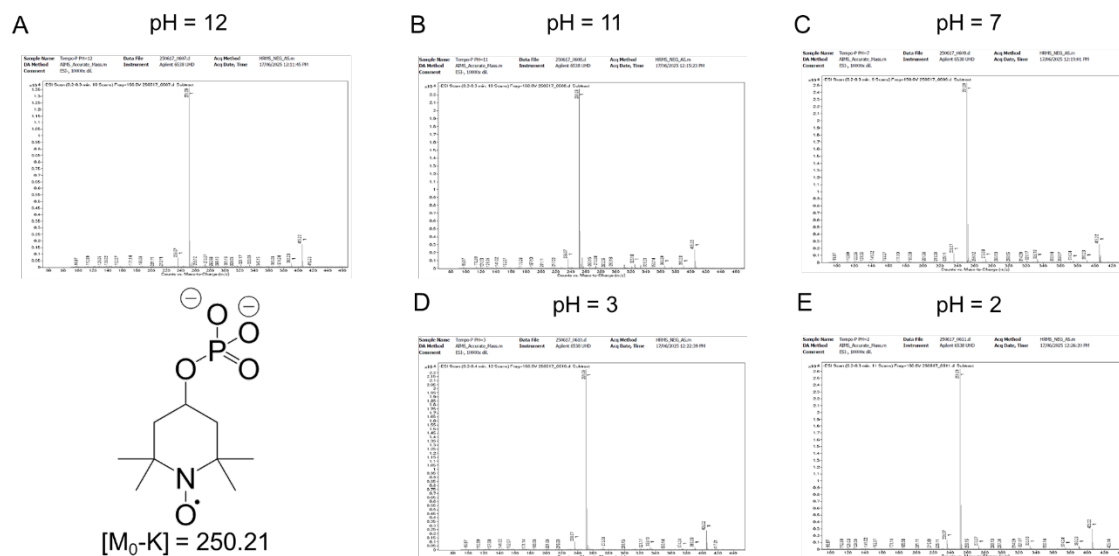
Supplementary Fig. 35: The UV-vis spectroscopy analysis of TEMPO-OH, TEMPO-S, and TEMPO-P under varying pH conditions (pH 12, 11, 7, 3, 2) after 6 days. (A) TEMPO-OH. (B) TEMPO-S. (C) TEMPO-P. The TEMPO signal is at 245 nm. No additional peaks were detected, indicating the chemical robustness of all TEMPO derivatives across the tested pH range.



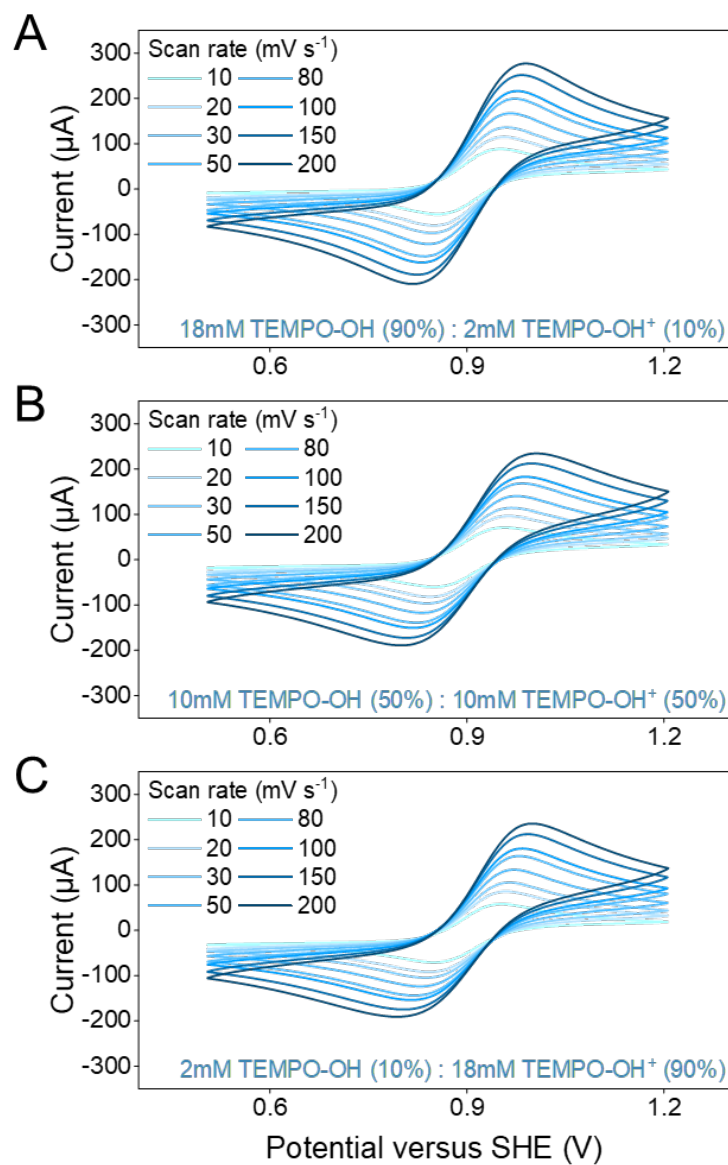
Supplementary Fig. 36: The high-resolution mass spectrometry profile of TEMPO-OH under varying pH conditions (pH 12, 11, 7, 3, 2) after 6 days. (A-E) Profile of TEMPO-OH at pH 12, 11, 7, 3, and 2, respectively. No additional peaks were detected, indicating the chemical robustness of TEMPO-OH across the tested pH range.



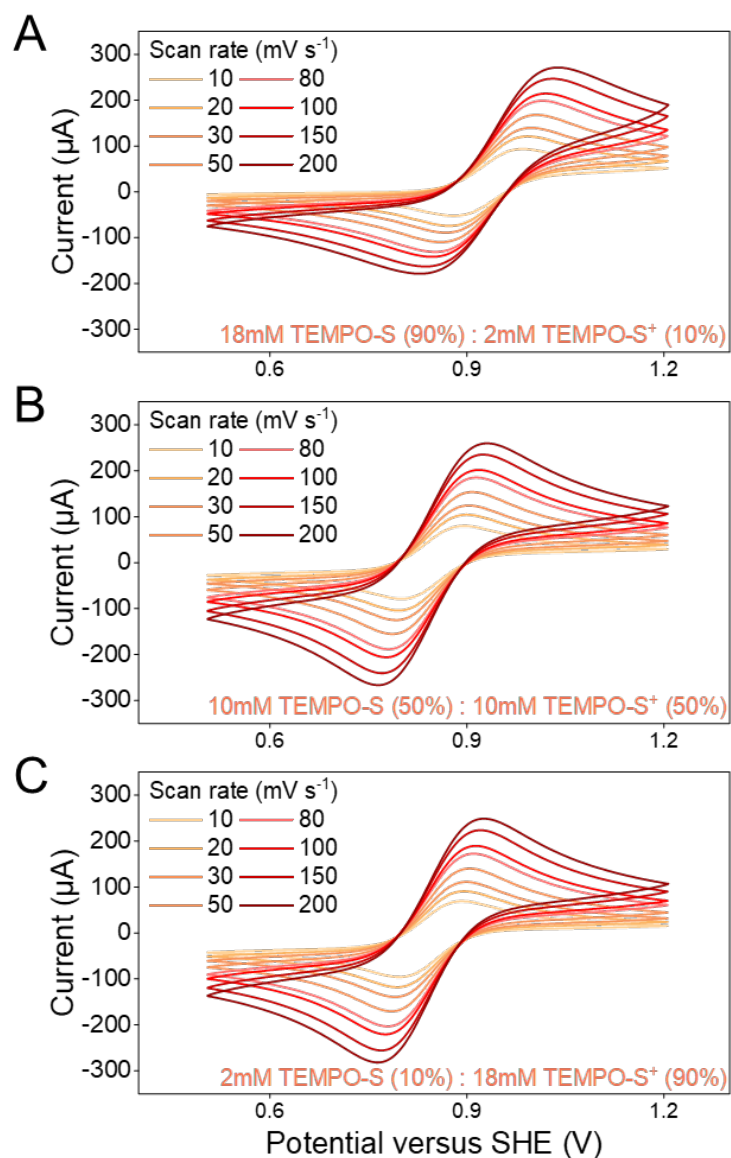
Supplementary Fig. 37: The high-resolution mass spectrometry profile of TEMPO-S under varying pH conditions (pH 12, 11, 7, 3, 2) after 6 days. (A-E) Profile of TEMPO-S at pH 12, 11, 7, 3, and 2, respectively. No additional peaks were detected, indicating the chemical robustness of TEMPO-S across the tested pH range.



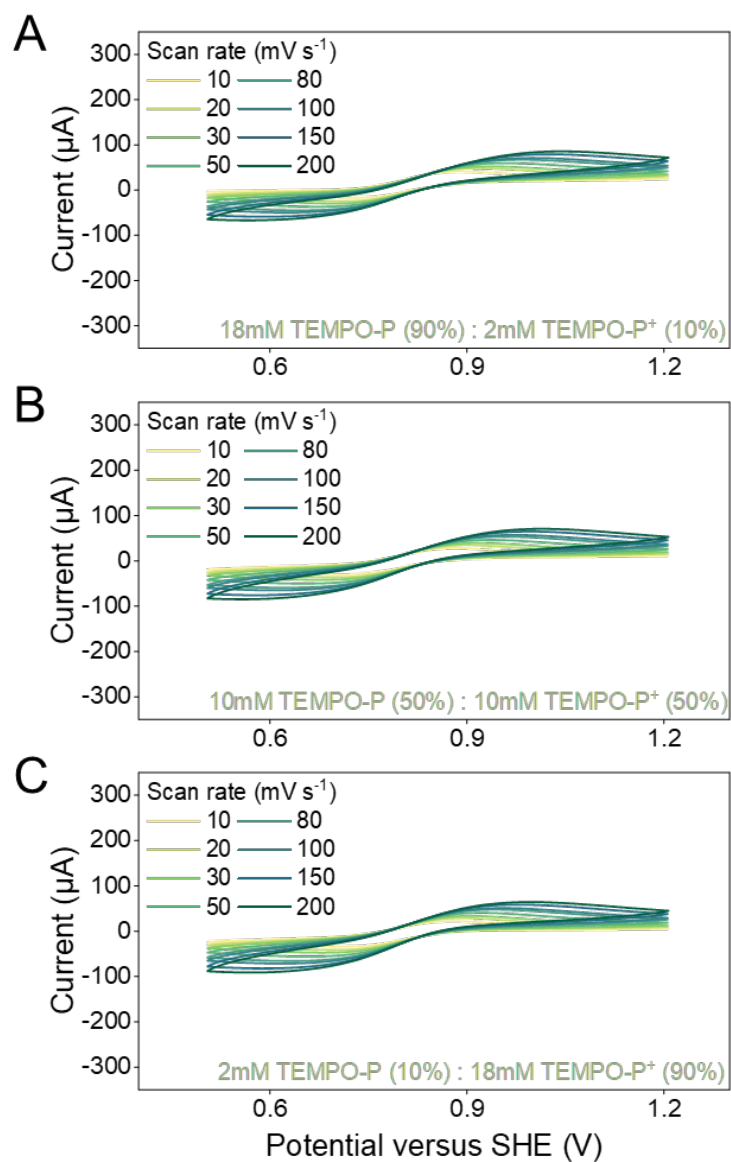
Supplementary Fig. 38: The high-resolution mass spectrometry profile of TEMPO-P under varying pH conditions (pH 12, 11, 7, 3, 2) after 6 days. (A-E) Profile of TEMPO-P at pH 12, 11, 7, 3, and 2, respectively. No additional peaks were detected, indicating the chemical robustness of TEMPO-P across the tested pH range.



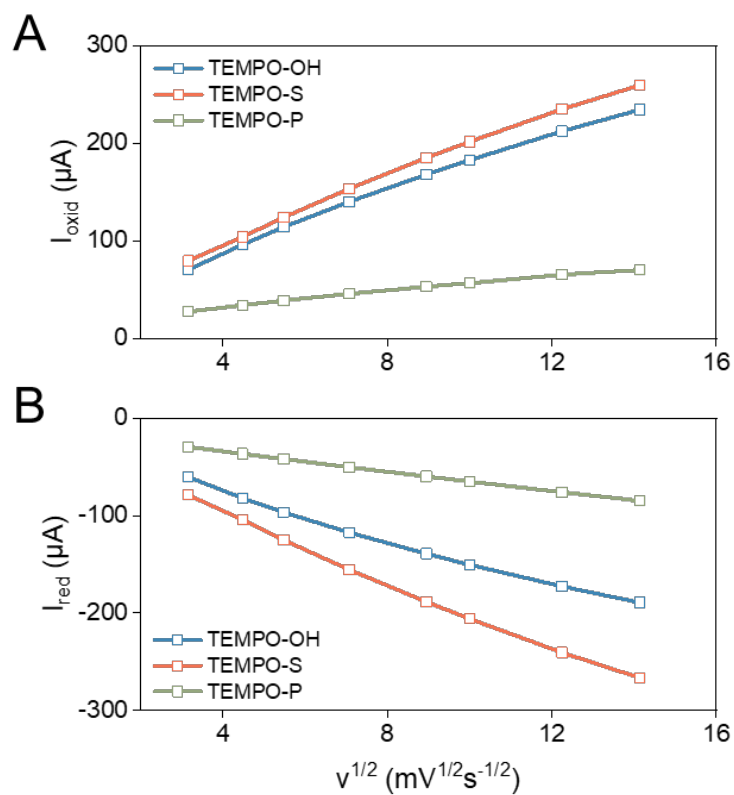
Supplementary Fig. 39: The cyclic voltammetry profiles of 20 mM TEMPO-OH at scan rates ranging from 10 to 200 mV s^{-1} at different TEMPO-OH: TEMPO-OH⁺ ratios. (A) 90% TEMPO-OH: 10% TEMPO-OH⁺. (B) 50% TEMPO-OH: 50% TEMPO-OH⁺. (C) 10% TEMPO-OH: 90% TEMPO-OH⁺.



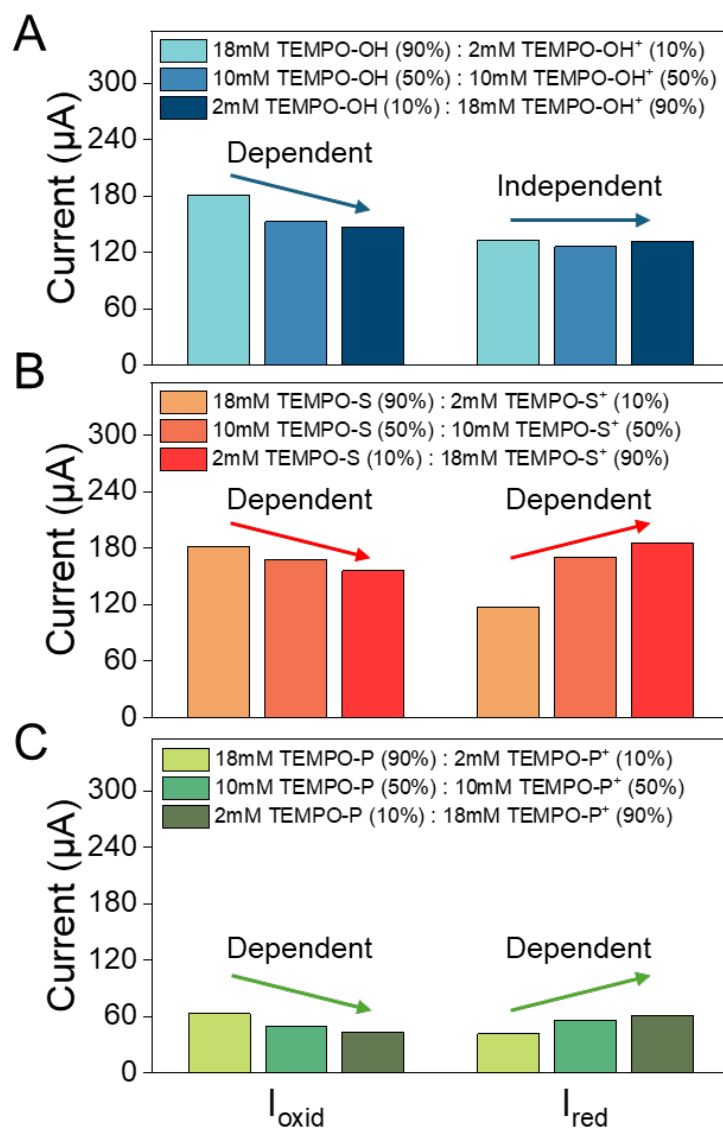
Supplementary Fig. 40: The cyclic voltammetry profiles of 20 mM TEMPO-S at scan rates ranging from 10 mV s^{-1} to 200 mV s^{-1} at different TEMPO-S: TEMPO-S⁺ ratios. (A) 90% TEMPO-S: 10% TEMPO-S⁺. (B) 50% TEMPO-S: 50% TEMPO-S⁺. (C) 10% TEMPO-S: 90% TEMPO-S⁺.



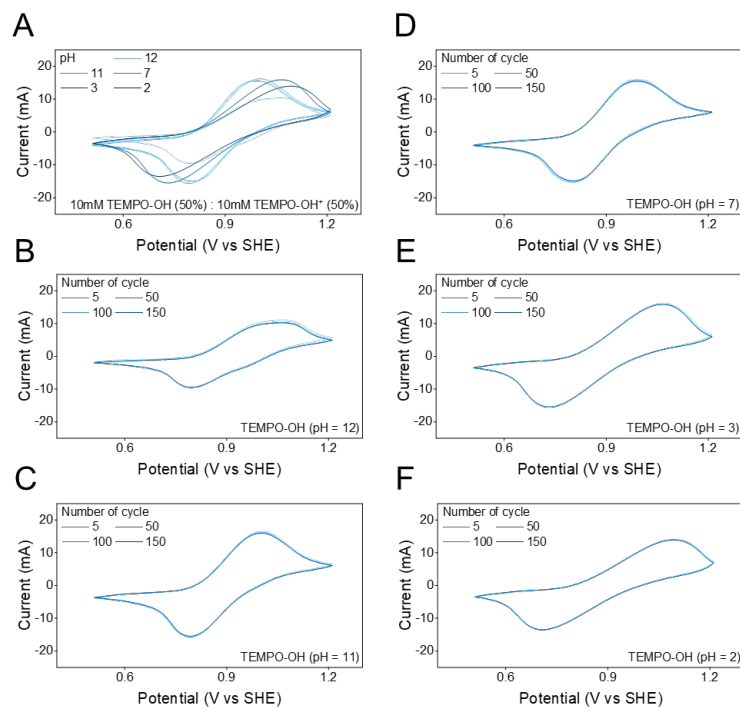
Supplementary Fig. 41: The cyclic voltammetry profiles of 20 mM TEMPO-P at scan rates ranging from 10 mV s^{-1} to 200 mV s^{-1} at different TEMPO-P: TEMPO-P⁺ ratios. (A) 90% TEMPO-P: 10% TEMPO-P⁺. (B) 50% TEMPO-P: 50% TEMPO-P⁺. (C) 10% TEMPO-P: 90% TEMPO-P⁺.



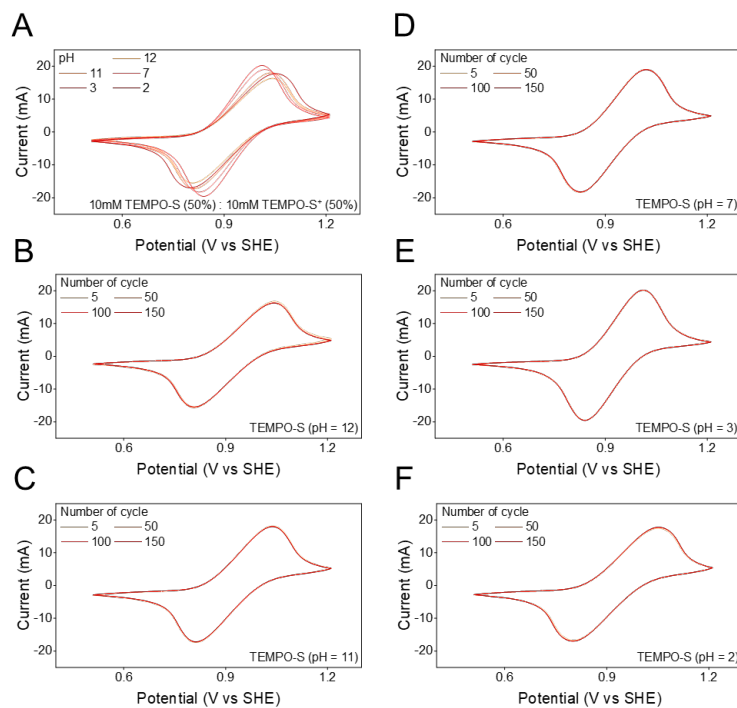
Supplementary Fig. 42: Summary of the peak TEMPO oxidation current (I_{oxid}) and the peak TEMPO⁺ reduction current (I_{red}) versus the square root of CV scan rate ($v^{1/2}$) of three TEMPO derivatives at ratio of 50% TEMPO: 50% TEMPO⁺, indicating they are under diffusion control region (A) I_{oxid} versus $v^{1/2}$. (B) I_{red} versus $v^{1/2}$.



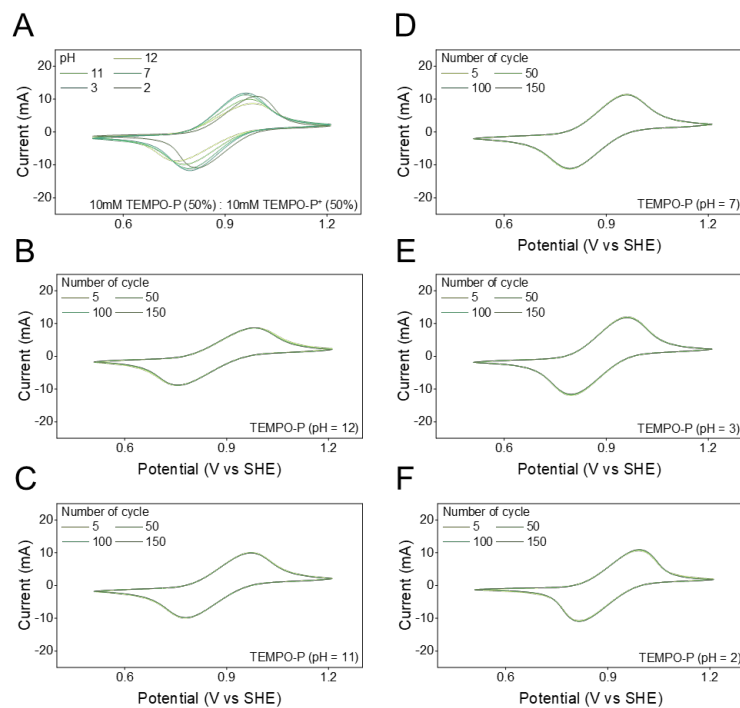
Supplementary Fig. 43: Summary of the I_{oxid} and I_{red} for three TEMPO derivatives at different TEMPO: TEMPO⁺ ratios at a scan rate of 100 mV s⁻¹. (A) TEMPO-OH. (B) TEMPO-S. (C) TEMPO-P.



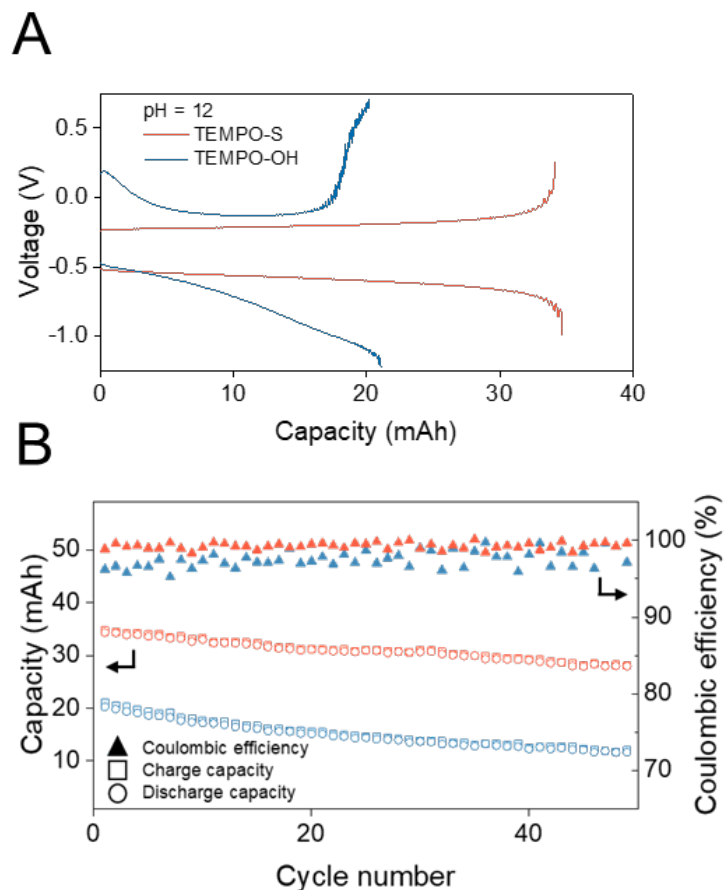
Supplementary Fig. 44: The cyclic voltammetry profiles of 20 mM TEMPO-OH (50% TEMPO-OH: 50% TEMPO-OH⁺ ratio) in 0.25 M K₂SO₄ aqueous electrolyte under varying pH conditions (pH 12, 11, 7, 3, 2). (A) Summary of CV profile for all pH conditions. (B-F) CV profiles of TEMPO-OH at pH 12, 11, 7, 3, and 2, respectively, indicating the electrochemical robustness of TEMPO-OH across a pH range of 11 to 2.



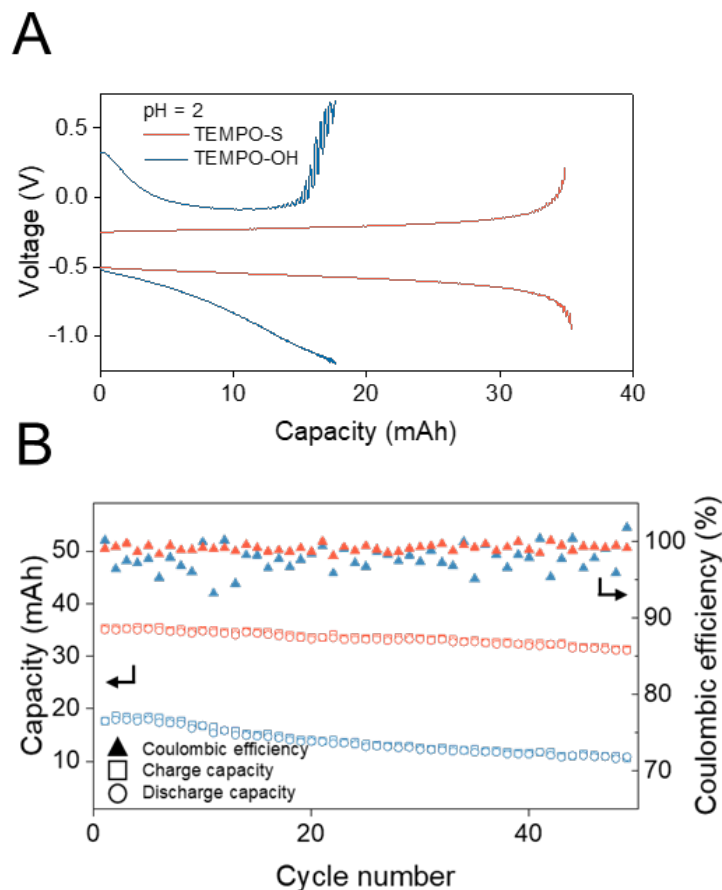
Supplementary Fig. 45: The cyclic voltammetry profiles of 20 mM TEMPO-S (50% TEMPO-S: 50% TEMPO-S⁺ ratio) in 0.25 M K₂SO₄ aqueous electrolyte under varying pH conditions (pH 12, 11, 7, 3, 2). (A) Summary of CV profile for all pH conditions. (B-F) CV profiles of TEMPO-S at pH 12, 11, 7, 3, and 2, respectively, indicating the electrochemical robustness of TEMPO-S across a pH range of 12 to 2.



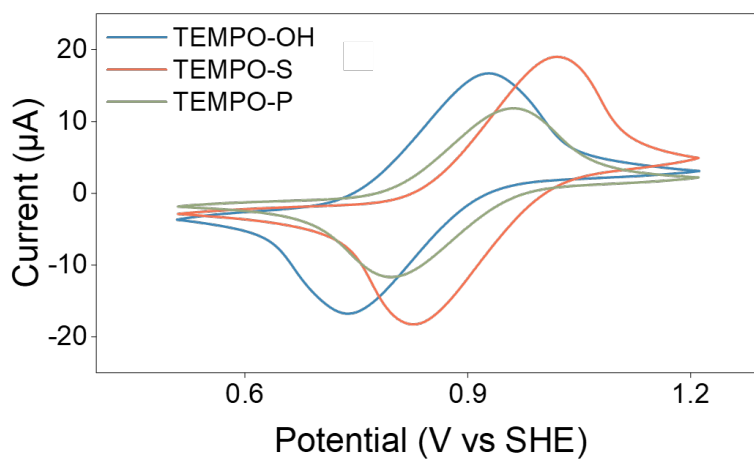
Supplementary Fig. 46: The cyclic voltammetry profiles of 20 mM TEMPO-P (50% TEMPO-P: 50% TEMPO-P⁺ ratio) in 0.25 M K₂SO₄ aqueous electrolyte under varying pH conditions (pH 12, 11, 7, 3, 2). (A) Summary of CV profile for all pH conditions. (B-H) CV profiles of TEMPO-P at pH 12, 11, 7, 3, and 2, respectively, indicating the electrochemical robustness of TEMPO-P across a pH range of 12 to 2.



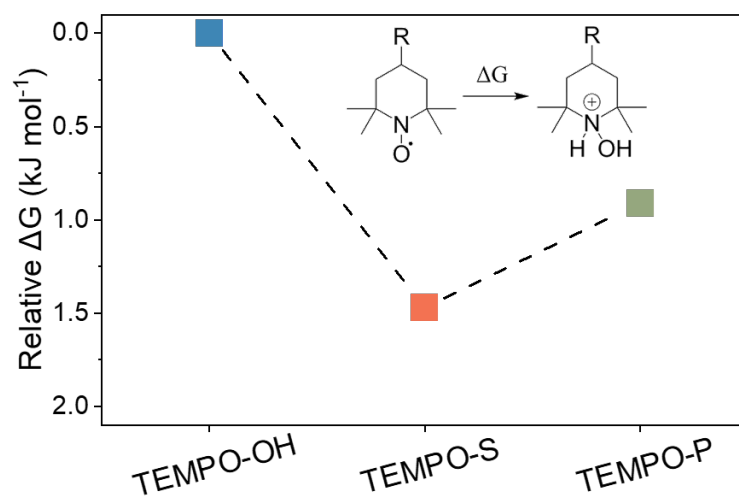
Supplementary Fig. 47: Galvanostatic charge-discharge profiles of TEMPO-OH and TEMPO-S in flow battery tests under basic (pH 12) condition. (A) Voltage profiles of TEMPO-OH and TEMPO-S flow cell under pH 12 condition at 50 mA cm^{-2} . (B) Charge capacity, discharge capacity, and columbic efficiency over prolonged cycling test at current density of 50 mA cm^{-2} . Note that, at pH 12, TEMPO-S has a decay rate of $0.37\% \text{ cycle}^{-1}$ ($5.83 \text{ mol L}^{-1} \text{ yr}^{-1}$), whereas TEMPO-OH showed a decay rate of $0.86\% \text{ per cycle}^{-1}$ ($18.08 \text{ mol L}^{-1} \text{ yr}^{-1}$). Note the volume of the working electrolyte was 5 mL 0.1 M TEMPO concentration. The counter electrolyte was 20 mL 0.2 M ferricyanide/ 0.2 M ferrocyanide in 0.25 M K_2SO_4 .



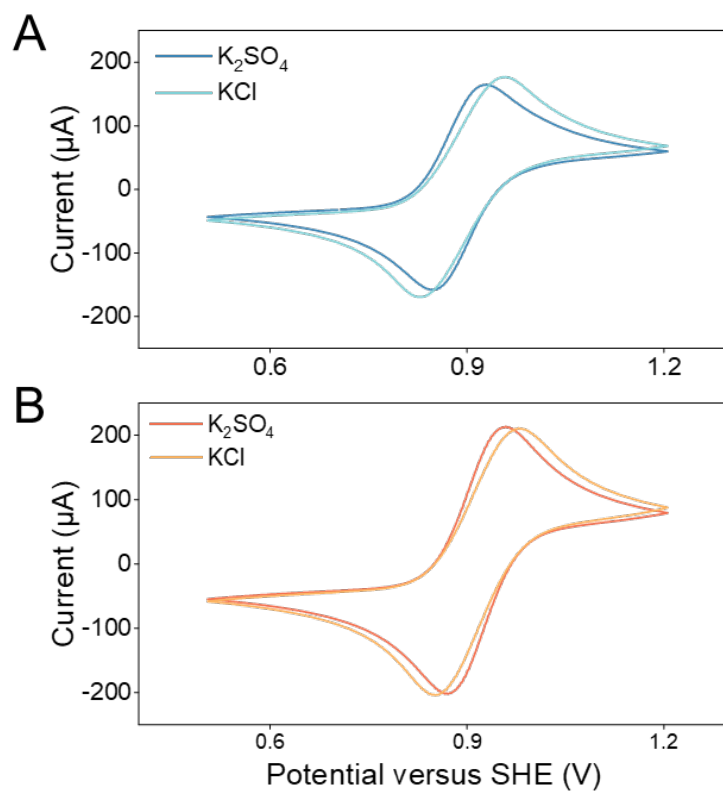
Supplementary Fig. 48: Galvanostatic charge-discharge profiles of TEMPO-OH and TEMPO-S in flow battery tests under acidic (pH 2) condition. (A) Voltage profiles of TEMPO-OH and TEMPO-S flow cell under pH 2 condition at 50 mA cm^{-2} . **(B)** Charge capacity, discharge capacity, and columbic efficiency over prolonged cycling test at current density of 50 mA cm^{-2} . Note that, at pH 2, TEMPO-S has a decay rate of $0.23\% \text{ cycle}^{-1}$ ($3.63 \text{ mol L}^{-1} \text{ yr}^{-1}$), whereas TEMPO-OH showed a decay rate of $0.79\% \text{ per cycle}^{-1}$ ($16.61 \text{ mol L}^{-1} \text{ yr}^{-1}$). Note the volume of the working electrolyte was 5 mL 0.1 M TEMPO concentration. The counter electrolyte was 20 mL 0.2 M ferricyanide/ 0.2 M ferrocyanide in 0.25 M K_2SO_4 .



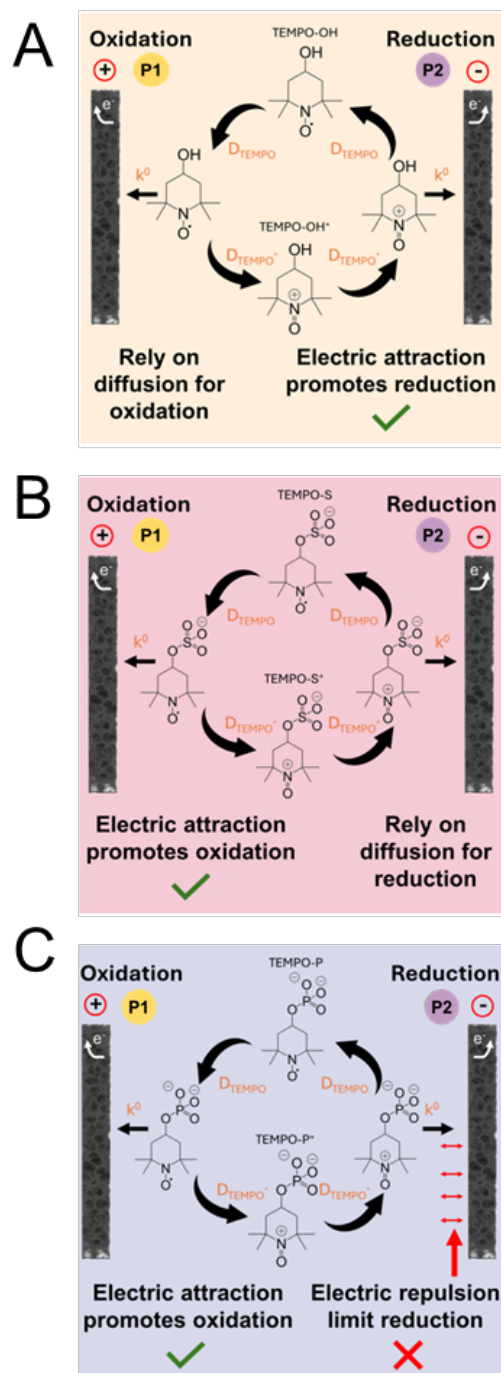
Supplementary Fig. 49: Summary cyclic voltammetry profiles of 20 mM TEMPO-OH, TEMPO-S and TEMPO-P at ratio of 50% TEMPO: 50% TEMPO⁺ in 0.25 M K₂SO₄ aqueous electrolyte under neutral conditions. Note that the working electrode is hydrophilic carbon paper.



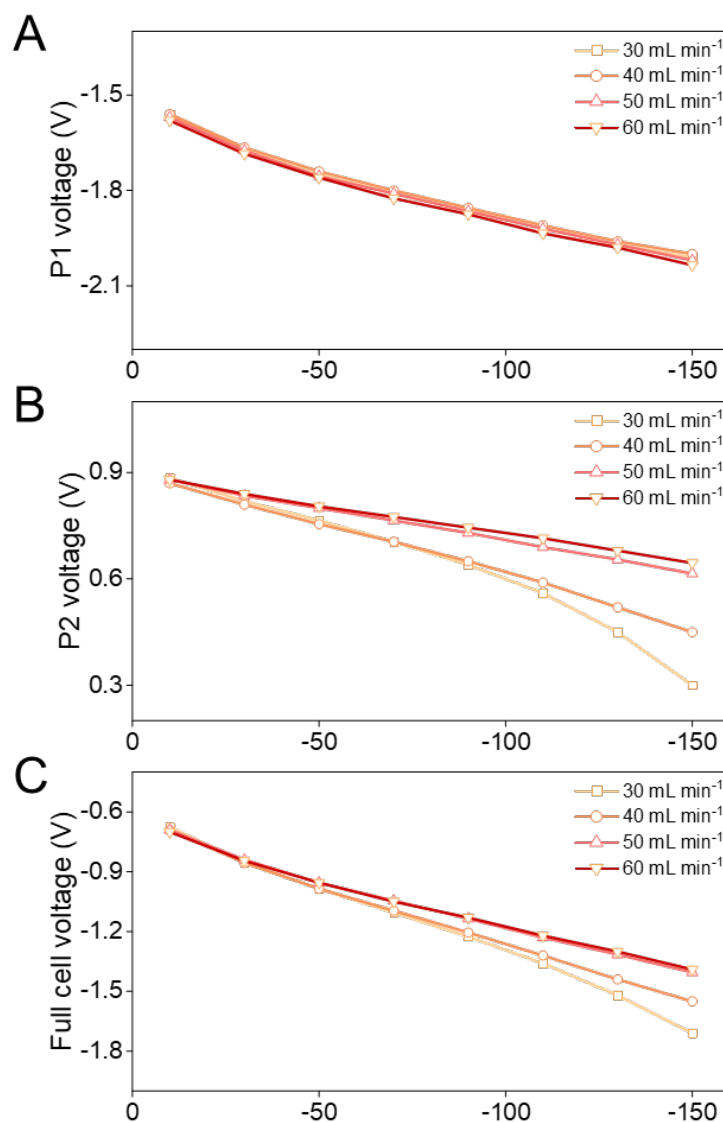
Supplementary Fig. 50: DFT calculations show that TEMPO-S exhibits the highest ΔG for side reactions compared to TEMPO-OH and TEMPO-P, indicating superior stability of TEMPO-S.



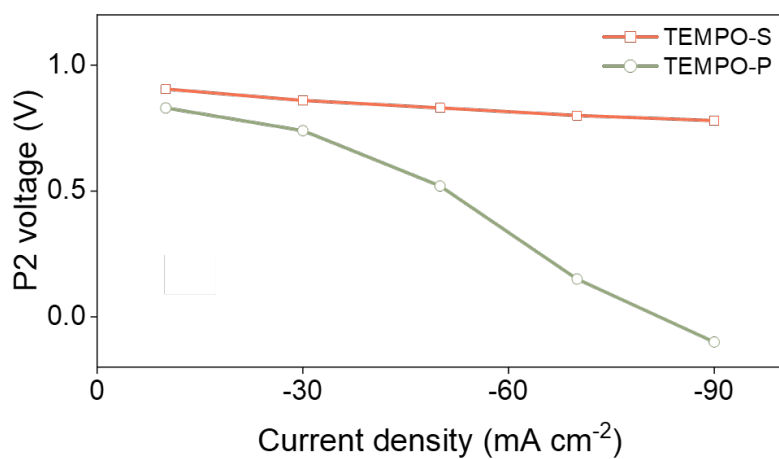
Supplementary Fig. 51: The cyclic voltammetry profiles of 20 mM TEMPO-OH and TEMPO-S at ratio of 50% TEMPO: 50% TEMPO⁺ in different supporting electrolyte. (A) TEMPO-OH in 0.25 M K_2SO_4 and 0.5 M KCl aqueous electrolyte under neutral conditions. (B) TEMPO-S in 0.25 M K_2SO_4 and 0.5 M KCl aqueous electrolyte under neutral conditions.



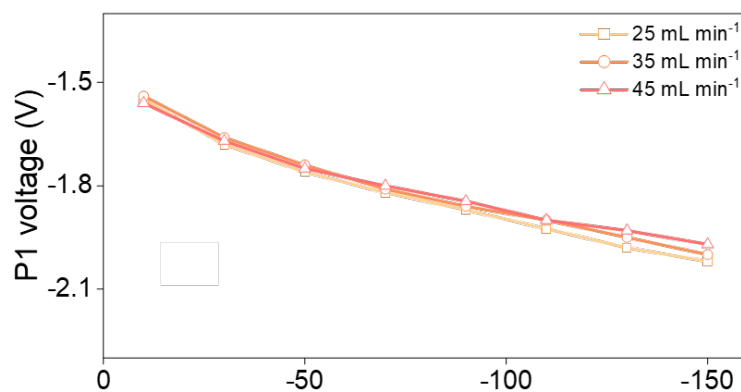
Supplementary Fig. 52: Schematic of electrochemical behaviour during oxidation and reduction of three different TEMPO derivatives. (A) Illustration of TEMPO-OH with overall charge of 0 and +1, achieving effective oxidation and reduction. **(B)** Illustration of TEMPO-S with overall charge of -1 and 0, achieving effective oxidation and reduction. **(C)** Illustration of TEMPO-P with overall charge of -2 and -1, limiting reduction due to negatively charged electric field repulsion.



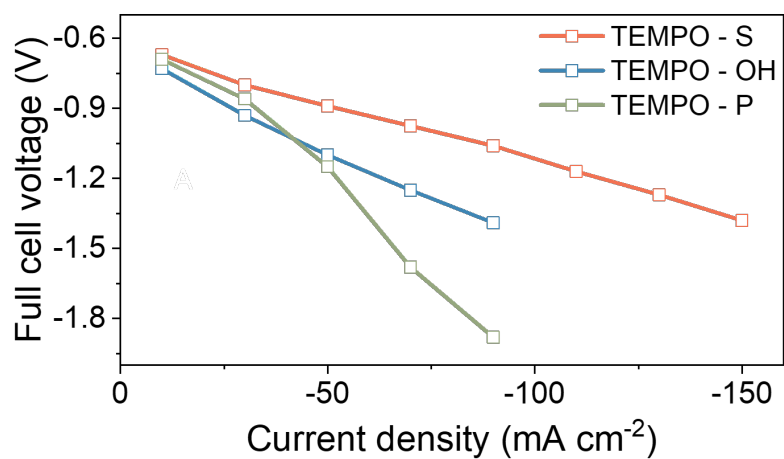
Supplementary Fig. 53: Effect of CO₂ liberation solution flow rate on full-cell voltage in a redox-decoupled system, using 0.1 M TEMPO-S (50% TEMPO-S: 50% TEMPO-S⁺ ratio) in 0.25 M K₂SO₄ aqueous electrolyte. (A) P1 voltage profile. (B) P2 voltage profile. (C) Full-cell voltage profile. The P1 (TEMPO oxidation) voltage remains constant across different CO₂ liberation solution flow rates, attributed to electric attraction. In contrast, the P2 (TEMPO reduction) voltage varies with flow rate, indicating a diffusion-controlled electrochemical process. The optimal flow rate of the CO₂ liberation solution was found to be 50 mL min⁻¹.



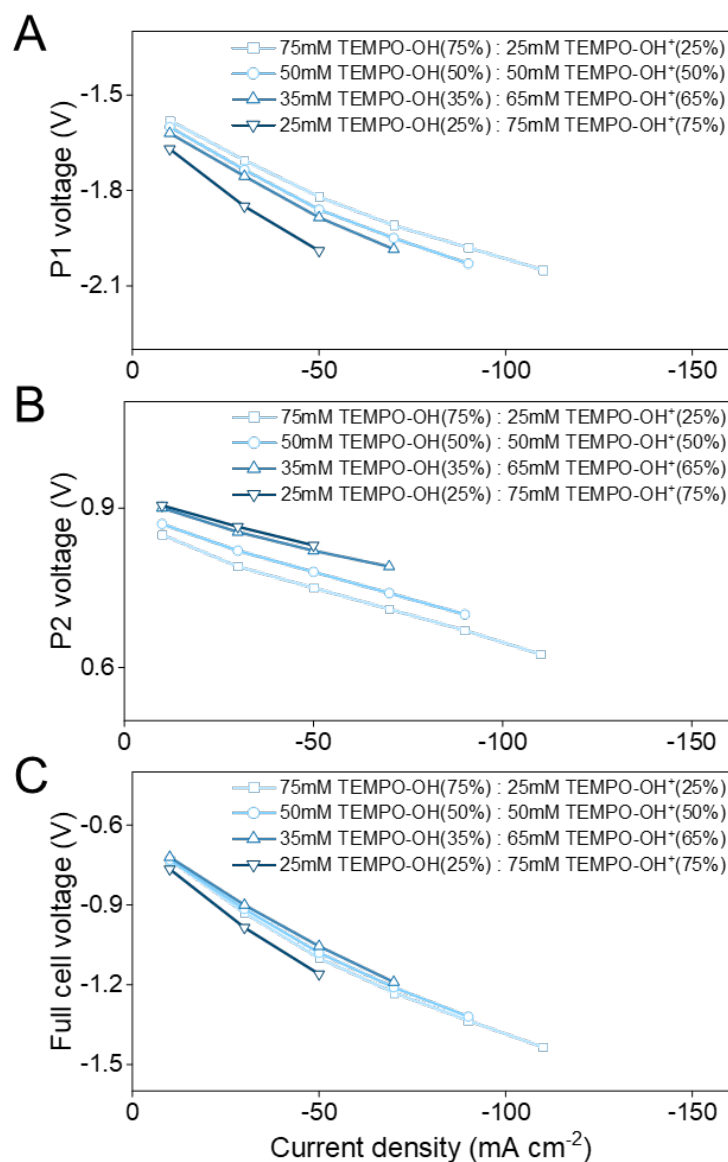
Supplementary Fig. 54: P2 voltage profile using 0.1 M TEMPO-S and TEMPO-P (50% TEMPO: 50% TEMPO⁺ ratio) in 0.25 M K₂SO₄ aqueous electrolyte during the redox-decoupled system at a 50 mL min⁻¹ flow rate. TEMPO-P⁺ carries a net charge of -1, which leads to electrostatic repulsion at the reduction electrode in P2.



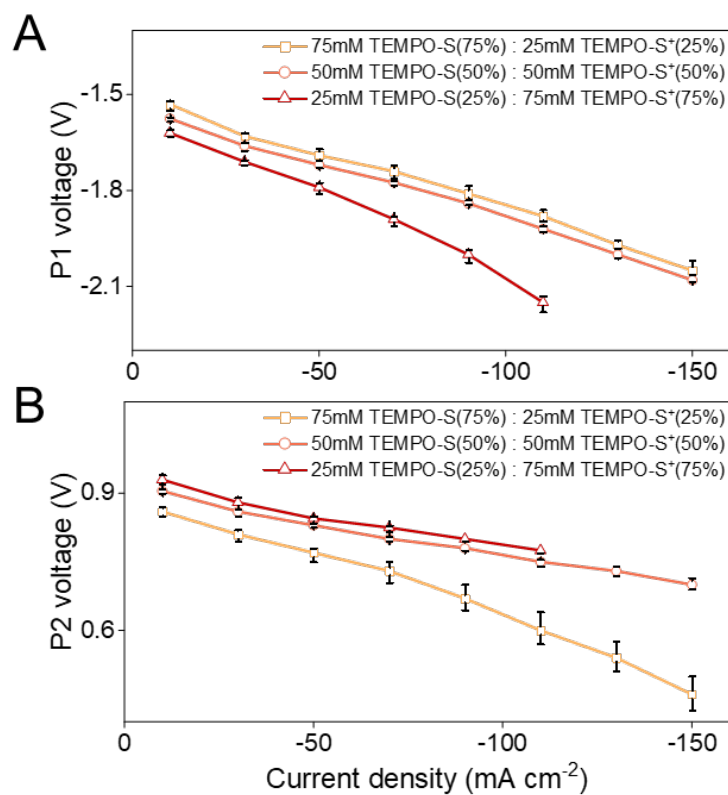
Supplementary Fig. 55: Effect of capture solution flow rate on P1 voltage in a redox-decoupled system, using 0.1 M TEMPO-S (50% TEMPO-S: 50% TEMPO-S⁺ ratio) in 0.25 M K₂SO₄ aqueous electrolyte. The P1 voltage remains constant across different capture solution flow rates, indicating 25 mL min⁻¹ is sufficient to remove the H₂ gas generated.



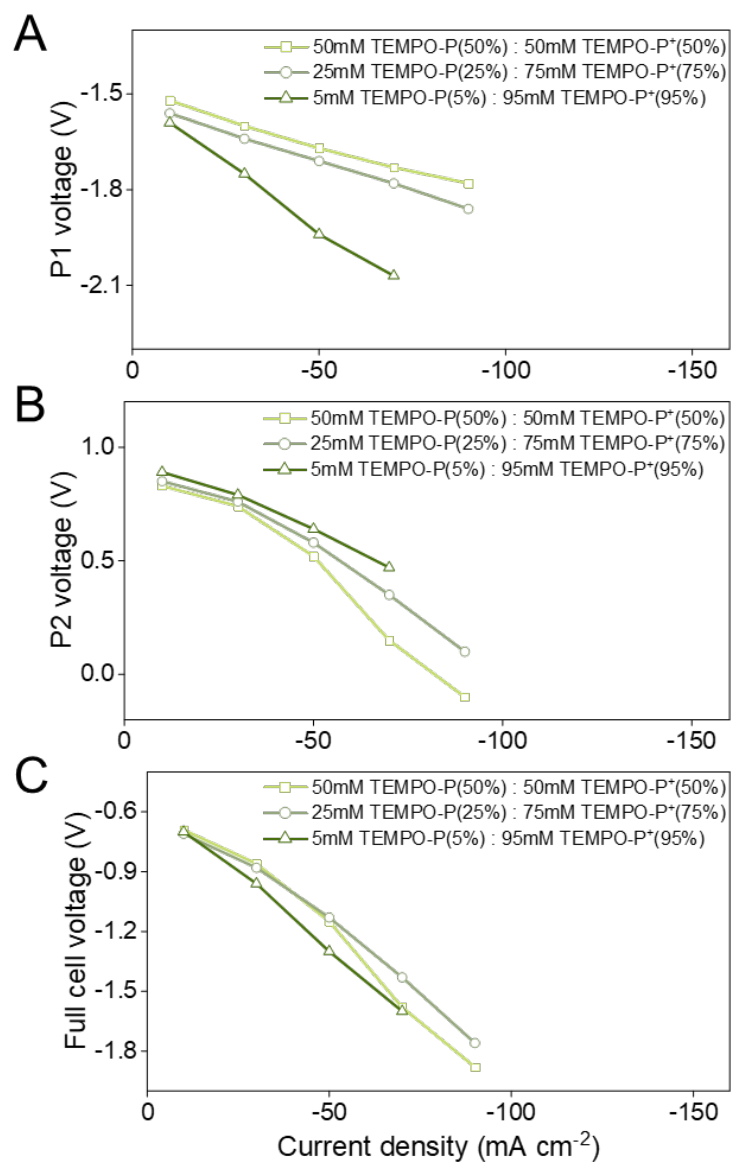
Supplementary Fig. 56: Summary of the full-cell voltage during redox-decoupled system using 0.1 M TEMPO in 0.25 M K₂SO₄ aqueous electrolyte at their highest efficiency ratios for TEMPO-OH, TEMPO-S, and TEMPO-P, respectively.



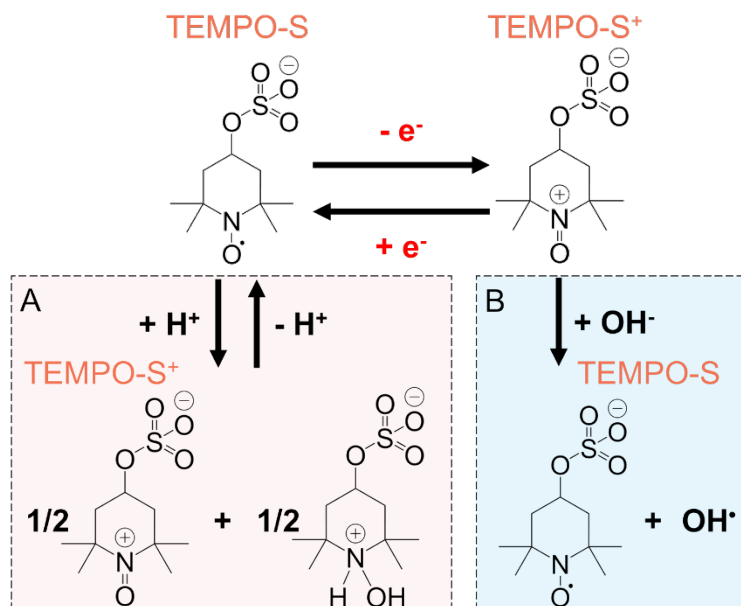
Supplementary Fig. 57: Performance using 0.1 M TEMPO-OH in 0.25 M K₂SO₄ aqueous electrolyte during the redox-decoupled system at varying TEMPO-OH: TEMPO-OH⁺ ratios. (A) P1 voltage profile. (B) P2 voltage profile. (C) Full-cell voltage profile.



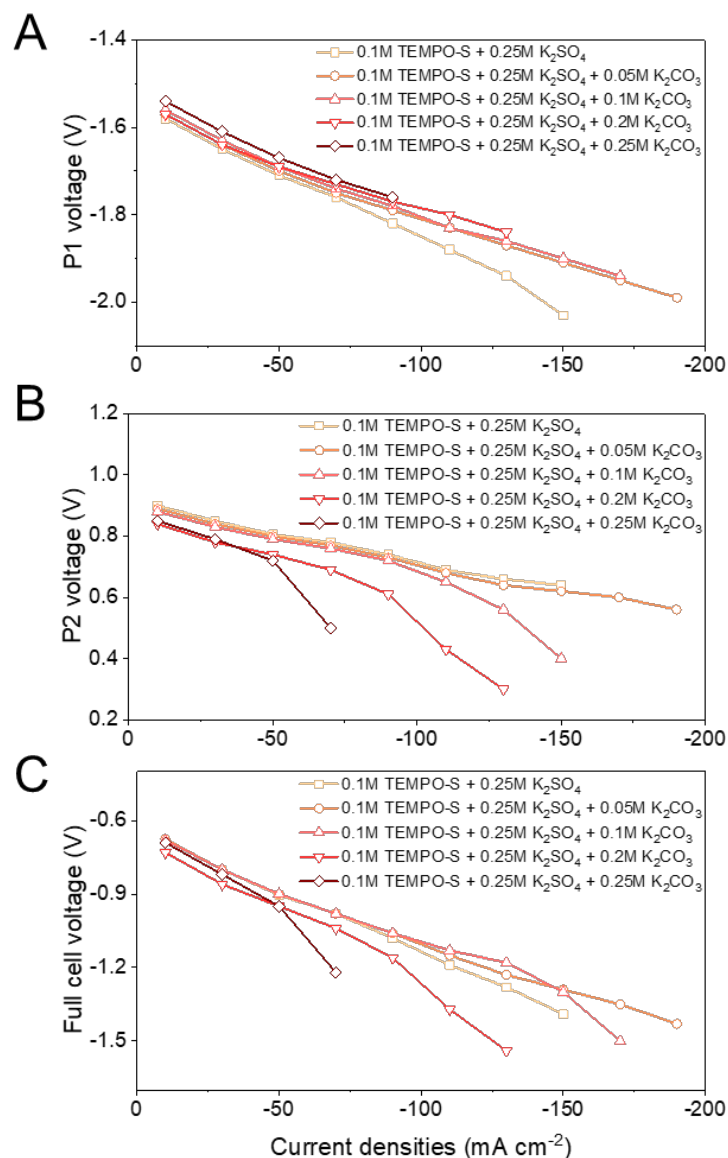
Supplementary Fig. 58: Performance using 0.1 M TEMPO-S in 0.25 M K₂SO₄ aqueous electrolyte during the redox-decoupled system at varying TEMPO-S: TEMPO-S⁺ ratios. (A) P1 voltage profile, data are presented as mean values $\pm$ 0.03 (n=3). (B) P2 voltage profile, data are presented as mean values $\pm$ 0.04 (n=3).



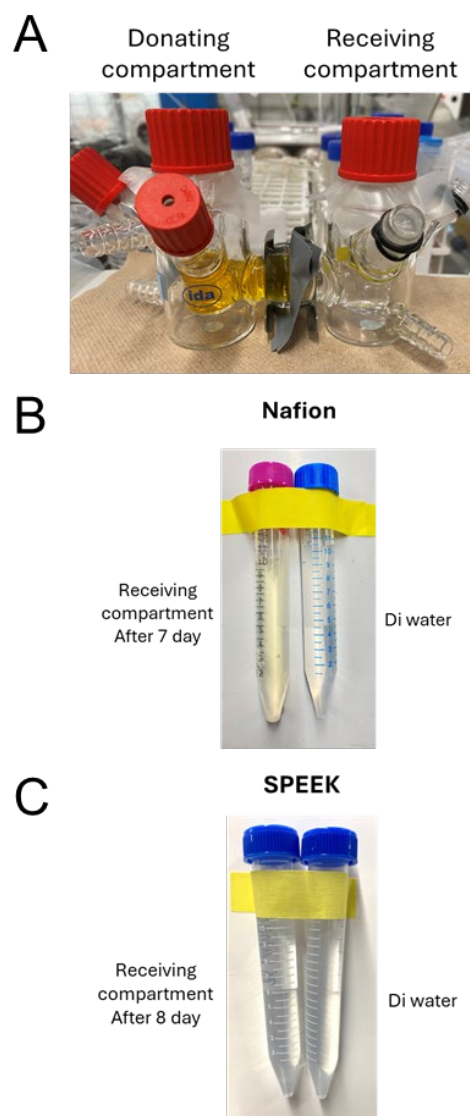
Supplementary Fig. 59: Performance using 0.1 M TEMPO-P in 0.25 M K₂SO₄ aqueous electrolyte during the redox-decoupled system at varying TEMPO-P: TEMPO-P⁺ ratios. (A) P1 voltage profile. (B) P2 voltage profile. (C) Full-cell voltage profile.



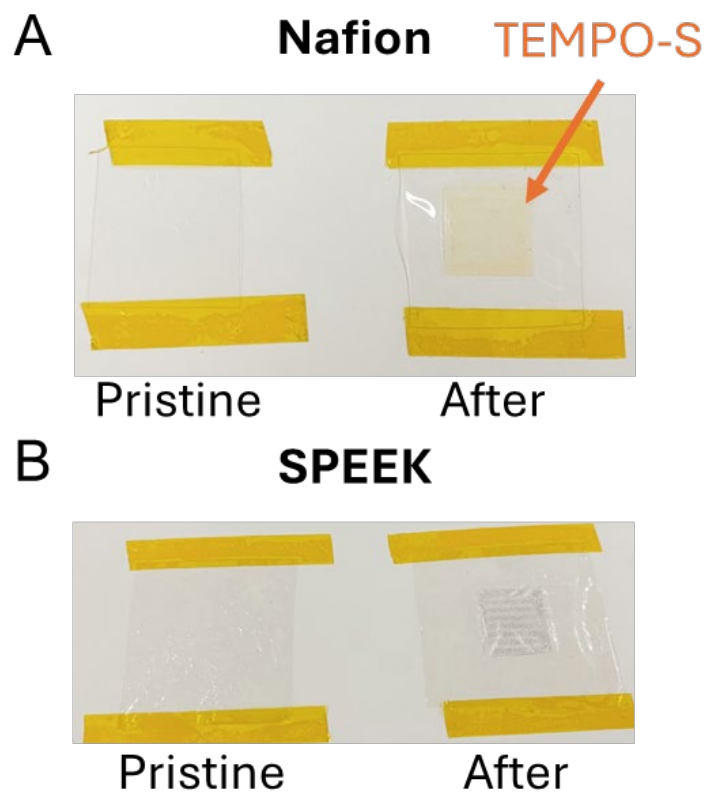
Supplementary Fig. 60: Reversible pH-dependent side reactions of TEMPO-S. (A) The disproportionation reaction of TEMPO-S with H⁺. (B) The chemical reduction of TEMPO-S⁺ with OH⁻.²⁶



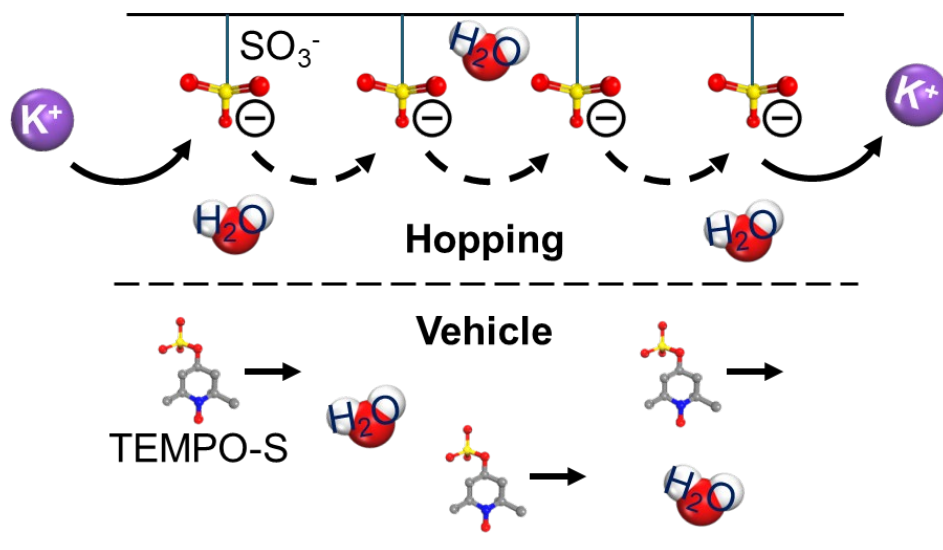
Supplementary Fig. 61: Performance using 0.1 M TEMPO-S (50% TEMPO-S: 50% TEMPO-S⁺ ratio) in 0.25 M K₂SO₄ aqueous electrolyte during the redox-decoupled system at varying carbonate concentration. (A) P1 voltage profile. (B) P2 voltage profile. (C) Full-cell voltage profile.



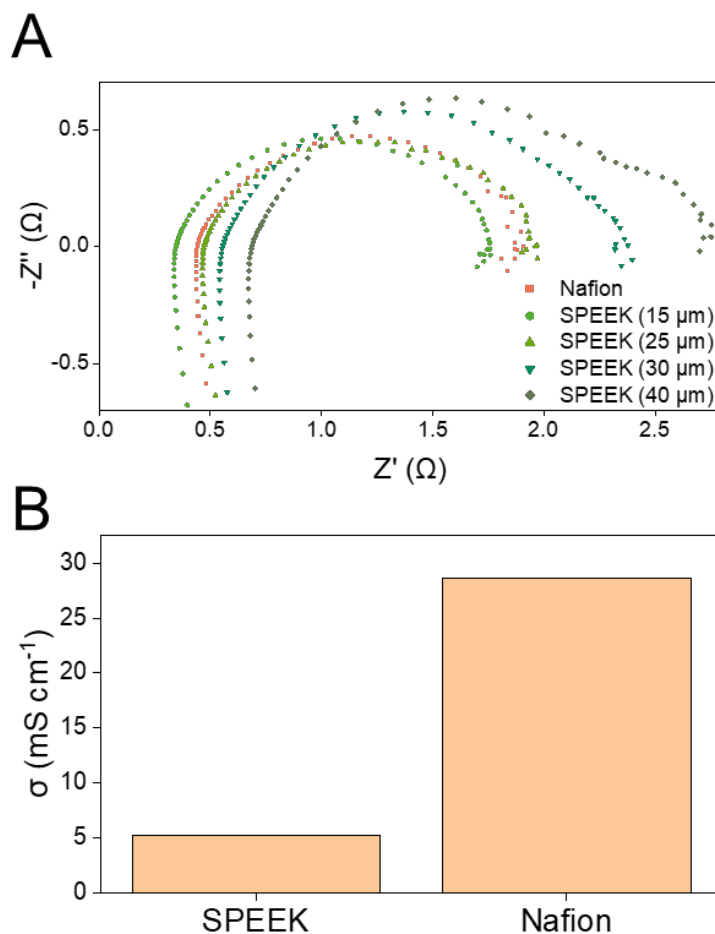
Supplementary Fig. 62: Experimental setup for TEMPO-S permeability experiment. (A) Image of setup for TEMPO-S permeability experiment. **(B)** Image of receiving compartment solution after 7 days, using Nafion membrane. **(C)** Image of receiving compartment solution after 8 days, using the cast sulfonated poly(ether ether ketone) (SPEEK) membrane.



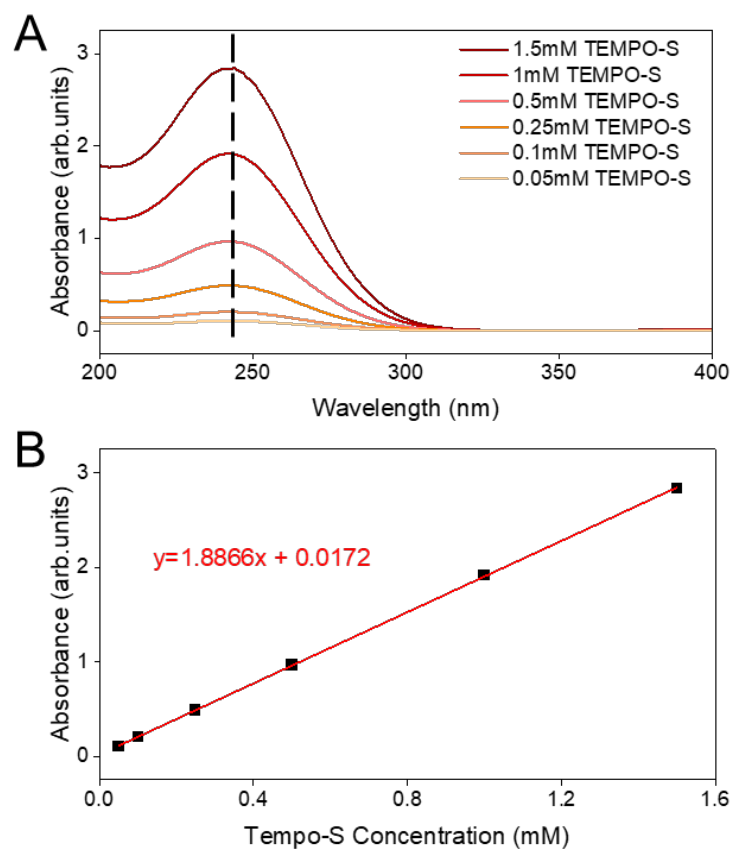
Supplementary Fig. 63: Image of the different CEMs employed in capture solution formation process (P1) in redox-decoupled system. (A) Pristine Nafion (left), TEMPO-S contained into the Nafion after P1 operation (right). (B) Pristine SPEEK (left), no visible TEMPO-S contained into the SPEEK membrane after P1 operation (right).



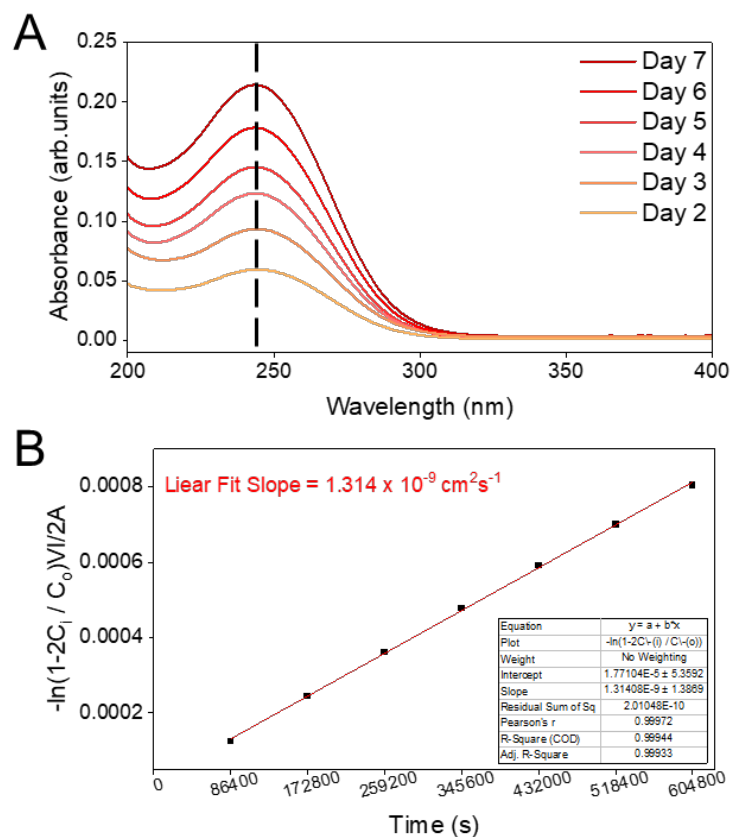
Supplementary Fig. 64: Schematics representation of K^+ transportation via the hopping mechanism (top), TEMPO-S and TEMPO-S⁺ transportation via the vehicle mechanism (bottom).



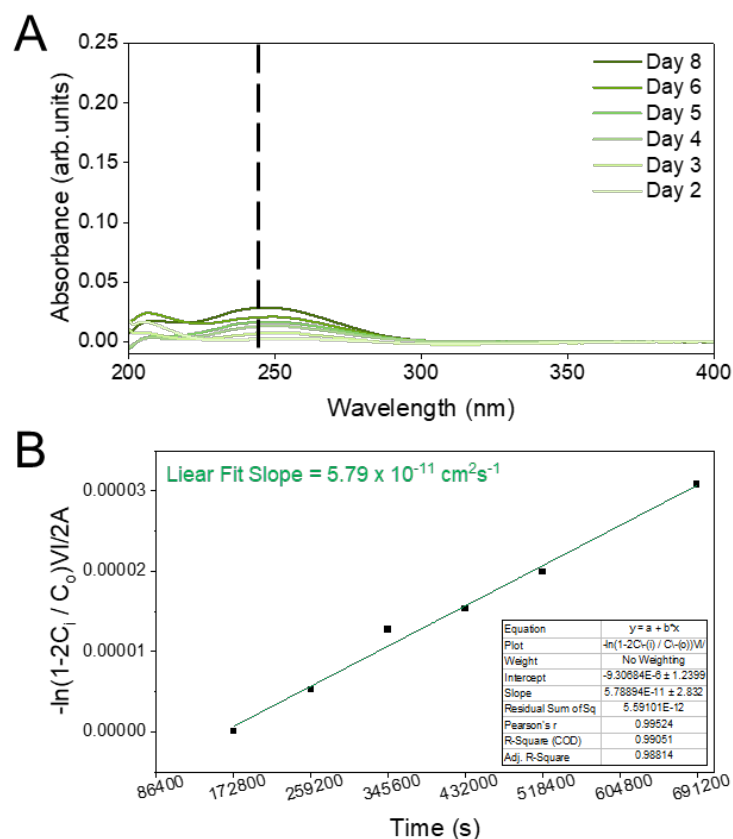
Supplementary Fig. 65: Electrochemical impedance spectroscopy (EIS) comparison of cast SPEEK and Nafion 115. (A) Full-cell EIS measurements for different thickness of casted SPEEK and conventional Nafion. (B) The membrane conductivity of the cast SPEEK is lower compared to the conventional Nafion. It is worth noting that the polymer structures of Nafion and SPEEK are inherently different and SPEEK exhibits a relatively lower ionic conductivity compared to Nafion²⁷.



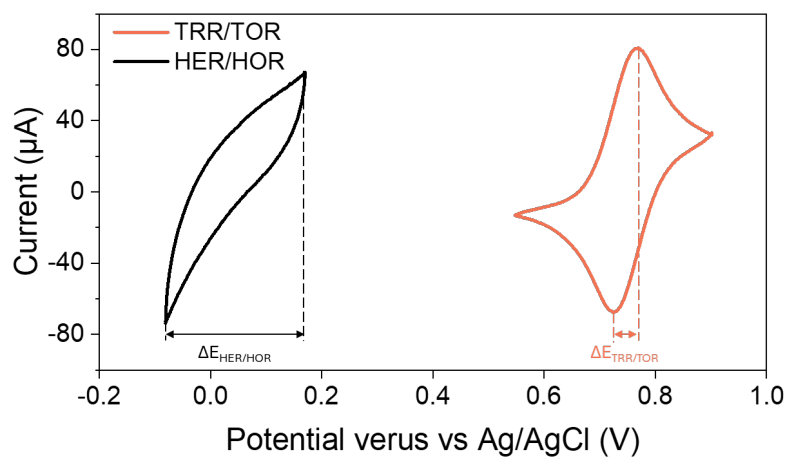
Supplementary Fig. 66: The UV-vis spectroscopy analysis of TEMPO-S standard. (A) UV-vis spectra of different TEMPO-S concentrations, 0.05, 0.1, 0.25, 0.5, 1, 1.5 mM. **(B)** Calibration line from measured TEMPO-S concentrations. The TEMPO-S signal at 245 nm.



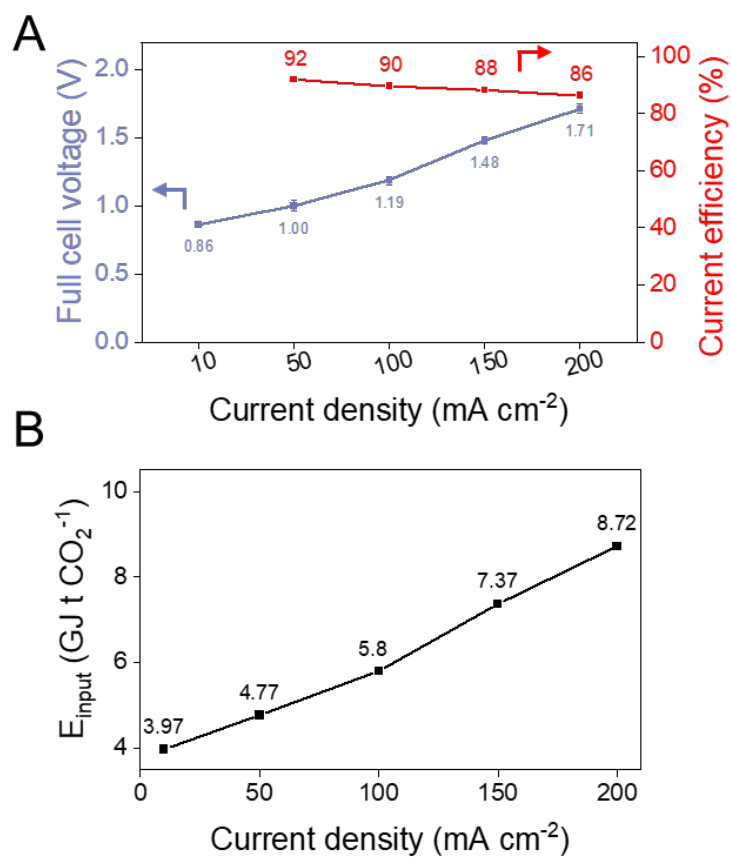
Supplementary Fig. 67: The UV-vis spectroscopy analysis for TEMPO-S permeability through the Nafion 115 membrane. (A) TEMPO-S concentration in the receiving compartment after 7 days. (B) Estimated permeability of TEMPO-S with a value of $1.31 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1}$. The TEMPO-S signal at 245 nm.



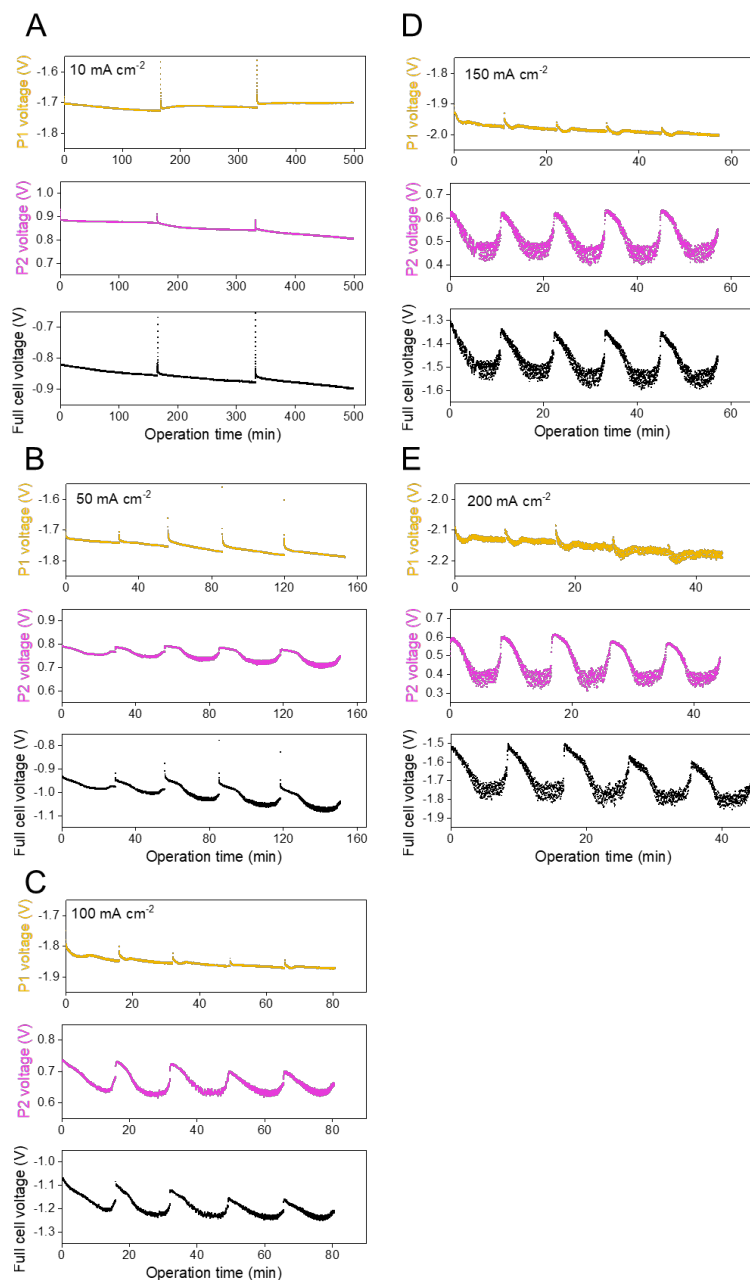
Supplementary Fig. 68: The UV-vis spectroscopy analysis for TEMPO-S permeability through the cast 15 μm SPEEK membrane. (A) TEMPO-S concentration in the receiving compartment after 8 days. (B) Estimated permeability of TEMPO-S with a value of $5.79 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$. The TEMPO-S signal at 245 nm.



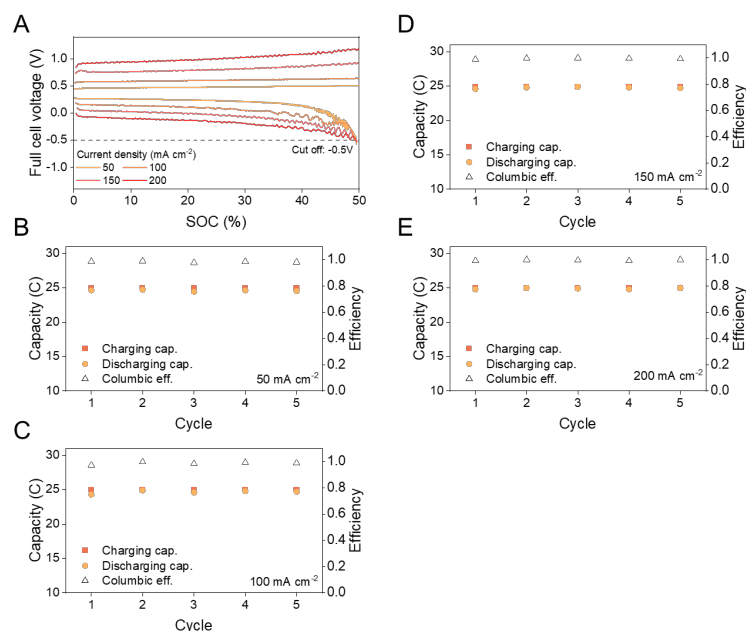
Supplementary Fig. 69: Cyclic voltammetry of DI water and diluted TEMPO-S at room temperature. The minimum energy was estimated based on the sum of $\Delta E_{\text{HER/HOR}}$ and $\Delta E_{\text{TRR/TOR}}$.



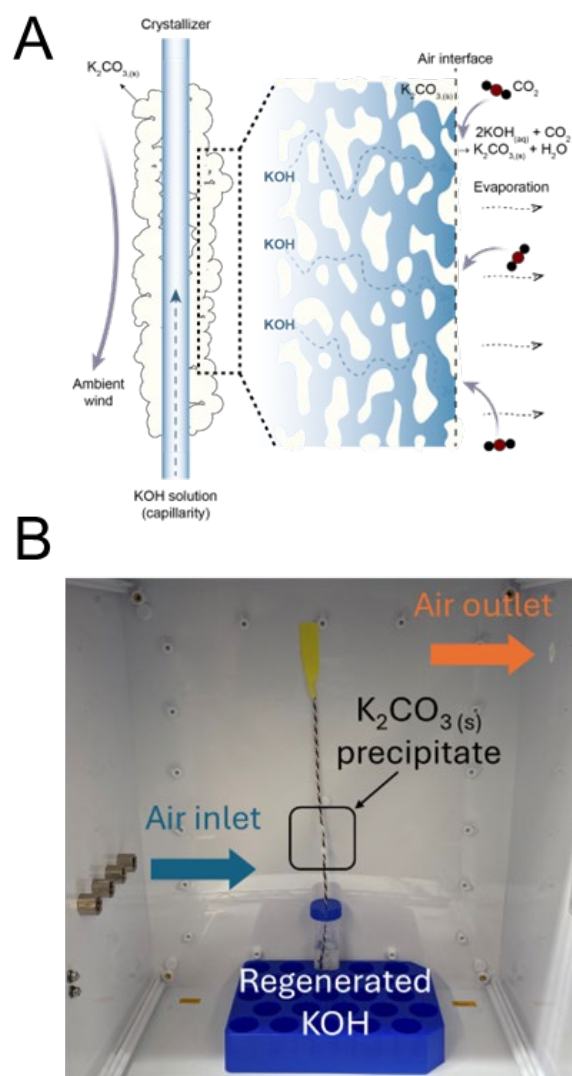
Supplementary Fig. 70: The redox-decoupled system voltage, current efficiency, and experimental energy consumption at current densities ranging from 10 to 200 mA cm⁻². (A) Full-cell voltage and current efficiency profiles. (B) Experimental energy input (E_{input}) profiles ($n=3$).



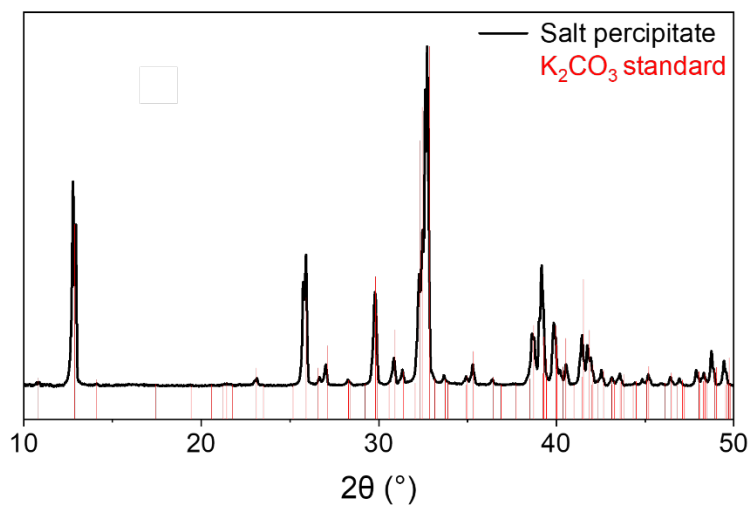
Supplementary Fig. 71: The redox-decoupled system P1, P2, and full cell voltage at current densities ranging from 10 to 200 mA cm⁻² to obtain a 1 M KOH capture solution. (A) Voltage profiles at 10 mA cm⁻². (B) Voltage profiles at 50 mA cm⁻². (C) Voltage profiles at 100 mA cm⁻². (D) Voltage profiles at 150 mA cm⁻². (E) Voltage profiles at 200 mA cm⁻². Note that initial electrolyte volume of capture solution was 5 mL Di water and CO₂ liberation solution was 10 mL 0.1 M TEMPO-S in 0.25 M K₂SO₄ with 0.05 M K₂CO₃ chemical. For the 10 mA cm⁻² case, we employed an initial 5 mL 1 M KOH capture solution to simulate the concentrated OH⁻ condition.



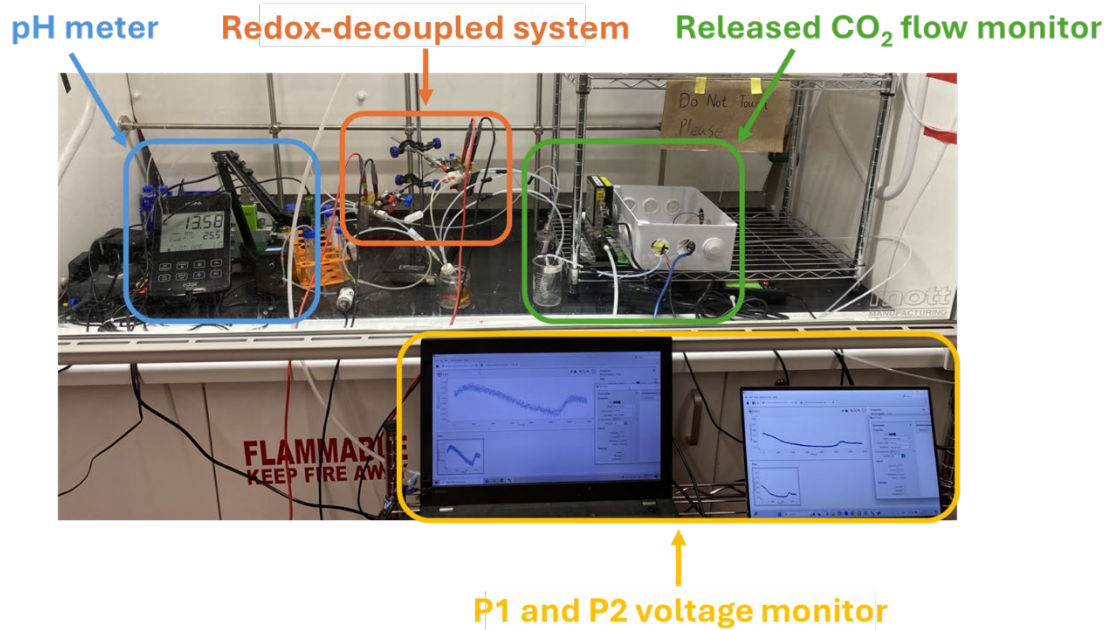
Supplementary Fig. 72: Electrochemical performance of TEMPO-S in flow battery tests. (A) Galvanostatic charge-discharge profiles at various current densities with a discharge cutoff voltage of -0.5 V. **(B-F)** Charge capacity, discharge capacity, and columbic efficiency over 5-round TEMPO-S charge-discharge experiment at current densities ranging from 50 to 200 mA cm⁻². Columbic efficiency exceeded 0.99 in all cases, indicating negligible oxygen evolution reaction across the tested current densities.



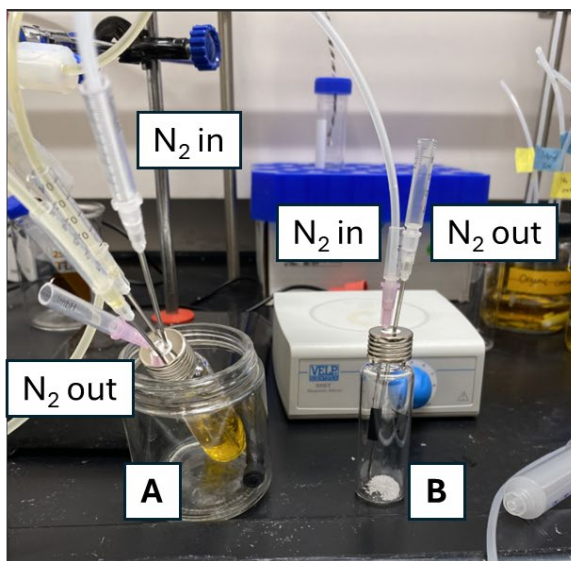
Supplementary Fig. 73: Experimental setup for CO₂ capture experiment in atmospheric condition using a crystallizing capture unit with the regenerated capture solution: (A) Schematic illustration of working principle of the crystallizing capture unit. **(B)** Image of setup for the capture experiment in atmospheric condition.



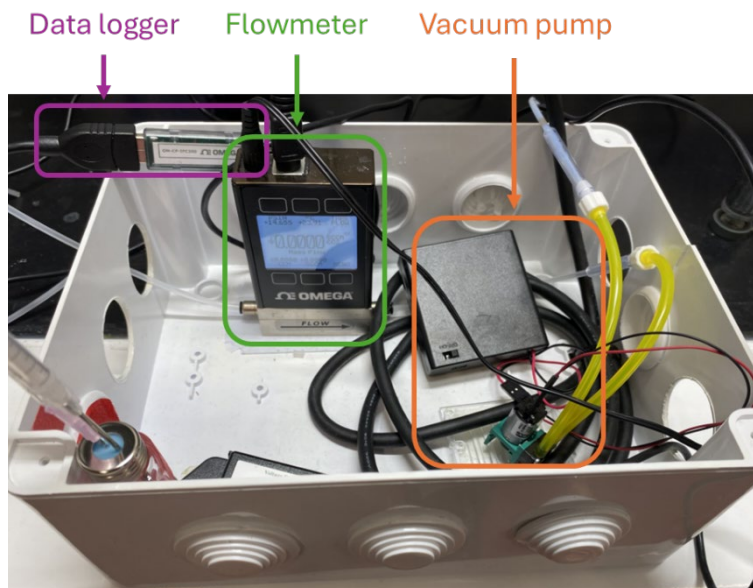
Supplementary Fig. 74: XRD measurement of salt precipitate and K_2CO_3 standard characteristic pattern. The characteristic pattern of salt precipitate matches the K_2CO_3 standard pattern.



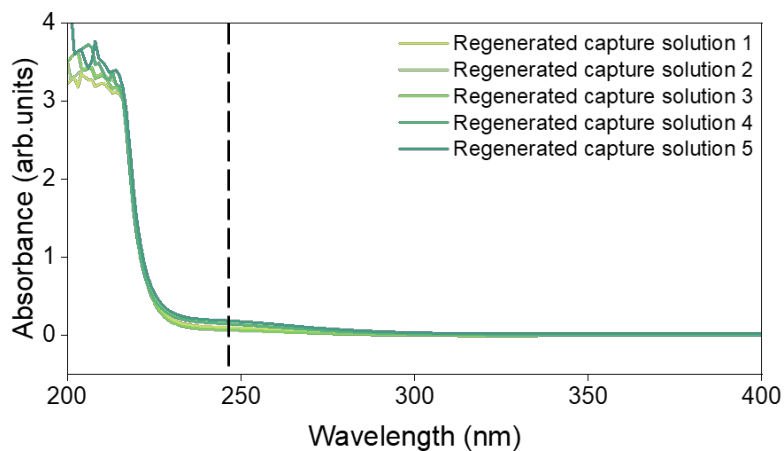
Supplementary Fig. 75: Experimental setup for redox-decoupled system. The redox-decoupled system operated and monitored by two electrochemical workstations. Both pH meter and flow monitor were used to record experiment data.



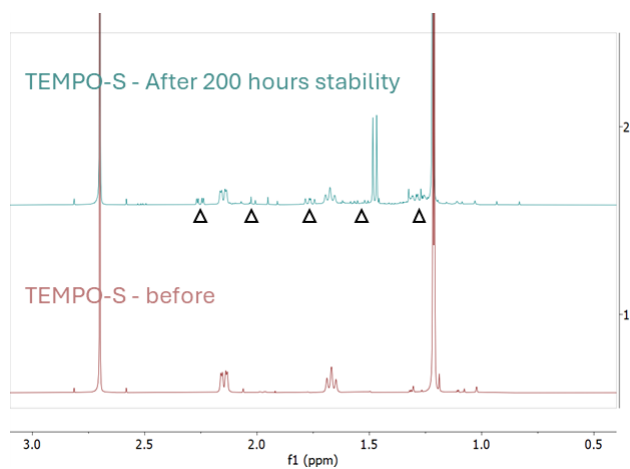
Supplementary Fig. 76: Experimental setup of the lab-scale K_2CO_3 transfer system. Solid K_2CO_3 was transferred using two sealed containers: A (CO_2 liberation solution tank) and B (K_2CO_3 transfer system), both maintained under N_2 protection throughout the experiment. At the end of each round, the electrolyte from container A was manually transferred to container B using a syringe. During this transfer, the N_2 supply to container B was temporarily turned off. Once the K_2CO_3 was fully dissolved, the solution was transferred back to container A for the next round of operation.



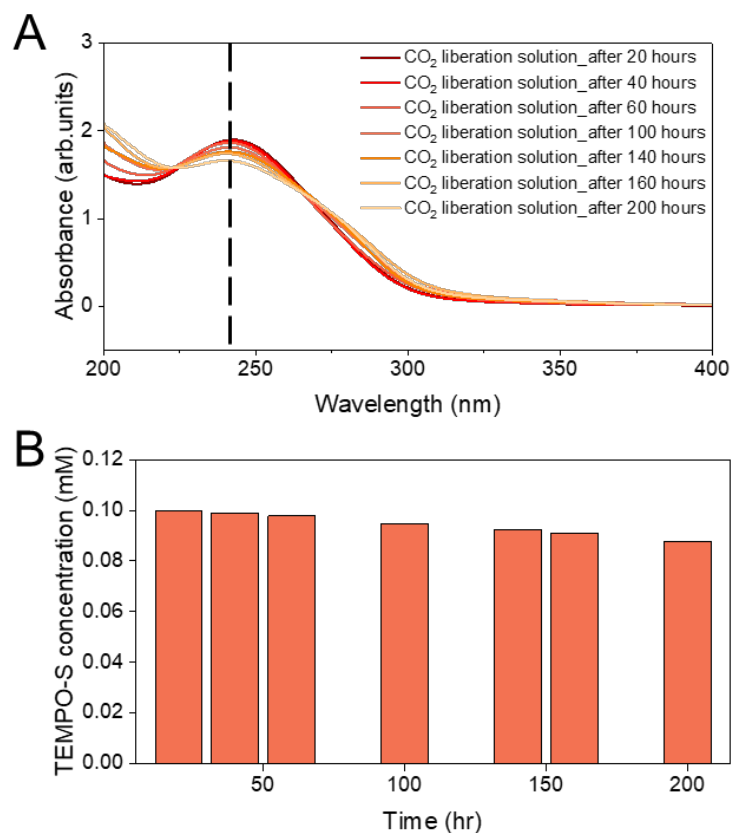
Supplementary Fig. 77: Experimental setup of the released CO₂ flow monitor. The monitoring system was connected to the outlet of the CO₂ liberation solution tank. To ensure that all recorded data represent only the released CO₂, a vacuum pump was continuously operated, and no external N₂ was introduced into the CO₂ liberation tank during the measurement period.



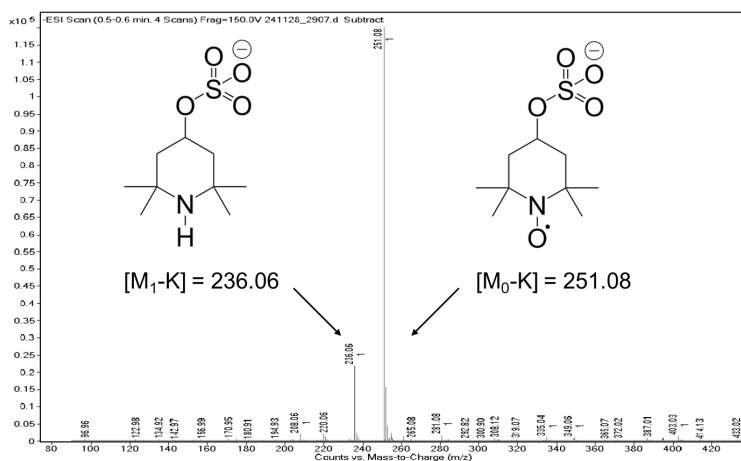
Supplementary Fig. 78: The UV-vis spectroscopy analysis for five different regenerated KOH capture solutions during the 200 hours stability test of redox-decoupled system. Negligible amount of TEMPO-S was detected at the 245 nm wavelength. Note that there is no dilution on these regenerated capture solutions for the UV-vis analysis; each regenerated capture solution was collected after > 10 hours of continuous electrochemical operation.



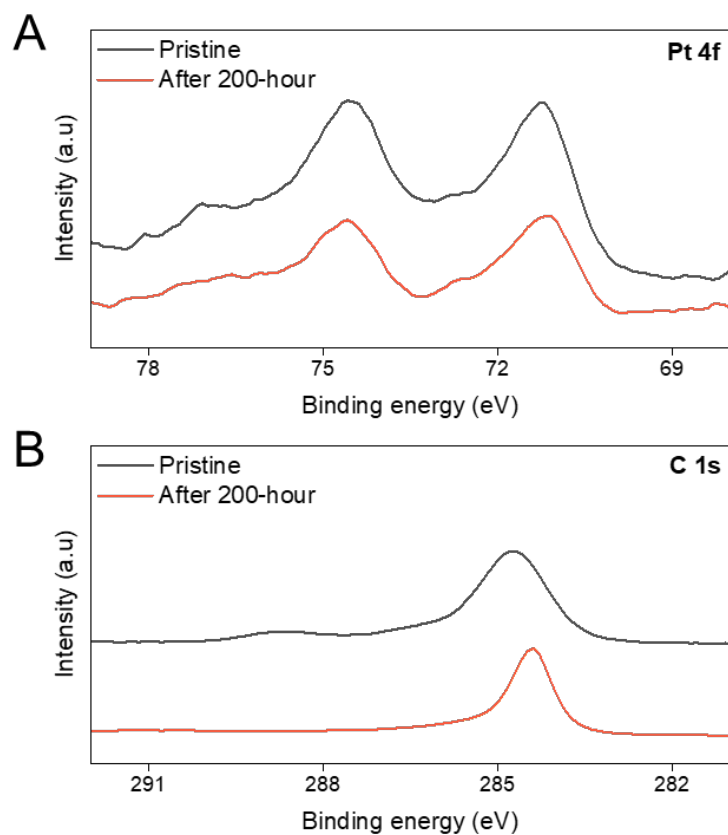
Supplementary Fig. 79: ¹H NMR spectra of the CO₂ liberation solution containing 0.1 M TEMPO-S in 0.25 M K₂SO₄ with 0.05 M K₂CO₃ chemical. Before and after 200 hours stability (Δ symbol indicates the potential TEMPO-S decomposition).



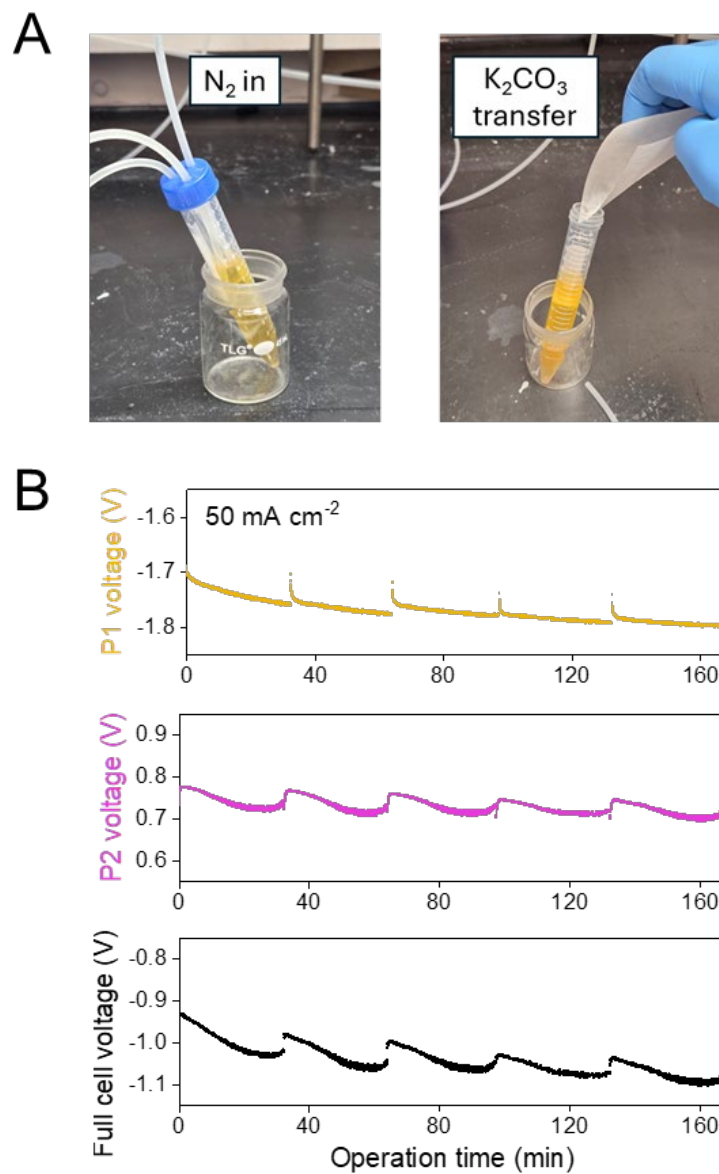
Supplementary Fig. 80: The UV-vis spectroscopy analysis for the CO₂ liberation solution during the redox-decoupled system 200 hours stability test. (A) UV-vis spectra of the CO₂ liberation solution over 200-hour operation. (B) Converting TEMPO-S concentration over 200-hour operation using the TEMPO-S calibration standard (Supplementary Fig. 66). The TEMPO-S signal at 245 nm.



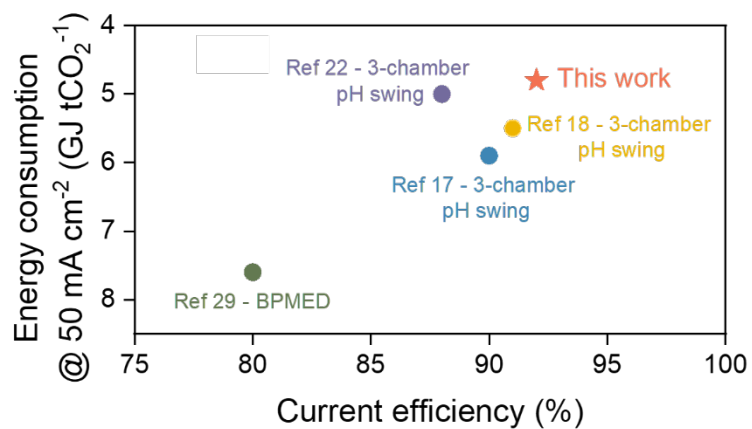
Supplementary Fig. 81: The high-resolution mass spectrometry profile of CO₂ liberation solution after 200-hour stability test. The m/z of calculated $[M_0-K]$: 251.08. The m/z of decomposition form calculated $[M_1-K]$: 236.06.



Supplementary Fig. 82: X-ray photoelectron spectroscopy (XPS) results of pristine and after 200-hour stability test electrodes. (A) Pt 4f spectra of PtC electrode. (B) C 1s spectra of carbon electrode.



Supplementary Fig. 83: The redox-decoupled system performance with a simplified setup. (A) Demonstration of simplified setup with less stringent protection. (B) Voltage profiles at 50 mA cm⁻².



Supplementary Fig. 84: Comparison of energy consumption at 50 mA cm⁻² and current efficiency in the benchmark eDAC approaches^{17,18,21,28}.

Supplementary Table 1. Summary of peak potential difference obtained from the cyclic voltammetry profiles of 20 mM TEMPO-OH at scan rates ranging from 10 mV s⁻¹ to 200 mV s⁻¹ at different TEMPO-OH:TEMPO-OH⁺ ratios

Molecules Scan rate [mV s ⁻¹]	18 mM TEMPO-OH (90%) : 2 mM TEMOM-OH ⁺ (10%)	10 mM TEMPO-OH (50%) : 10 mM TEMOM-OH ⁺ (50%)	2 mM TEMPO-OH (10%) : 18 mM TEMOM-OH ⁺ (90%)
	ΔE_p [mV]	ΔE_p [mV]	ΔE_p [mV]
10	96	106	102
20	107	119	117
30	115	129	127
50	125	144	142
80	139	161	160
100	147	173	171
150	164	192	190
200	172	207	206

Supplementary Table 2. Summary of peak potential difference obtained from the cyclic voltammetry profiles of 20 mM TEMPO-S at scan rates ranging from 10 mV s⁻¹ to 200 mV s⁻¹ at different TEMPO-S:TEMPO-S⁺ ratios

Molecules Scan rate [mV s ⁻¹]	18 mM TEMPO-S (90%) : 2 mM TEMOM-S ⁺ (10%)	10 mM TEMPO-S (50%) : 10 mM TEMOM-S ⁺ (50%)	2 mM TEMPO-S (10%) : 18 mM TEMOM-S ⁺ (90%)
	ΔE_p [mV]	ΔE_p [mV]	ΔE_p [mV]
10	102	96	95
20	117	104	102
30	131	111	109
50	145	122	120
80	162	134	132
100	173	139	137
150	193	154	152
200	207	165	162

Supplementary Table 3. Summary of peak potential difference obtained from the cyclic voltammetry profiles of 20 mM TEMPO-P at scan rates ranging from 10 mV s⁻¹ to 200 mV s⁻¹ at different TEMPO-P:TEMPO-P⁺ ratios

Molecules Scan rate [mV s ⁻¹]	18 mM TEMPO-P (90%) : 2 mM TEMOM-P ⁺ (10%)	10 mM TEMPO-P (50%) : 10 mM TEMOM-P ⁺ (50%)	2 mM TEMPO-P (10%) : 18 mM TEMOM-P ⁺ (90%)
	ΔE_p [mV]	ΔE_p [mV]	ΔE_p [mV]
10	137	144	141
20	174	173	170
30	201	198	189
50	241	234	224
80	286	278	264
100	320	306	285
150	367	344	332
200	406	378	361

Supplementary Table 4. Summary of peak reduction and oxidation current densities obtained from the cyclic voltammetry profiles of 20 mM TEMPO-OH at scan rates ranging from 10 mV s⁻¹ to 200 mV s⁻¹ at different TEMPO-OH:TEMPO-OH⁺ ratios

Molecules Scan rate [mV s⁻¹]	18 mM TEMPO-OH (90%) : 2 mM TEMOM-OH ⁺ (10%)		10 mM TEMPO-OH (50%) : 10 mM TEMOM-OH ⁺ (50%)		2 mM TEMPO-OH (10%) : 18 mM TEMOM-OH ⁺ (90%)	
	I _{p, red} [μA]	I _{p, oxid} [μA]	I _{p, red} [μA]	I _{p, oxid} [μA]	I _{p, red} [μA]	I _{p, oxid} [μA]
10	-55.71	88.26	-60.37	70.80	-70.71	58.17
20	-80.69	114.96	-81.97	96.34	-92.04	85.82
30	-97.29	136.23	-96.53	114.35	-104.95	105.38
50	-121.46	167.24	-117.37	140.32	-123.90	133.73
80	-148.62	199.49	-139.10	168.21	-143.83	163.94
100	-162.29	216.10	-150.42	182.89	-153.69	180.57
150	-189.06	251.53	-172.58	212.46	-174.77	212.28
200	-209.50	277.25	-189.24	234.59	-190.89	235.66

Supplementary Table 5. Summary of peak reduction and oxidation current densities obtained from the cyclic voltammetry profiles of 20 mM TEMPO-S at scan rates ranging from 10 mV s⁻¹ to 200 mV s⁻¹ at different TEMPO-S:TEMPO-S⁺ ratios

Scan rate [mV s ⁻¹]	Molecules	18 mM TEMPO-S (90%) : 2 mM TEMOM-S ⁺ (10%)		10 mM TEMPO-S (50%) : 10 mM TEMOM-S ⁺ (50%)		2 mM TEMPO-S (10%) : 18 mM TEMOM-S ⁺ (90%)	
		I _{p, red} [μA]	I _{p, oxid} [μA]	I _{p, red} [μA]	I _{p, oxid} [μA]	I _{p, red} [μA]	I _{p, oxid} [μA]
10		-52.54	93.52	-78.89	79.86	-95.94	69.09
20		-74.19	121.02	-104.20	104.30	-118.00	90.84
30		-89.10	139.75	-125.05	124.28	-139.45	110.95
50		-109.84	168.05	-155.43	153.62	-170.48	140.02
80		-131.06	198.60	-188.36	185.24	-203.20	172.48
100		-141.86	214.48	-205.88	201.72	-220.58	189.57
150		-163.01	246.87	-240.52	235.05	-255.56	223.61
200		-178.72	271.24	-266.57	259.63	-281.75	248.72

Supplementary Table 6. Summary of peak reduction and oxidation current densities obtained from the cyclic voltammetry profiles of 20mM TEMPO-P at scan rates ranging from 10 mV s⁻¹ to 200 mV s⁻¹ at different TEMPO-P:TEMPO-P⁺ ratios

Molecules Scan rate [mV s ⁻¹]	18 mM TEMPO-P (90%) : 2 mM TEMOM-P ⁺ (10%)		10 mM TEMPO-P (50%) : 10 mM TEMOM-P ⁺ (50%)		2 mM TEMPO-P (10%) : 18 mM TEMOM-P ⁺ (90%)	
	I _{p, red} [μA]	I _{p, oxid} [μA]	I _{p, red} [μA]	I _{p, oxid} [μA]	I _{p, red} [μA]	I _{p, oxid} [μA]
10	-17.22	41.25	-29.40	27.85	-33.96	22.25
20	-24.99	47.93	-36.44	34.44	-41.58	28.80
30	-29.87	52.24	-41.95	39.16	-47.52	33.58
50	-37.00	59.44	-50.30	46.11	-55.98	40.38
80	-45.11	66.99	-59.71	53.37	-65.83	47.61
100	-49.59	70.80	-65.00	57.04	-71.44	50.85
150	-59.11	79.16	-75.87	65.67	-85.34	58.45
200	-67.11	85.42	-84.81	70.53	-91.16	64.12

Supplementary Table 7. Summary of D_{TEMPO} , D_{TEMPO^+} , and k^0 of three TEMPO derivatives were determined by the Randles-Sevcik and Nicholson method

Molecule	D_{TEMPO} [$\text{cm}^2 \text{s}^{-1}$]	D_{TEMPO^+} [$\text{cm}^2 \text{s}^{-1}$]	k^0 [cm s^{-1}]
TEMPO-OH	5.60×10^{-6}	3.74×10^{-6}	2.41×10^{-3}
TEMPO-S	6.14×10^{-6}	5.56×10^{-6}	2.72×10^{-3}
TEMPO-P	6.22×10^{-7}	6.17×10^{-7}	1.01×10^{-4}

Supplementary Table 8. Summary of crystallizing air capture unit experimental results

Sample #	Before OH ⁻ concentration / volume (M / ml)	After OH ⁻ concentration / volume (M / ml)	Theoretical K ₂ CO ₃ formation (g)	Experimental K ₂ CO ₃ formation (g)	$\eta_{\text{precipitate}}$
1	0.948 / 30	1.005 / 22.8	0.3807	0.3609	0.948
2	0.922 / 30	0.977 / 23.3	0.3381	0.3119	0.923
3	0.922 / 30	0.977 / 21.2	0.4798	0.4468	0.931

Supplementary Table 9. Summary of theoretical H₂ production rate at the P1 cathode, experimental H₂ production rate at the P1 cathode, and experimental H₂ consumption rate at the P2 anode

Current density [mA cm⁻²]	Theoretical H₂ production rate at P1 cathode [sccm]	Experimental H₂ production rate at P1 cathode [sccm]	Experimental H₂ consumption rate at P1 cathode [sccm]
100	0.748	0.741	0.763
150	1.121	1.113	1.153
200	1.495	1.487	1.527
250	1.869	1.863	1.903
300	2.243	2.237	2.289
350	2.616	2.610	2.675
400	2.990	2.985	3.034

Supplementary Table 10. Summary of four major DAC/DOC capture and release systems

Capture system	CO ₂ source	Current density (mA cm ⁻²)	Capture solution (M / pH)	Current efficiency (%)	Stability (hour)	Energy inputs (GJ tCO ₂ ⁻¹)	Capture rate-normalized energy intensity (GJ m ² yr t ⁻²)
BPMED	HCO ₃ ⁻	5 – 80	0.1 M / 13	~50	/	3.4 – 7.4	1.5 – 3.4 ²⁹
	HCO ₃ ⁻	20 – 140	0.1 M / 13	~80	/	4.5 – 9.1	2.0 – 4.1 ²⁸
PCET pH-swing	CO ₃ ²⁻	10	0.05M / 12.7	99	22	2.9	1.3 ³⁰
PCET non-aqueous	ROCO ₂ ⁻	10	/	~80	55	4.4	2.0 ³¹
	CO ₃ ²⁻	0.5 – 3	/	~40-100	72	4.5 – 6.8	2.0 – 3.1 ³²
	CO ₃ ²⁻	1 – 200	0.5 M / 13.7	~80	100	2.3 – 9.5	0.5 – 2.2 ²¹
3-chamber pH-swing	46% v/v HCO ₃ ⁻ + 54% v/v CO ₃ ²⁻	2.5 – 15	0.2 M / 13.3	80	/	4.9 – 10.2	2.2 – 4.6 ³³
	HCO ₃ ⁻	2 – 50	/	91	1.7	3.5 – 8.8	1.6 – 4.0 ³⁴
	CO ₃ ²⁻	10 – 200	0.1 M / 13.1	~90	100	6.1 – 7.7	2.8 – 3.5 ¹⁷
	CO ₃ ²⁻	10 – 120	0.2 M / 13.3	~90	200	3.8 – 8.5	1.7 – 3.9 ¹⁸
This work	HCO ₃ ⁻	10 – 200	1 M / 14	86 – 92	200	2.0 – 4.4	0.1 – 0.2
	CO ₃ ²⁻	10 – 200	1 M / 14		200	4.0 – 8.7	0.2 – 0.4

Note: the DAC capture rate for a 1 M OH⁻ capture solution was estimated at 22 tCO₂·m⁻²·yr⁻¹, while a 0.5 M OH⁻ solution yielded an estimated rate of 4.4 tCO₂·m⁻²·yr⁻¹, and < 0.2 M OH⁻ solutions estimated 2.2 tCO₂·m⁻²·yr⁻¹^{24,35}.

Supplementary Table 11. Benchmark reported TEMPO derivatives employed in aqueous organic redox flow batteries.

Redox-active molecules	Operation pH range	Maximum current density (mA cm ⁻²)	Concentration (mol L ⁻¹)	Capacity fade rate (% per day)	Reference
TEMPMA	Neutral	200	2	1.1 @ 80 mA cm ⁻²	36
TMAP-TEMPO	Neutral	280	1.5	0.6 @ 100 mA cm ⁻²	37
PSS-TEMPO	Neutral	900	0.15	0.7 @ 40 mA cm ⁻²	38
Pyr-TEMPO	Neutral	325	0.5	3.47 @ 40 mA cm ⁻²	39
This work	3 – 11	200	0.1	1.1 @ 50 mA cm⁻²	

Supplementary Table 12. Benchmark electrochemically mediated CO₂ capture systems that employ redox-active organics and other mechanisms.

Mechanism	Redox-active molecules	CO ₂ sources	Current density (mA cm ⁻²)	Stability (hr)	Capacity fade rate (% per day)	Reference
EMCC	1-aminopyridium nitrate	air	50	8.4	6.6	40
EMCC	Anthraquinone	Simulated flue gas (15% CO ₂ + 3% O ₂)	0.5	34	15	41
EMAR	50% ethylenediamine + 50% aminoethylethanol-amine	Pure CO ₂	2.2	100	0.72	42
Redox-mediated HER/HOR pH-swing	1,1'-bis(3-phosphonopropyl)-[4,4'-bipyridine]-1,1'-diium	air	40	200	0.7	18
HER/HOR pH-swing	N/A	air	100	100	N/A	21
ORR/OER pH-swing	N/A	Simulated flue gas (14% CO ₂ + 8% O ₂)	100	72	N/A	32
HER/HOR pH-swing	N/A	air	50	2	N/A	43
Redox-mediated HER/HOR pH-swing	TEMPO-S	air	50	200	1.1	

Note: the EMCC stands for Electrochemically mediated carbon capture; EMAR stands for electrochemically mediated amine regeneration.

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